

MINISTRY OF EDUCATION AND TRAINING

STATE BANK OF VIETNAM

HO CHI MINH UNIVERSITY OF BANKING



BACHELOR THESIS

**IMPACT OF CAPITAL STRUCTURE ON FINANCIAL
PERFORMANCE OF THE CONSTRUCTION COMPANIES
LISTED ON THE STOCK EXCHANGE OF VIETNAM**

MAJOR: FINANCE AND BANKING
CODE: 7340201

HOANG BAO NGOC

HO CHI MINH CITY, 2025

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HO CHI MINH CITY, 2025

ABSTRACT

“The Impact of Capital Structure on the Financial Performance of Construction Companies Listed on the Vietnam Stock Exchange” is conducted to analyze and evaluate the effects of capital structure and factors influencing financial performance of construction companies listed on the Vietnam Stock Exchange during the period 2015–2024. All data used in this thesis are secondary data, collected from two main sources: (1) The Economic-Financial-Macro Database of Vietnam (FiinPro-X), to collect data from audited financial statements of 42 construction companies listed on the Vietnam Stock Exchange; (2) The World Bank, to collect macroeconomic data. In the research model, the target variable representing financial performance of listed construction companies on the Vietnam Stock Exchange is Return on Assets (ROA). Meanwhile, the feature variables reflecting capital structure include: Short-term Debt Ratio (STDR), Long-term Debt Ratio (LTDR), and Total Debt Ratio (TDR). In addition to the main feature variables, the thesis incorporates control variables into the research model to provide a more comprehensive understanding of the capital structure and financial performance of listed construction companies on the Vietnam Stock Exchange, while also considering firms’s financial performance within the broader macroeconomic context.

By applying machine learning algorithms, specifically Random Forest, the study identified structural capital factors such as STDR, LTDR, and TDR as having a negative impact on corporate financial performance. Additionally, variables such as TANG, SIZE, and LIQ also indicate an inverse relationship with ROA. Meanwhile, variables GDP and INF have a positive impact on ROA, indicating their positive influence on the financial performance of the company.

Keywords: capital structure, financial performance

TÓM TẮT

Bài khóa luận với tên đề tài “ Tác động của cấu trúc vốn đến hiệu quả tài chính của các doanh nghiệp ngành xây dựng được niêm yết trên Sở giao dịch Chứng khoán Việt Nam” thực hiện phân tích và đánh giá tác động của cấu trúc vốn đến hiệu quả tài chính của các doanh nghiệp ngành xây dựng được niêm yết trên Sở giao dịch chứng khoán Việt Nam trong giai đoạn từ năm 2015 đến 2024. Dữ liệu nghiên cứu được sử dụng trong bài khóa luận trên là dữ liệu thứ cấp, được tác giả tổng hợp từ hai nguồn: (1) The Economic-Financial-Macro Database of Vietnam (FiinPro-X), nhằm thu thập dữ liệu từ các báo cáo tài chính đã qua kiểm toán của 42 doanh nghiệp ngành xây dựng được niêm yết trên Sở Giao dịch Chứng khoán Việt Nam; (2) Ngân hàng thế giới nhằm thu thập các dữ liệu kinh tế vĩ mô. Trong nghiên cứu này, biến mục tiêu được sử dụng để đo lường hiệu quả tài chính của các doanh nghiệp ngành xây dựng là Tỷ suất sinh lợi trên tổng tài sản (ROA); bên cạnh đó, các biến đầu vào đại diện cho cấu trúc vốn là Tỷ lệ nợ ngắn hạn (STDR), Tỷ lệ nợ dài hạn (LTDR), Tỷ lệ nợ trên tổng tài sản (TDR).

Kết quả sau khi nghiên cứu áp dụng các thuật toán học máy, thuật toán học máy được sử dụng trong bài là Random Forest, đã cho thấy kết quả rằng các yếu tố cấu trúc vốn như STDR, LTDR và TDR có tác động tiêu cực đến hiệu quả tài chính của các doanh nghiệp. Kế đến, kết quả cũng đã chỉ ra rằng các biến như TANG, SIZE, LIQ có tác động ngược chiều với ROA. Cuối cùng các biến GDP, INF, LIQ đã chỉ ra tác động cùng chiều đối với hiệu quả tài chính của doanh nghiệp. Từ kết quả nghiên cứu, tác giả sẽ đưa ra các hàm ý quản trị nhằm tối ưu cấu trúc vốn của doanh nghiệp, nâng cao hiệu quả tài chính của các doanh nghiệp xây dựng.

Từ khóa: Cấu trúc vốn, Hiệu quả tài chính

DECLARATION OF AUTHENTICITY

As the author of this bachelor thesis, I would like to make the following commitments: My name is Hoang Bao Ngoc, currently a student in the Finance and Banking department, Banking University of Ho Chi Minh City with the graduation thesis title: " The Impact of Capital Structure on the Financial Performance of Construction Companies Listed on the Vietnam Stock Exchange ". This thesis is entirely the author's own research work, and the research results presented are completely honest. In the research paper, there is absolutely no previously published content or content made by others except for citations that are fully cited in the thesis.

I take full responsibility for my statement.

Ho Chi Minh City, July 4th, 2025

Author

Hoàng Bảo Ngọc

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Finally, I would also like to send my sincere thanks to ***my family*** for always supporting me during the process of completing my research.

Because the time to complete my graduation thesis is limited, it is difficult for me to avoid shortcomings during the research process. Therefore, I would like to receive suggestions and comments from teachers. The suggestions provided will be of maximum value, assisting in enhancing your skills and knowledge in the future.

I sincerely thank you!

Ho Chi Minh City, July 4th, 2025

Author

Hoàng Bảo Ngọc

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LIST OF ABBREVIATIONS

ABBREVIATIONS	MEANING
CS	Capital Structure
CV	Cross-validation
Fiinpro - X	The Economic-Financial-Macro Database (Vietnam)
FP	Financial Performance
FS	Financial Statement(s)
GDP	GDP Growth Index
INF	Inflation Rate
LIQ	Liquidity Ratio
LTDR	Long-Term Debt Ratio
ROA	Return on Assets
ROE	Return on Equity
SIZE	Firm Size
STDR	Short-Term Debt Ratio
TANG	Fixed Asset Ratio
WB	WorldBank

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CHAPTER 1: INTRODUCTION

In the first chapter, the thesis will provide an overview of the research topic, including its relevance, research objectives, research questions, scope and subject of the study, research methods to be used, and the structure of the research. This will establish a solid foundation for the research and ensure consistency throughout all stages of the study.

1.1 Rationale of the Study

Overview of the development potential of the construction industry in Vietnam: Despite ongoing fluctuations in the residential market, the industrial construction sector and the construction industry as a whole have shown significant improvement and recorded strong growth trends over the years. At the same time, the trend toward industrialization and the benefits from the continuous increase in foreign direct investment have also been driving forces behind the development of this sector. Additionally, the trend toward increasing industrial contracts by civil contractors remains strong. Contractors have intensified their strategies to increase the proportion of industrial projects in their portfolios. The construction industry is also recognizing the strong growth potential of the industrial real estate segment in the coming period, supported by the trend toward green industrial zones from 2025 to 2030, accompanied by strict construction requirements. This presents significant opportunities for construction companies with experience in implementing projects that meet international standards such as LEED and EDGE.

After the economic downturn caused by the Covid-19 pandemic, Vietnam's economy has entered a period of strong recovery in a new economic cycle. In this context, the construction industry plays an important role in the strong recovery of the economy.

The current construction industry is experiencing positive growth prospects, with foreign direct investment (FDI) reaching 15.35 billion USD in 2024, an increase of 9.4% compared to the same period last year. Additionally, the industry is currently benefiting

from government economic support policies, including the issuance of Decision 2227/QĐ-TTg and Resolution 39/2021/QH15, which aim to develop 210,000 hectares of industrial land. Based on these favorable conditions, construction companies listed on the Vietnam Stock Exchange have reported positive results related to profit growth. Despite many positive growth signals, the industry is facing numerous challenges such as cash flow fluctuations and new stringent standards and policies from the government. This has posed threats to the financial performance and profits of construction companies. Macroeconomic fluctuations have exerted pressure on the capital structure and financial performance of companies in the industry. Capital structure is being impacted by high debt ratios and difficulties in disbursing funds for long-term projects. These issues underscore the importance of capital structure management, alongside the need to study the impact of capital structure on financial performance to enhance financial efficiency.

Capital structure (CS) is considered one of the most important factors that play a key role in directly influencing business results and ensuring the stability of business operations. An appropriate capital structure is the key to optimizing the allocation of financial resources, helping managers minimize financial risks.

Currently, in Vietnam, business managers have not yet determined how to manage and analyze CS fluctuations. This has led to many businesses currently facing difficulties in mobilizing and using capital effectively. Additionally, the lack of comprehensive analysis of CS can result in inefficient production and operations, inability to preserve capital, and may directly threaten the survival of the business.

CS is one of the key factors influencing a company's profitability and risk management, particularly in the construction industry, which is characterized by short project cycles, high profit margins, and significant capital requirements. However, there has been no clear methodology for identifying, measuring, and managing CS in the construction industry. Additionally, the use of diverse data sources and estimation methods has led to inconsistencies in previous empirical studies on the impact of capital structure on the financial performance of listed construction companies on the Vietnam Stock Exchange. This inconsistency poses limitations in formulating capital structure

allocation policies for managers. Furthermore, through a review of previous studies in Vietnam, the author found that there are few empirical studies related to the industry updated to 2024, and that new industry information and data have not been updated to date. From this, the author identifies the research gap of the topic, combined with the potential application of machine learning models to analyze the relationship between capital structure and financial performance in a more comprehensive manner.

In the context of the industrial revolution 4.0 along with the explosion of science and technology, artificial intelligence (AI) has contributed to solving many problems in society, especially in the process of scientific research. In addition, the application of machine learning models enables research to be applied to larger data sets, while also improving model performance and increasing the ability to explain correlations between target variables and feature variables. Finally, the application of machine learning models helps improve the quality of research and ensure the accuracy of predictions from research.

In addition, the capital structure of construction companies in Vietnam has some unique characteristics, which are influenced by the nature of the industry as follows: The project cycle in the construction industry is prolonged due to many factors affecting the project situation. During the project investment process, companies invest a significant amount of capital, but this capital is spread out as it must be disbursed in accordance with the progress of the project. As a result, the capital used in the construction industry requires large investments spread out over a long period of time. This leads to delayed capital recovery as payments often extend beyond project completion, creating a high demand for working capital to ensure timely payment capabilities for businesses.

Based on the above reasons, the author has chosen the topic: “The Impact of Capital Structure on the Financial Performance of Listed Industrial Construction Companies on the Vietnam Stock Exchange.” From there, the study analyzes the relationship between capital structure and financial performance of firms, while proposing optimal solutions for capital structure optimization. The research findings will

contribute to enhancing the competitiveness and sustainable development of construction companies in the context of economic integration and green transition.

1.2 Objectives of the study

1.2.1 General Objectives

The general objective of this study is to determine and applying Machine Learning models to measure the impact of capital structure on the financial performance of listed construction companies in the Vietnamese stock market. Based on this, the study proposes management implications to help managers effectively allocate capital structure to enhance the financial performance of construction company in the future.

1.2.2 Specific Objectives

From the general objective, the study identifies the following specific research objectives:

First, determine the capital structure of the capital structure that affect the financial performance of construction companies listed on the Vietnam Stock Exchange.

Second, using Machine Learning algorithms to measure the extent and determine the direction of the impact of capital structure on the financial performance of construction companies listed on the Vietnam Stock Exchange.

Third, providing management implications about capital structure to enhance the financial performance of construction companies listed on the Vietnam Stock Exchange.

1.2.3 Research questions

The study will address the following questions:

First, does the capital structure and its components affect the financial performance of listed construction companies on the Vietnam Stock Exchange?

Second, what is the extent and direction of the impact of capital structure on the financial performance of construction companies listed on the Vietnam Stock Exchange?

Third, what are the management implications for the use and management of capital structure to improve the financial performance of construction companies listed on the Vietnam Stock Exchange?

1.4 Research Objectives and Scope

1.4.1 Research Objectives

This study focuses on analyzing the impact of capital structure on the financial performance of 42 listed construction companies on the Vietnam Stock Exchange. The financial performance measures used in this study are ROA.

1.4.2 Scope of the study

- Spatial scope:

The study sample will be analyzed based on 42 construction companies listed on the Vietnam Stock Exchange (Appendix 1).

According to business databases, there are a total of 104,732 companies operating in the construction industry in Vietnam. This study uses data from 42 leading companies in the construction industry. The selection criteria are based on the top companies with the largest market capitalization in the industry.

- In terms of time:

This study uses secondary data from the Fiinpro-X software and macroeconomic data from the World Bank, with an observation period from 2015 to 2024. The reason the author selected the 2015–2024 period is that during this time, the construction industry in Vietnam, as well as the Vietnamese economy as a whole, experienced significant periods of volatility.

For example, during the 2014–2018 period, alongside the recovery of the real estate market, the construction, installation, and infrastructure development sectors experienced robust growth. According to the socio-economic development strategy report, the construction industry played a strategic role and held an important position in Vietnam's economic development during this period. Additionally, during the 2019–2021 period, the construction industry was severely impacted by the COVID-19 pandemic, which caused significant and direct fluctuations in profits, leading to a decline in business performance for companies. After overcoming this phase, from 2021 to 2024, construction companies and the real estate market in Vietnam are expected to rebound strongly, driven by the increasing inflow of foreign direct investment (FDI). Moreover,

despite the rapid increase in industrial zone occupancy rates in key provinces, Vietnam still has an abundant supply of industrial land, which is one of the positive factors contributing to the development potential of the construction industry in Vietnam during the 2015–2024 period. The period from 2015 to 2024 was selected for the study because the data from this period is closest to the study's target year of 2025, enabling the research to provide the most accurate assessments and insights aligned with current realities.

1.5 Research Methods

- Regarding data collection and synthesis:

The data used in this study is secondary data. Financial variable data was provided by the Fiinpro – X data software platform, enabling the study to collect and compile data from audited consolidated financial statements of 42 listed construction companies on the Vietnam Stock Exchange from 2015 to 2024. Additionally, data on macroeconomic variables were collected by the author from the World Bank.

After collection, the data will be compiled into panel data. Subsequently, to achieve optimal performance for the research model, this panel data will be further processed using the Python programming language.

- Regarding research methods:

The thesis uses both qualitative and quantitative methods in the research process.

Qualitative method

First, the study uses a synthetic analysis method in constructing the theoretical framework, identifying research gaps, and developing a research model through a review of relevant empirical studies.

Next, the descriptive statistical method is also applied in this thesis to describe the characteristics and changes in the financial performance of the construction industry under the management and use of capital structure by listed construction companies on the Vietnam Stock Exchange. From there, a qualitative analysis of the impact of capital structure on the financial performance of construction companies listed on the Vietnam Stock Exchange is conducted.

Finally, the comparative method is applied to compare relative and absolute data in the research data. This enables an analysis and evaluation of the current status of capital structure utilization and management at construction companies listed on the Vietnam Stock Exchange.

Quantitative method

This thesis employs quantitative methods to analyze the impact of capital structure on the financial performance of construction companies listed on the Vietnam Stock Exchange. To enhance the understanding of the research data and optimize the model's performance, the study conducted an additional data cleaning process based on the collected tabular database and integrated previously available data. The entire process of data processing and analysis are carried out using the Python programming language. Specifically, the thesis checked for outliers, examined the distribution of the data, and then standardized the data by applying the Robust Scaler data standardization method. Next, the thesis applies methods such as descriptive statistics, Pearson correlation analysis, model analysis using the Random Forest algorithm. Among these, the study is found that the algorithm with the best performance is Random Forest algorithm. Based on the above, this thesis will apply the algorithm in the regression analysis to examine the research hypotheses.

1.6 Contribution of the study

In terms of theory, the study clearly identifies the relationship between capital structure and financial performance of firms. The research topic is significant in systematizing the theoretical basis for the relationship between capital structure and financial performance. From there, it clarifies the extent of the impact of capital structure on the financial performance of construction companies.

In practical terms, the study provides empirical evidence that helps managers understand the role and influence of capital structure on the financial performance of construction companies listed on the Vietnam Stock Exchange. From this, managers can develop more effective capital management policies to optimize the financial performance of their companies. The information and results from this study will also

provide a comprehensive view of capital structure and financial performance, thereby contributing to the development of financial policies for companies in the industry. In addition, managers can use the information from the study to update and supplement management policies to optimize capital structure, enhance business efficiency, and minimize financial risks. The capital structure analyses in this study can be used to adjust financial strategies, optimize capital structure, enhance financial efficiency, and guide the sustainable development of construction companies listed on the Vietnam Stock Exchange.

1.7 Research structure

The study is structured into five chapters, with the general content of each chapter presented as follows:

CHAPTER 1: INTRODUCTION

In the first chapter, this chapter will provide an overview of the research topic, including an introduction to the urgency of the topic, the reasons for selecting the topic, the objectives of the research, the research questions, the scope and subject of the research, the methods used in the research, the contributions of the topic, and the overall structure of the research paper.

CHAPTER 2: RESEARCH OVERVIEW

In Chapter 2, the author will present the theoretical foundations related to capital structure, financial performance of firms, and the impact of capital structure on financial performance of firms; alongside a review of domestic and international studies.

CHAPTER 3: RESEARCH METHOD

In Chapter 3, the author will present the research methods, procedures, and scale construction in the study, the data collection process, and the data analysis techniques used in the research and model development to provide a basis for comparing the research results in Chapter 4.

CHAPTER 4: RESEARCH RESULTS ANALYSIS AND DISCUSSION

Within the scope of Chapter 4, the study will present descriptive statistics of the research variables, standardize data using Robust Scaler, analyze Pearson correlation

matrices, and then apply the Random Forest algorithm in regression analysis. Next, the study will present and discuss the research results, comparing them with the previously established research hypotheses and the research results of previous authors. From there, the impact of capital structure on the financial performance of construction companies listed on the Vietnam Stock Exchange will be clarified.

CHAPTER 5: CONCLUSION AND MANAGEMENT IMPLICATIONS

Based on the results above, Chapter 5 will provide an overview of the study, its limitations, and recommendations to propose management implications, thereby building an appropriate capital structure to enhance the financial performance of construction companies listed on the Vietnam Stock Exchange.

SUMMARY OF CHAPTER 1

The introductory chapter clarified the urgency of conducting this study, emphasizing that the research objective was to analyze the impact of capital structure on the financial performance of construction companies listed on the Vietnam Stock Exchange. In addition, the study established research questions and defined the scope of the study as an analysis of 42 construction companies listed on the Vietnam Stock Exchange from 2015 to 2024. The study also highlighted the contribution of the topic to business managers and future research in the same field.

Finally, this chapter provided an overview of the research structure, which was divided into five main chapters, from introduction to conclusion and management implications.

Based on this, the thesis will develop a theoretical framework related to the research subject and conduct a review of related studies in the next chapter.

CHAPTER 2: RESEARCH OVERVIEW

In Chapter 2, the author will provide a detailed explanation of the concepts and measurement methods for capital structure and financial performance of construction companies listed on the Vietnam Stock Exchange. This chapter will also introduce and analyzes theories on capital structure, including the Modigliani & Miller theory, the trade-off theory, and the pecking order theory. Additionally, this chapter will provide an overview of empirical evidence from both domestic and international sources to offer a comprehensive view of the current state of research and identify new insights from gaps in previous studies.

2.1 Theory of Capital Structure

2.1.1 Capital Structure

According to Niu (2008), capital structure is the combination of long-term financial resources, including common stock, preferred stock, bonds, long-term loans, and retained earnings, which a company uses to finance its investment and production activities. Capital structure is the combination of two elements, equity capital and debt capital, which the company maintains through financial decisions.

According to Paramasivan (2009), capital structure refers to the relationship between long-term financial sources such as equity and debt. Capital structure plays a crucial role in determining a firm's value. The decisions made by a firm's managers regarding capital structure play a vital role in determining the company's financial performance, as this directly affects the company's cost of capital and profitability. In the infrastructure construction industry, where projects often face high risks due to market fluctuations, a well-managed capital structure is a key factor in helping businesses optimize financial efficiency and maintain financial stability.

Gitman and Zutter (2012) describe capital structure as the combination of long-term debt and equity that a company maintains to finance its operations. Companies typically strive to maintain a reasonable capital structure, in which the ratio of debt to equity is adjusted to align with the company's financial objectives. For the construction industry in Vietnam, maintaining this target capital structure is a matter of survival for

businesses, serving as a critical balancing act between financial risk and profit, particularly in the face of frequent economic fluctuations such as interest rates and inflation.

2.1.2 Capital structure indicators

Debt-to-total assets ratio (TDR)

According to Bui et al. (2023), the capital structure includes short-term debt ratio (STDR), long-term debt ratio (LTDR), and total debt-to-total assets ratio (TDR), which play a decisive role in balancing financial risk and profitability. Companies can flexibly adjust their capital structure by issuing stocks, bonds, or utilizing loans to align with long-term objectives and market conditions.

According to Gill et al. (2011), the lower the debt-to-total-assets ratio, the better the company's ability to repay its debts. For companies, a lower debt-to-total-assets ratio increases the likelihood of repaying debts. Conversely, the higher this ratio, the lower the company's ability to repay its debts. A low debt-to-total assets ratio indicates that the company is managing its capital well and is able to repay its debts on time.

$$\text{Debt} - \text{to} - \text{total Assets Ratio (TDR)} = \frac{\text{Total Debt}}{\text{Total Assets}}$$

In this study, the author uses the short-term debt ratio (STDR); the long-term debt ratio (LTDR) to measure capital structure and the debt-to-total assets ratio (TDR). According to Abor (2005); Olaniyi et al. (2015); Hang et al. (2022).

Short-term debt ratio (STDR)

According to Tailab (2014), short-term debt is a financial indicator reflecting the percentage of short-term liabilities relative to total assets of a firm. This indicator helps assess the extent of short-term debt usage in the capital structure and the firm's ability to cope with short-term payment pressures.

$$\text{Short} - \text{term Debt Ratio (STDR)} = \frac{\text{Short} - \text{term Debt}}{\text{Total Assets}}$$

Long-term debt ratio (LTDR)

According to Jannati (2014) and Broyles (2003), the long-term debt ratio reflects the extent to which assets are financed by long-term debt, and serves as a tool to assess financial risk and profitability of a company.

According to El-Sayed Ebaid (2009), Nassar (2016), and Nguyen and Nguyen (2020), the long-term debt ratio was used to analyze its impact on a company's financial performance, utilizing indicators such as ROA and ROE in their studies.

$$\text{Long-term Debt Ratio (LTDR)} = \frac{\text{Long-term Debt}}{\text{Total Assets}}$$

2.2 Theory of Financial Performance

2.2.1 Concept of Financial Performance

According to Bui et al. (2023), financial performance is the ability of a firm to use its resources to achieve optimal results. This is reflected in indicators such as return on equity (ROE), return on assets (ROA), and performance ratios for asset, capital, and cost utilization.

According to Egbunike & Okerekeoti (2018), financial performance reflects the degree of effectiveness and efficiency in using resources to achieve profit objectives and increase economic value for the business.

Suhadak and colleagues (2019) add that financial performance reflects a company's ability to achieve its financial objectives through the optimization of operational efficiency and resource management, and is an important indicator of financial health for investors and stakeholders.

The author derives the concept of financial performance as follows: “Financial performance is a concept that reflects the degree of success of a firm in using financial resources to generate profits for that firm.”

2.2.2 Financial performance indicators

The above performance indicators are evaluated by previous studies based on the results of financial statements over a specific period of time. There are various ways to measure a company's profitability, and each indicator has a different meaning depending

on the purpose of the evaluation. The indicator frequently used in research is return on assets (ROA). The choice of indicators for evaluation depends on the nature, purpose, and requirements of the research; however, the main objective of financial performance improvement strategies is to maximize profits.

Financial performance is measured by various criteria. There are many ratios used to measure a company's financial performance, among which the most commonly used is Return on total assets (ROA). Numerous studies both domestically and internationally have utilized these ratios to evaluate corporate financial performance, including those by Hang et al. (2022), Olaniyi et al. (2015), Aljamaan (2018), and Sharifa (2017).

Return on total assets (ROA)

According to Damodaran (2014), the return on assets (ROA) of a company measures the efficiency of its operations in generating profits from its assets, before the effects of financing.

According to Hang et al. (2022), Return on Assets (ROA) is a financial ratio that reflects financial performance. This ratio is measured by dividing net income for the year by total assets. ROA measures the effectiveness of a company's management in converting available assets into profits. Thus, ROA indicates the efficiency of a company's use of assets to generate income, measuring the extent to which the company converts assets into profits, thereby showing investors the company's asset management capabilities and operational efficiency. According to Ravinder Singh (2024), a higher ROA indicates that the company is using its assets efficiently, generating more profit per unit of assets owned. According to Bui et al. (2023), Hang et al. (2022), Nguyen and Nguyen (2020), ROA is determined by:

$$\text{Return on Assets (ROA)} = \frac{\text{Net Profit}}{\text{Total Assets}}$$

Based on a review of empirical studies focusing on the impact of capital structure on firms' financial performance, the author found that the ROA ratio is the most commonly used in research models. Building on the aforementioned empirical studies,

this thesis will select the ROA ratio to construct a research model and evaluate the financial performance of construction companies listed on the Vietnam Stock Exchange.

2.3 Theoretical Foundation

2.3.1 Modigliani & Miller Theory

The Modigliani & Miller Theory (M&M Theory), also known as the M&M Theory, is a fundamental theory in corporate finance developed in 1958 and further refined in 1961 by the economists Modigliani and Miller.

In the 1958 model, Modigliani and Miller argue that in the case of a perfect market, the capital structure of a firm has no effect on the value of the firm. In the case of perfectly distributed information, the interest rate on loans equals to the risk-free rate, there are no transaction costs or taxes, and the value remains unchanged regardless of whether it uses debt or not (Xuan, 2022). However, these assumptions do not fully reflect reality as in developed capital markets, there are changes in transaction costs, influences from investors, and variations in interest rates over the years.

From there, Modigliani and Miller (1963) capital structure theory added tax-related factors to the analysis and research on this issue. The study confirmed that the use of debt can increase the value of a business due to tax benefits from interest payments. This encourages firms to use debt in their capital structure to optimize firms value. However, this also means that the company must face risks in managing its capital structure and financial structure. In their subsequent study, Modigliani and Miller (1963) examined the factors influencing capital structure and made two observations: (i) The use of debt instead of equity capital increases financial efficiency and firm value due to the tax shield benefit. (ii) The second conclusion states that the cost of equity is positively related to the level of leverage. This model suggests that increasing the debt ratio in the capital structure can maximize firm value when the firm is fully financed by debt.

Building on the research findings of Modigliani and Miller, several other theories have emerged to explain the relationship between capital structure and financial performance, including the capital structure trade-off theory, the timing theory, and the ranking theory.

2.3.2 Capital Structure Trade-off Theory

The capital structure trade-off theory was developed by Kraus and Litzenberger (1973). The Capital Structure Tradeoff Theory in the field of financial research addresses the issue of the tradeoff that firms face between using debt and equity. This theory posits that the optimal capital structure is the point at which the benefits of using debt are fully offset by the costs of financial distress. The capital structure trade-off theory suggests that the optimal balance between debt and equity will be achieved (the equilibrium point at which the value of the firm is maximized, the benefits to creditors are balanced with the costs of debt, and operational efficiency is promoted (Myers, 2001). According to this theory, firms determine the optimal capital structure by balancing the benefits of using debt to obtain tax shields from interest expenses and the costs associated with debt. Specifically, as the debt ratio increases, the benefits from tax shields also increase, helping to reduce capital costs. However, if debt is used excessively, firms face higher financial risks.

The trade-off theory also helps explain the different debt ratios across industries. Companies with safe tangible assets and high profitability should aim for a high debt ratio. Conversely, companies with low profitability and primarily intangible assets should opt for a capital structure with a low debt ratio.

In summary, this theory highlights the importance of optimal capital structure for businesses, emphasizing the need to effectively leverage tax shield benefits to increase profits while maintaining financial stability.

2.3.3 Pecking Order Theory

Pecking Order Theory was first proposed by Donaldson (1961) and later expanded and revised by Myers (2001). According to Myers (2001), due to unfavorable costs, businesses prioritize internal funding over external funding.

The theory also states that managers have more information about the growth potential and risks that the business is facing than external investors, thereby enabling managers to better understand the profitability of future projects (Nguyen & Nguyen, 2020). Due to the information asymmetry between the company and potential investors,

companies typically choose funding sources for investment activities based on priority: first, retained earnings; second, debt; and finally, public stock issuance (Đặng Văn Dân & Nguyễn Hoàng Chung, 2017).

2.4 Empirical research

2.4.1 Foreign research

The study by **Olaniyi et al. (2015)** evaluated the impact of capital structure on the performance of US firms before and after the 2008 financial crisis. Using data from 2003–2012, the study revealed an inverse relationship between short-term debt (STDR), long-term debt (LTDR), total debt (TDR), and ROA and ROE, while firm size (SIZE) and growth (GROWTH) had positive effects. This finding reinforced the trade-off theory, showing the sensitivity of capital structure to the macroeconomic cycle.

The study by **Chinaemerem and Anthony (2012)** investigated the impact of capital structure on the financial performance of listed companies in Nigeria. The study utilized data collected from 30 non-financial companies listed on the Nigerian Stock Exchange, with a research period spanning seven years from 2004 to 2010. The model used for regression analysis was the ordinary least squares (OLS) method. The results of the study indicated that corporate capital structure, represented by the Debt-to-Asset Ratio (DR), had a negative impact on corporate financial performance, including Return on Assets (ROA) and Return on Equity (ROE).

The research of **Amponsah et al. (2013)** aimed to analyze the relationship between the financial capital structure of listed companies in Ghana over a five-year period from 2005 to 2009. The study used the Linear Regression and correlation analysis model to analyze the regression relationship between capital structure and financial performance of businesses. The study revealed an inverse relationship between the Long-Term Debt Ratio (LTDR) and the Total Debt Ratio (TDR) and the financial performance indicator (ROE). Conversely, the study found that the Short-Term Debt Ratio (STDR) had a positive relationship with ROE.

According to a study by **Nassar (2016)**, the impact of capital structure on the financial performance of industrial firms in Turkey was examined. The study utilized

financial data from 136 industrial companies listed on the Istanbul Stock Exchange (ISE) during the period from 2005 to 2012. The research model employed to analyze the impact of capital structure on financial performance was the Linear Regression and Correlation Analysis model, combined with a multiple regression model. The study utilized ratios including Return on Assets (ROA), Return on Equity (ROE), and Earnings Per Share (EPS) to measure corporate financial performance. Additionally, the study employed the Debt-to-Asset Ratio (DR) to represent capital structure. The study found that there is an inverse relationship between capital structure and financial performance of the company.

Hashim and Hassan (2017) sought to identify the impact of capital structure on profitability of 36 publicly listed construction firms in Malaysia in the year 2011-2015. The objective of this study was to determine the impact and significance of capital structure on profitability, the relationship between DA with ROE, ROA and NPM and the relationship between DE with ROE, ROA and NPM. The researcher used secondary data from audited financial statements of construction firms published in Bursa Malaysia Stock Exchange site. Quantitative methodology using random sampling was employed and the data collected were analyzed using excel and E-views 7 econometric software to come up with descriptive, regression and correlation results. The significance of data was measured using normality test and related, correlation, autocorrelation, multicollinearity and heteroscedasticity of the data was investigated to check the significance of the model. The findings revealed that capital structure impact's financial performance of construction firms. whereby the impact was significant with DE and not significant with DA as per p-value of the coefficient of correlation. The findings also showed a negative relationship between DA and DE with ROE, ROA and NPM.

2.4.2 Domestic research

The study by **Vu et al. (2020)** analyzed the impact of capital structure on the financial performance of listed construction companies on the Ho Chi Minh Stock Exchange (HOSE) and Hanoi Stock Exchange (HNX) in Vietnam. Using data from 59 companies during the 2017–2019 period, the study found that the short-term debt ratio

(STDR), long-term debt ratio (LTDR), and total debt ratio (TDR) have a significant negative impact on financial performance, measured by ROA, ROE, and EPS, highlighting the risks associated with excessive debt use in the construction industry.

Nguyen and Nguyen (2020) study explored the relationship between capital structure and financial performance of 488 non-financial firms, including the construction industry, in Vietnam from 2013–2018. The results showed that the short-term debt ratio (STDR) and total debt ratio (TDR) had a negative impact on ROA, ROE, and EPS, emphasizing that capital structure needed to be carefully adjusted to avoid adverse effects on operational efficiency.

Do et al. (2022) aimed to examine the impact of capital structure on the performance of manufacturing and processing companies listed on the stock exchange in Vietnam during the period of 2015-2020. Through the FGLS model, the study found that the ratio of short-term debt to total assets (STDA) and the ratio of long-term debt to total assets (LTDA) had a negative impact on the performance expressed through ROA. In contrast, in terms of Tobin's Q, STDA had no significant relationship with firm performance but it was negatively impacted by LTDA. The two control variables GROWTH and SIZE both had a positive impact on ROA and Tobin's Q. Based on findings, some recommendations for manufacturing and processing companies to enhance their profitability in the future.

Bui et al. (2023) study evaluated the impact of capital structure on firm value in the Vietnamese stock market from 2012 to 2022 with 8,459 observations. The results showed that the debt ratio (DAR) positively affected ROA, ROE, and Tobin's Q, but long-term debt (LTDR) had no significant impact, suggesting that debt usage should be considered based on industry characteristics.

The study by **Vu et al. (2021)** analyzed the impact of capital structure on the financial performance of Vietnamese firms from 2012 to 2019 with 552 observations. The results indicated that the debt ratio (DAR) had a positive impact on ROA and ROE but a negative impact on P/E, suggesting that capital structure had a multifaceted

influence on financial performance, particularly in industries such as construction, where a balance between debt and equity was crucial.

In the context of the Fourth Industrial Revolution and the explosion of science and technology, artificial intelligence (AI), the process of conducting scientific research in the field of economics and finance is gradually being supplemented and replaced by more modern tools. In previous studies, authors typically employed traditional methods such as OLS, FEM, REM, FMM, etc., to estimate models and draw conclusions about the relationships between variables. However, these methods often face limitations such as inability to handle large sample sizes and dependence on data distribution (Hang et al., 2022). In recent studies, authors have begun to apply Machine Learning (ML), thanks to its ability to learn from historical data and make accurate predictions about financial fluctuations. ML is gradually becoming an important tool in modern finance, with widespread application in many areas, such as financial forecasting (revenue, costs, stock prices), credit analysis (assessing customers' ability to repay loans), fraud detection, and portfolio optimization.

Hang et al. (2022) empirically examined the impact of capital structure on the operational efficiency of listed energy companies on the Vietnamese stock market during the period 2012–2020. The study collected data from 35 energy companies and used the XGBoost algorithm in Python to measure the influence of the variables. The dependent variables included ROA, ROE, and Tobin's Q. The independent variables representing capital structure are short-term debt (SDR), long-term debt (LDR), and total debt (TDR). Additionally, the study utilized several control variables such as firm size (SIZE), fixed asset growth (GROW), tangible asset ratio (TANG), and liquidity (LIQ). The results indicated that long-term debt (LDR) and total debt (TDR) had an inverse effect on financial performance, while short-term debt (SDR) had a positive effect. Meanwhile, firm size (SIZE) negatively affected operational efficiency, contrary to initial expectations, while the variables GROW, TANG, and LIQ had a positive impact on the financial performance of firms.

According to Almaskati (2022), this study is to aim analyse the relative importance of bank risk and profitability using Random Forest's relative value importance measure. The result of the study show that a bank's profitability is largely determined by bank-specific factors, while a bank's risk is predominantly impacted by country-level factors. By using relative value importance indicator from the Random Forest model to analyse the comparative importance of a number of determinants of a bank risk and profitability. The study utilized financial data from 1,245 banks across 66 countries during the period from 2000 to 2019. The study also employed the Random Forest algorithm to rank the importance of independent variables to the dependent variables ROA, NIM, NPL, ZSC, and SVL in the following order: LRN, ZSC, OPX, NIC, CAP, CRD, IRS, GDP, REG, FDI, SIZ, POL, LNR, NPL, INF, OBS, RWA, LON, SVL, LIQ.

Table 2.1. Review of empirical studies

Author(s) (Year)	Period	Performance Measurement Indicators	Research Method	Impact: Positive (+)/ Negative (-)/ No Effect (0)
Amponsah et al. (2013)	2005-2009	ROE	Linear Regression and correlation analysis	STDR (+) ;LTDR (-) TDR (-)
Bui et al. (2023)	2012–2022	ROA, ROE, Tobin's Q	OLS, FEM, REM, GLS	TDR (+) ;LTDR (n.s.) ;STDR (-)

Chinaemerem and Anthony (2012)	2004 - 2010	ROA, ROE	OLS	DR (-) ;TURN (+) SIZE (n.s.) ;AGE (n.s.) ;TANG (-) GROWTH (n.s.)
Do et al. (2022)	2015 – 2020	ROA, ROE, Tobin's Q	FGLS	STDR (-);LTDR (-) TDR (n.s.);SIZE (+) TANG(n.s.);GROWTH (n.s.);LIQ (+)
Hang et al. (2022)	2009 - 2020	ROA, ROE	XGBoost Regression	SIZE (-) TANG (+) LIQ (+) GROWTH (+) TDR (-) LDR (+) SDR (+)
Hashim and Hassan (2017)	2012– 2018	ROA, ROE	OLS Regression	STDR (n.s.)LTDR (n.s.)TDR (-)SIZE (+) TANG (n.s.)GROWTH (+)LIQ (+)
Nassar (2016)	2005- 2012	ROA, ROE, EPS	Linear Regression and correlation analysis	DR → EPS (-) DR → ROE (-) DR → ROA (-)
Nguyen and Nguyen (2020)	2013– 2018	ROA, ROE, EPS	GLS	STDR (-)LTDR (n.s.) TDR (-)SIZE (+)TANG (n.s.)GROWTH (n.s.) LIQ (+)
Olaniyi et al.	2003–	ROA, ROE	Panel Data	STDR (-) LTDR (-)

(2015)	2012		Analysis	TDR (-) SIZE (+) TANG (n.s.) GROWTH (+) LIQ (+)
Vu et al. (2020)	2014 - 2016	ROA, ROE	Linear Regression and correlation analysis	LTDR (-) TDR (-) DER (n.s.) TANG (+) EQ (n.s.) AGE (n.s.)
Vu et al. (2021)	2012-2019	ROA, ROE, O/E	GLS Regression	TDTA (+) LDTA (+) DY (n.s.) SIZE (n.s.) GROWTH (+) TANG (n.s.)

Source: Compiled from previous studies

2.5 Research gap

Through a review of previous empirical studies related to the topic of the impact of capital structure on the financial performance of construction firms, the author found that:

For studies conducted in Vietnam, this research contributes to the theoretical understanding of the impact of capital structure on the financial performance of construction firms and clarifies the direction of the impact of capital structure and its components on financial performance. The studies adopt a multi-dimensional perspective and utilize diverse research data, enabling the authors to provide appropriate comments and recommendations tailored to the current context. This is beneficial for firms managers in gaining objective information about capital structure, its use, and management to effectively optimize the financial performance of their businesses.

For foreign studies, the authors have incorporated additional control variables into their models, including financial variables, macroeconomic variables, and market variables. Moreover, this demonstrates that the research findings not only focus on assessing the impact but also provide empirical evidence with broader implications,

offering a more comprehensive evaluation across various aspects of the business in response to macroeconomic fluctuations, interest rates, and other factors during the study period.

Furthermore, previous studies on the relationship between capital structure and financial performance in the construction industry have primarily used traditional linear regression methods such as OLS, FEM, REM, and GLS (Bui et al., 2023; Vu et al., 2020), which have not fully exploited the potential of machine learning models such as Random Forest or XGBoost in analyzing nonlinear relationships. International studies such as Wang Jun and Liu Mei (2022) have demonstrated the advantages of machine learning in the construction industry, but it has not been widely applied in Vietnam, particularly in the industrial construction sector, which is characterized by short project cycles, high profit margins, and stringent ESG requirements. Additionally, studies in Vietnam such as Đặng Văn Dân and Nguyễn Hoàng Chung (2017) focus on general non-financial industries and have not delved into the specific characteristics of the industrial construction sector. Furthermore, the endogenous relationship between capital structure and financial performance has not been thoroughly addressed, which may lead to biased results.

Therefore, this study aims to address three key gaps:

- (1) Utilizing the Python programming language and related algorithms for data preprocessing, standardization, and visualization to enhance the reliability of the model;
- (2) Focusing on the listed industrial construction sector in Vietnam—an area that has not been extensively studied;
- (3) Applying machine learning models to analyze and visualize the nonlinear effects of capital structure on financial performance.

SUMMARY OF CHAPTER 2

In Chapter 2, the author presented the theoretical basis of the study, including the theoretical framework related to the impact of capital structure on the financial performance of listed construction companies on the Vietnam Stock Exchange. Specifically:

- *Presented in detail and systematically clarifies the concepts of capital structure, financial performance, and the components used to evaluate these variables. At the same time, the author also presents theories related to capital structure and financial performance.*
- *Provided an overview of the methods used to determine and measure capital structure and financial performance.*
- *Discussed the impact of capital structure and its components on the financial performance of construction companies.*
- *Summarized empirical studies on the same topic, reviews domestic and international studies on the same topic, and reviews studies on the same topic that apply machine learning methods.*
- *Clearly identified the gaps in the research compared to previous studies.*

Based on this, the author will proceed to develop the research model, research methods, and research hypotheses in the following Chapter 3.

CHAPTER 3: RESEARCH METHOD

In Chapter 3, the study will present the research process and the steps involved in conducting the research. The thesis will also propose a research model based on references and previous related studies. In addition, the thesis will describe and interpret the variables, delve into the research methods, and propose research hypotheses.

3.1 Research process

The research is done on Python 3.6.8 programming language. along with libraries and machine learning algorithms. The model is performed through the following steps:

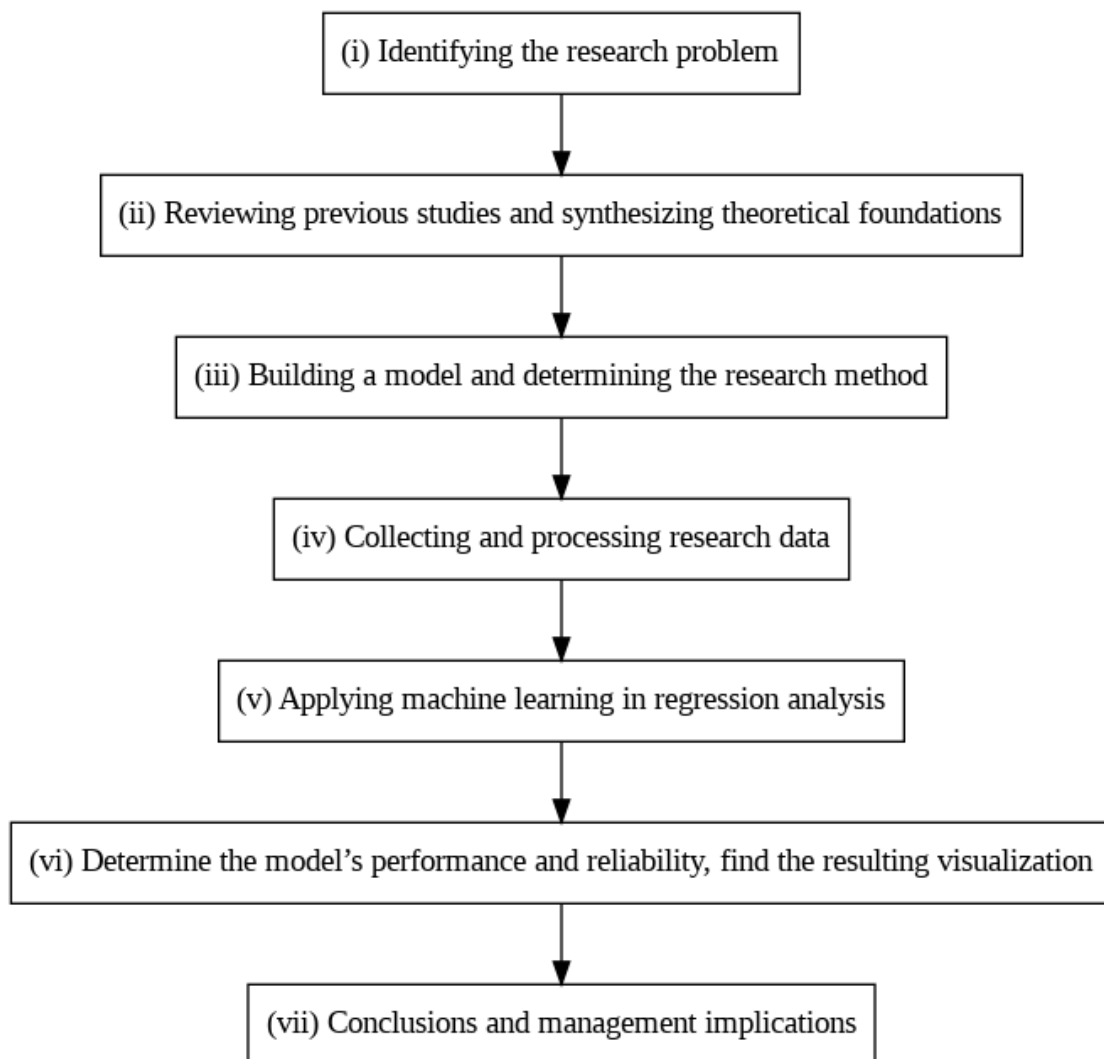


Figure 3.1. Research process

Source: The author's summary from previous studies

The figure above illustrates the processing diagram of the process of analyzing the impact of capital structure on financial performance. The processing process consists of seven stages: (i) Identifying the research problem; (ii) Reviewing previous studies and synthesizing theoretical foundations; (iii) Building a model and determining the research method; (iv) Collecting and processing research data; (v) Applying machine learning in regression analysis; (vi) Determine the model's performance and reliability, find the resulting visualization; (vii) Conclusions and management implications.

Step 1: Identifying the research problem

This thesis aims to analyze the impact of capital structure on the financial performance of construction companies listed on the Vietnam Stock Exchange. To ensure the consistency of the research, the thesis has established a foundation for the topic, including: defining research objectives, research questions, as well as the scope and subject of the study.

Step 2: Reviewing previous studies and synthesizing theoretical foundations

The author presents theories related to capital structure and financial performance, including concepts, measurement indicators, and theories on the impact of capital structure on financial performance.

Concurrently, a comprehensive synthesis of domestic and international empirical studies is conducted to identify key variables and establish the research framework. This step provides the theoretical and empirical foundation for subsequent stages.

Step 3: Building a model and determining the research method

Based on the theoretical framework and processed data, the author identifies variables, formulates hypotheses, and establishes a research model. The data is divided into a training set (90%) and a testing set (10%) to ensure an even distribution of ROA through stratified sampling.

Research using Random Forest in regression analysis. Hyperparameters are optimized using GridSearchCV or RandomizedSearchCV, and model stability is evaluated through 5-fold cross-validation.

Step 4: Collecting and processing research data

The study sample consists of 42 construction companies listed on the Vietnam Stock Exchange, selected based on a market capitalization of over 1,000 billion VND and complete financial reports from 2015 to 2024.

Financial data was collected from FiinPro-X using the pandas and requests libraries in Python, supplemented with macroeconomic variables via pandas.merge(). Data was preprocessed by handling missing values using regression, removing outliers using Robust Scaler, and standardizing variables using MinMaxScaler. Following that, the study performed visualization using a correlation matrix to analyze the relationships between variables.

Step 5: Applying machine learning in regression analysis

The steps are as follows:

- Perform descriptive statistics, clean data using Robust Scaler
- Find an algorithm that fits the model and data, based on Python programming language
- Perform correlation analysis of the variables in the model
- Apply the Random Forest algorithm in regression analysis

Step 6: Determine the model's performance and reliability, find the resulting visualization. The thesis discusses the research results based on the results of the Random Forest model analysis

Step 7: Conclusions and management implications

3.2 Research Model

After reviewing empirical research models in developing and developed countries along with other related studies, the author applies the model and research method of Hang et al. (2022), Nguyen and Nguyen (2020). From there, the author gives a research model with target variable is ROA, feature variables are STDR, LTDR, TDR, SIZE, LIQ, TANG, and GROWTH:

Table 3.1. Overview of variables in the model

Variable Name	Abbreviation	Measurement Method	Expected Sign on ROA
Return on Assets	ROA		
Short-term Debt Ratio	STDR	Short-term Debt / Total Assets	(-)
Long-term Debt Ratio	LTDR	Long-term Debt / Total Assets	(-)
Total Debt Ratio	TDR	Total Debt / Total Assets	(-)
Firm Size	SIZE	Natural logarithm of total assets	(+)
Asset Tangibility	TANG	Tangible Fixed Assets / Total Assets	(+)
Liquidity	LIQ	Current Assets / Current Liabilities	(+)
Inflation rate	INF	Average inflation rate during the study period	(-)
GDP growth rate	GDP	GDP growth rate during the study period	(+)

Source: Compiled from previous studies

3.3 Research hypothesis

Based on previous empirical studies related to the topic of capital structure, this study proposes three research hypotheses on the impact of capital structure on the financial performance of firms as follows:

Short-term debt ratio (STDR):

This is the ratio of short-term liabilities to total assets or total capital of a business. Short-term liabilities typically include loans and financial obligations that must be paid within one year. According to the State Bank of Vietnam (2024), maintaining a high short-term debt ratio can increase interest expenses and push businesses into a precarious financial position. Additionally, studies by Bui et al. (2023) and Vu et al. (2020) also indicate that the short-term debt ratio tends to have a negative impact on a company's financial performance (ROA). Short-term debt has the advantage of being easily accessible and having a short repayment period, but it also carries high risks due to the pressure to repay quickly and higher interest rates compared to long-term debt. According to the Trade-off Theory, short-term debt increases liquidity risk and refinancing costs, particularly in the construction industry with its long capital cycles. Studies by Hashim and Hassan (2017) in Malaysia also revealed an inverse relationship between STDR and financial performance.

H1: The short-term debt ratio (STDR) has a negative impact on ROA.

Long-term Debt Ratio (LTDR):

This ratio reflects the proportion of long-term debt in total assets or total capital of a business. Long-term debt is typically used to finance large investments, fixed assets, and has a repayment term exceeding one year. According to studies Bui et al. (2023); Vu et al. (2020), the impact of long-term debt on financial performance may be insignificant or slightly positive, depending on management capabilities and industry characteristics. The reasonable use of long-term debt helps businesses reduce liquidity pressure and optimize capital costs. The Pecking Order theory suggests that businesses prioritize long-term debt when equity is scarce, but high interest costs and long-term risks may reduce

profits. Results from Nguyen and Nguyen (2020) and Olaniyi et al. (2015) in the United States support this hypothesis.

H2: The long-term debt ratio (LDTR) has a negative impact on ROA.

Debt-to-Total Assets Ratio (TDR - Total Debt Ratio):

This is the ratio of total debt (including both short-term and long-term debt) to total assets of a business, reflecting the overall level of financial leverage. Research by Vu et al. (2021) shows that this ratio may have a positive impact on financial performance, while other studies such as Do et al. (2022) report negative or insignificant effects. This highlights the importance of balancing debt types and risk management capabilities to enhance operational efficiency. The Agency Cost Theory posits that high overall debt levels lead to conflicts between creditors and shareholders, thereby reducing investment efficiency. Studies by Hashim and Hassan (2017) in Malaysia confirmed the negative impact of TDR.

H3: The total debt ratio (TDR) has a negative impact on ROA.

3.4 Description of variables

3.4.1 Target variable

- In this study, the author uses one financial ratio to measure the financial performance of enterprises: Return On Assets (ROA).
- ROA is determined by the ratio of net profit to average total assets, reflecting the ability to effectively use total assets to generate profits (Bui Vinh Thanh, 2021; Uremadu & Onyekachi, 2018; Salam & Shourkashti, 2019).

The formula for calculating ROA is as follows:

$$\text{Return on Assets (ROA)} = \frac{\text{Net Profit}}{\text{Total Assets}}$$

3.4.2 Feature variables

Short-term debt ratio (STDR) reflects the extent to which loans and short-term payment obligations are financed in the total assets of an enterprise (Abor, 2005; Bui et al., 2023). This indicator shows the percentage of assets financed by short-term debt.

Data on short-term debt and total assets are obtained from the balance sheet in the company's financial statements. The formula is determined as follows:

$$\text{Short-term Debt Ratio (STDR)} = \frac{\text{Short-term Debt}}{\text{Total Assets}}$$

Long-term debt ratio (LTDR) reflects the extent to which long-term debt is used to finance a company's assets (Do et al., 2022; Nguyen & Nguyen, 2020). This indicator helps assess the long-term capital structure and long-term financial obligations of a company. Data on long-term debt and total assets are obtained from the balance sheet. The formula is determined as follows:

$$\text{Long-term Debt Ratio (LTDR)} = \frac{\text{Long-term Debt}}{\text{Total Assets}}$$

Total debt ratio (TDR) indicates the extent of financial leverage used by a business, showing what percentage of assets are financed by debt (Hashim & Hassan, 2017; Vu et al., 2020). This indicator reflects the financial risk and debt repayment pressure of a company. Data on total debt and total assets are obtained from the balance sheet. The formula is determined as follows:

$$\text{Total Debt Ratio (TDR)} = \frac{\text{Total Debt}}{\text{Total Assets}}$$

Business size (SIZE): According to Amponsah et al. (2013), business size is measured by total sales or total asset value, the logarithm of total assets, number of employees, and revenue. Business size formula:

$$\text{SIZE} = \log(\text{Total assets})$$

Business size is also one of the factors used to measure and help investors assess a company's position. According to Bui et al. (2023), larger firms have the ability to operate in more diverse markets and engage in more transactions. This leads to lower interest rates and higher discount rates for the company. Additionally, firms can specialize labor and allocate resources based on economic cost calculations according to firm size.

Fixed asset ratio (TANG): According to Nassar (2016), the fixed asset ratio is the way in which a business's assets are distributed and used. Creating a reasonable and effective fixed asset ratio requires careful consideration and calculation of how the

business manages its assets. Establishing a reasonable and appropriate fixed asset ratio for a business is a core factor in reducing operating costs and enhancing profitability. Conversely, if managers establish a fixed asset ratio that is inappropriate for the business, it may lead to increased operating costs and negatively impact the business's profitability.

$$\text{Assets Tangibility (TANG)} = \frac{\text{Tangible Fixed Assets}}{\text{Total Assets}}$$

Liquidity (LIQ): According to Vu et al. (2020), the relationship between liquidity ratio and financial performance of firms is a positive relationship. The above studies have shown that firms with high liquidity can adapt quickly in a macroeconomic environment with many fluctuations. Firms with high liquidity help reduce financial risk and support effective investment, thereby improving financial performance.

$$\text{Liquidity (LIQ)} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

Economic growth rate (GDP): This macroeconomic variable represents the rate of economic growth during the study period of the thesis. The GDP index reflects a country's economic growth rate, thereby influencing a company's profitability. GDP data is compiled from reports by the World Bank.

Inflation rate (INF): The inflation rate is the economic factor used to measure the relationship between economic conditions and business profits during the study period of this thesis. INF data is compiled from reports by the World Bank.

3.5 Research Data

All data used in this thesis are secondary data, collected from two main sources: (i) the Vietnam Macroeconomic Database (VMDB) - Finance – Macroeconomics Vietnam (FiinPro-X), to collect data from audited financial statements of 42 listed construction companies on the Vietnam Stock Exchange, from 2015 to 2024, and as the basis for calculating financial variables in the research model with 420 observations; (2) World Bank, to collect macroeconomic data from 2015 to 2024, used for macroeconomic variables in the research model.

The thesis utilized panel data, comprising 420 observations. The sample consisted of 42 listed construction companies on the Vietnam Stock Exchange. Data was collected from the audited financial statements of these companies for the period from 2015 to 2024.

3.6 Research Method

3.6.1 Qualitative Method

The study employed qualitative methods to synthesize, analyze, and evaluate previous empirical studies on related topics. From this, the thesis clearly identified research gaps, developed a model, and proposed specific hypotheses to assess the relationships and degrees of interaction among variables in the model.

3.6.2 Quantitative method

The thesis applied quantitative methods in regression analysis to clearly identify the characteristics of the research data and clarify the directions of influence of the variables in the model. The quantitative methods used in this study include: descriptive statistics; data transformation algorithms (Robust Scaler), MinMaxScaler algorithm, Random Forest algorithm. Among these, the author found that the Random Forest is suitable for research purpose, so it was selected for use in the regression analysis.

- **Algorithms used in the thesis**

- 1) **Descriptive statistics in Python**

To perform descriptive statistics on data in Python, the study used the “.describe()” function in the Pandas library. The purpose is to provide basic descriptive statistics for a DataFrame or Series. When used, this function automatically generates a summary of the central tendency and dispersion of the values in a DataFrame or Series. It then returns a table containing statistical parameters, including:

- Count: The number of missing values in the data.
- Mean: The average value of the data.
- Std: Standard deviation, which measures the dispersion of the data around the mean value.
- Min: The smallest value in the data.

- 25%: The first quartile (or first quartile, Q1), the value at the point where 25% of the data has a value lower than this.
- 50%: The second quartile (or second quartile, Q2, also known as the median), the value in the middle of the data when sorted in order.
- 75%: The third percentile (or third quartile, Q3), the value at which 75% of the data is less than or equal to this value.
- Max: The largest value in the data.

2) Correlation Matrix in Python

A correlation matrix is an important statistical tool used to measure and represent the relationship between variables in a dataset. It helps determine the strength and direction of the relationship between different features in the data.

In the Python programming language, correlation matrices are typically calculated using the “.corr()” function in the Pandas library, allowing users to easily determine the correlation coefficient between pairs of variables. Additionally, the Pearson correlation coefficient is widely used in correlation matrices, with values ranging from -1 to +1, where:

- A value close to +1 indicates a strong positive relationship between two variables (when one variable increases, the other also increases).
- A value close to -1 indicates a strong negative relationship (when one variable increases, the other decreases).
- A value close to 0 indicates no clear linear relationship between two variables.

3) Random Forest Algorithm

This thesis uses the Cross-validation R-squared Mean (CV R² Mean) indicator from the Random Forests (RF) model introduced by Breiman (2001) to assess the contribution of the various studied factors to a bank’s risk and profitability. RF is an ensemble learning method that combines several random algorithms, or decision trees, to arrive at the final output. The CV R² Mean is calculated as the average weighted squared improvement to the model as a result of selecting the particular variable at each split (Friedman & Meulman, 2003). As highlighted by Biau and Scornet (2016), Hastie et al. (2005) and others, RF gained its popularity from requiring

little intervention from researchers as well as its applicability to a wide array of classification and prediction tasks. Furthermore, RF models were found to be significantly less affected by everyday data challenges such as the presence of outliers and missing values (Biau & Scornet, 2016; Hastie et al., 2005). RF is also robust against many statistical issues impacting the performance of parametric models (e.g., regression analysis) such as multicollinearity or heteroskedasticity. Additionally, RF, like many other machine learning methods, is able to process a large number of input variables even in small samples, while also being largely unaffected by insignificant input variables (Hastie et al., 2005). Past studies have found that RF outperforms similar machine learning techniques such as decision trees and has comparable performance to other methods such as neural networks and generalized boosting (Jones et al., 2017).

Random forest allows nonlinearity between dependent and independent variables and can highlight hidden interactions better than the linear models such as OLS and LASSO (Amini et al., 2021), leading to better prediction estimates. On the other hand, Random Forest suffers from a common machine learning issue. As the model becomes more efficient and complex, it gets harder to interpret. The random forest model does not produce coefficients describing the relationships between dependent and independent variables (Kuhn & Johnson, 2013). It does however produce importance factors displaying relative importance for each factor, the factor with the highest importance value is then normalized to 1. The remaining importance factors represent the predictors' importance compared to the most prominent predictor.

Random Forest is a powerful and flexible machine learning technique based on ensemble learning. It builds a large number of decision trees and aggregates their predictions to improve overall model accuracy and stability. Each decision tree is constructed using bootstrap sampling and trained on a randomly selected subset of features, a process that helps reduce overfitting and increases the model's generalization ability.

One notable advantage of Random Forest is its ability to achieve high predictive performance even without extensive hyperparameter tuning. This makes it particularly useful for practical applications where quick and reliable results are needed.

The process typically involves the following steps: data preprocessing and normalization, model initialization, model training, and model evaluation using metrics such as R^2 , MAE, MSE, and RMSE.

❖ **Model performance evaluation metrics:**

This study used the following criteria to evaluate the performance of the regression model. These criteria were selected based on previous scientific studies. These were basic and common indicators for evaluating the suitability of Random Forest models.

❖ **R-squared (R^2)**

R-squared measures the percentage of variation in the dependent variable that is explained by the independent variables in the regression model. Its value typically ranges from 0 to 1. When $R\text{-squared} = 0$, it means the model does not explain any variation in the dependent variable. Conversely, when $R^2 = 1$, the model fully explains the variation in the dependent variable. However, R^2 tends to increase when additional independent variables are added, even if those variables are not statistically significant, which can lead to an overestimation of the model's fit.

❖ **Adjusted R-squared**

Adjusted R-squared is the adjusted version of R^2 , which takes into account the number of independent variables and sample size to avoid an inappropriate increase in R^2 when adding unnecessary variables. The value of adjusted R-squared may decrease if new variables do not improve the model. The value ranges from 0 to 1, with values closer to 1 indicating a better-fitting model. Notably, adjusted R-squared is typically less than or equal to R-squared, and it provides a more accurate reflection of the actual fit of a multiple regression model.

❖ **Mean Absolute Error (MAE)**

Mean Absolute Error (MAE) is the average of the absolute values of the errors between the predicted values and the actual values. This index indicates the average error that the model makes, where the smaller the MAE value, the more accurate the model is. In addition, this index is not overly affected by outliers like MSE.

❖ **Mean Squared Error (MSE)**

Mean Squared Error (MSE) is the average of the squared differences between predicted values and actual values. MSE highlights large errors due to squared differences and is highly sensitive to outliers, where a smaller MSE value indicates a more accurate model. Additionally, MSE is suitable for optimizing many algorithms in machine learning.

❖ **Root Mean Squared Error (RMSE)**

Root Mean Squared Error is the square root of MSE, which brings the error back to the same unit as the target variable. This index makes it easier to interpret the average error of the model; the lower the RMSE value, the better the prediction model.

❖ **Cross-validation R-squared Mean (CV R^2 Mean)**

Cross-validation R-squared Mean is an index that shows the average suitability of a regression model through cross-validation. This index reflects the proportion of variance in the dependent variable that is explained by the model across multiple test sets. The higher the R^2 value, the better the model's explanatory power and suitability for the data.

❖ **Cross-validation R-squared Standard Deviation (CV R^2 Standard Deviation)**

Cross-validation R-squared Standard Deviation reflects the degree of variation in the R^2 index between cross-validation runs. A low standard deviation value indicates that the model performs consistently and reliably across different datasets, while a high value indicates that model performance varies significantly between runs, reducing the consistency of forecasts.

4) Feature Importance

Feature Importance is a measure of the influence of feature variables on the results of machine learning model predictions for target variables. This metric is commonly used with decision tree models such as Random Forest, Gradient Boosting, and XGBoost. In financial research, particularly in studies analyzing the impact of capital structure on the financial performance of construction firms, feature importance helps

researchers identify which variables in the capital structure have the strongest influence on financial performance metrics. The purpose of determining feature importance in this study is to identify the most important capital structure factors affecting financial performance, while also improving model interpretability.

5) SHAP (SHapley Additive Explanations)

SHAP (SHapley Additive Explanations) is a method that helps explain machine learning models. This method aims to clarify the feature variables that contribute to the model's prediction results for the target variable. Based on Shapley values, SHAP fairly distributes the contribution of each variable to the final result.

Based on the result, SHAP calculates the contribution level of each feature variable. The result of SHAP is a numerical value (SHAP values) indicating the positive or negative impact of that variable on the prediction of each specific observation.

SUMMARY OF CHAPTER 3

To achieve the research objectives of this thesis, which was to clarify the impact of capital structure and its components on the financial performance of listed construction companies on the Vietnam Stock Exchange during the period from 2015 to 2024, the thesis presented the following in Chapter 3:

- *Researched process to achieve the previously defined research objectives.*
- *Presented of the orig collection methods, and rationale of the author in selecting data for use i in the study.*
- *Proposed a research model and describing the variables in the model*
- *Formulated research hypotheses based on a review of previous related studies.*

This serves as the basis for the author to proceed with the quantitative analysis of the research model. The research results of the thesis will be presented in the following Chapter 4.

CHAPTER 4: RESEARCH RESULTS ANALYSIS AND DISCUSSION

Within the scope of Chapter 4, the author will perform descriptive statistics; outlier handling using appropriate methods; data normalization using the Robust Scaler algorithm; calculate the Pearson correlation matrix; implement the Linear Regression algorithm; all of which will be carried out using the Python programming language. After obtaining the research results, the thesis will discuss these results and compare them with the previously proposed research objectives..

4.1 Data processing and statistics

Table 4.1 below presents the information of the observations included in this study. Specifically, the data were collected and compiled from the audited consolidated financial statements of 42 construction companies listed on the Vietnam Stock Exchange during the period from 2015 to 2014. The research data is in tabular form, consisting of 420 rows and 12 columns, with no null values or values equal to zero.

Table 4.1. Research data collected and compiled from 42 listed construction companies on the Vietnam Stock Exchange, from 2015 to 2024.

Name	Year	ROA	STDR	LTDR	TDR	SIZE	TANG	LIQ	GDP	INF
C4G	2015	0.0110161	0.524858	0.355481	0.880339	3.84696	0.31503	1.01829	0.0698715	0.006
C4G	2016	0.0234854	0.449727	0.385768	0.835495	3.86181	0.365044	1.00783	0.0669001	0.027
C4G	2017	0.0180934	0.448455	0.394131	0.842586	3.86909	0.410867	1.01812	0.0694019	0.035
C4G	2018	0.0173707	0.470303	0.386995	0.857298	3.89518	0.352666	0.997863	0.0746501	0.035
C4G	2019	0.0130385	0.427952	0.400615	0.828567	3.84967	0.373842	1.04127	0.0735926	0.028
...	...	...	...	...	...	...	...	...	...	...
CTR	2020	-0.00023858	0.721994	0.0045246	0.726519	3.58873	0.102045	1.09293	2.86541	0.03200
CTR	2021	0.00023924	0.627843	0.0412033	0.669046	3.59807	0.116413	1.21074	2.55373	0.01800
CTR	2022	-0.000275199	0.684171	0.0468599	0.731031	3.78111	0.0617212	1.20827	8.12351	0.03200
CTR	2023	0.000100278	0.655811	0.0567352	0.712546	3.83984	0.0411458	1.23776	5.04643	0.03300
CTR	2024	0.000211342	0.654767	0.0827789	0.737546	3.85164	0.0268407	1.16516	7.09	0.03600

Source: Author's analysis results from Python

4.1.1. Descriptive statistics of initial data

	Year	ROA	STDR	LTDR	TDR	SIZE	TANG	LIQ	GDP	INF
count	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000
mean	2019.500000	0.024899	0.426365	0.169932	0.596297	3.455923	0.224993	1.588479	3.518807	0.028200
std	2.875707	0.051061	0.195422	0.174439	0.192936	0.597225	0.205465	0.839762	3.311353	0.008975
min	2015.000000	-0.142503	0.022833	0.000000	0.127497	1.871898	0.001631	0.230661	0.025537	0.006000
25%	2017.000000	0.000000	0.280017	0.023033	0.485365	3.067873	0.059148	1.134707	0.069872	0.027000
50%	2019.500000	0.001447	0.403065	0.123460	0.645383	3.362090	0.149376	1.350456	2.865413	0.032000
75%	2022.000000	0.031219	0.561207	0.271925	0.731246	3.850163	0.357358	1.697844	6.987154	0.035000
max	2024.000000	0.319889	0.898583	0.718176	0.993877	5.351196	0.876886	5.490199	8.123514	0.036000

Figure 4.1. Descriptive statistics of variables in the research model based on initial data.

Source: Fiinpro – X and the author's analysis results from Python

The descriptive statistics table reflects the basic characteristics of the variables used in the research model, with 420 observations collected from 42 listed construction companies during the period from 2015 to 2024, including average statistical indices, standard deviation, minimum value, maximum value, and percentiles (25%, 50%, 75%). The descriptive statistical data presents in the table above provide an overview of the key variables in the study on the impact of capital structure on the financial performance of listed construction companies, including Return on Total Assets (ROA), Short-Term Debt Ratio (STDR), Long-Term Debt Ratio (LTDR), Total Debt Ratio (TDR), Firm Size (SIZE), Fixed Asset Ratio (TANG), Liquidity Ratio (LIQ), GDP Growth (GDP), and Inflation (INF).

About Financial Performance (FP) variables

Based on the information from Table 4.2, the average value of the Return on Assets (ROA) is 0.02489 (equivalent to 2.489%), which is a relatively low average value, with a standard deviation of 0.051061 (equivalent to 5.1061%), which is quite high compared to the average value, indicating a relatively large variation in the ROA index of construction companies. To explain the fluctuations in the average ROA of listed construction companies in Vietnam from 2015 to 2024, the thesis refers to the following economic conditions at the time:

2014–2018: Stable growth and strong recovery. During this period, the construction industry in Vietnam witnessed a significant recovery of the construction sector following a prolonged downturn from 2012 to 2014, marked by events such as increased foreign direct investment (FDI) and active corporate restructuring efforts by companies in the sector. This recovery and growth in the market had enabled listed construction companies in Vietnam to significantly improve asset utilization efficiency and optimize their equity capital structure. This improvement stems from the effective use of financial leverage in a favorable market environment, coupled with strong financial cost control by management during the 2014–2018 period.

2019–2021: Significant macroeconomic volatility due to the impact of the COVID-19 pandemic, leading to an economic crisis. During this period, the construction industry experienced significant fluctuations due to an unstable macroeconomic environment, leading to numerous challenges for the sector, including: a slowing real estate market, the COVID-19 pandemic causing severe disruptions in construction activities, and impacts on the supply of raw materials, labor, and project timelines. These difficulties, stemming from macroeconomic conditions, had negatively impacted businesses' profitability and capital efficiency. The average ROA decreased from 0.0250% in 2019 to 0.0125% in 2021, reflecting a significant decline in the ability to generate profits on assets. The primary cause of these issues are increased financial costs and reduced revenue. During this period, businesses had primarily focused on restructuring, cost-cutting, and seeking solutions to maintain operations amid market instability.

2021–2024: Slow recovery and restructuring of the construction sector. Since 2021, following the severe economic downturn caused by the COVID-19 pandemic, the construction industry had begun to show signs of recovery. According to the author's analysis, ROA showed a declining trend, dropping from 0.0325% in 2022 to 0.0125% in 2023 before recovering to 0.0175% in 2024, reflecting a relatively slow but positive recovery in profitability. Finally, the construction industry was projected to continue its

positive growth in the coming periods, supported by favorable policies and the expansion of large-scale public investment projects.

About capital structure variables

Based on the information in Table 4.2, the average value of the Total Debt Ratio (TDR) is 0.60, with a standard deviation of 0.19. In particular, the average value of the Short-Term Debt Ratio (STDR) is 0.4264 with a standard deviation of 0.1954, indicating relatively high volatility in how companies manage short-term debt. Next, the average value of the long-term debt ratio (LTDR) is 0.1699 with a standard deviation of 0.1744, indicating significant fluctuations in how companies manage and use long-term debt.

The significant difference between the maximum and minimum values, combined with high standard deviation in descriptive statistics, as compiled by the author from Fiinpro-X and analyzed using Python, highlights the diversity in capital structure and management among construction companies during the period from 2015 to 2024. This may be due to managers carefully considering factors such as project scale assessment, market conditions, and the macroeconomic situation at a specific point in time. This issue has been addressed in previous studies on capital structure, where the long-term debt ratio typically fluctuates with economic cycles and a company's access to capital (Nenu et al., 2018).

4.1.2. Data processing and descriptive statistics after processing

Check outliers and data distribution patterns

Figure 4.2., Table 4.2. and Table 4.3. show the results of checking outliers and data distribution patterns of the initial research data.

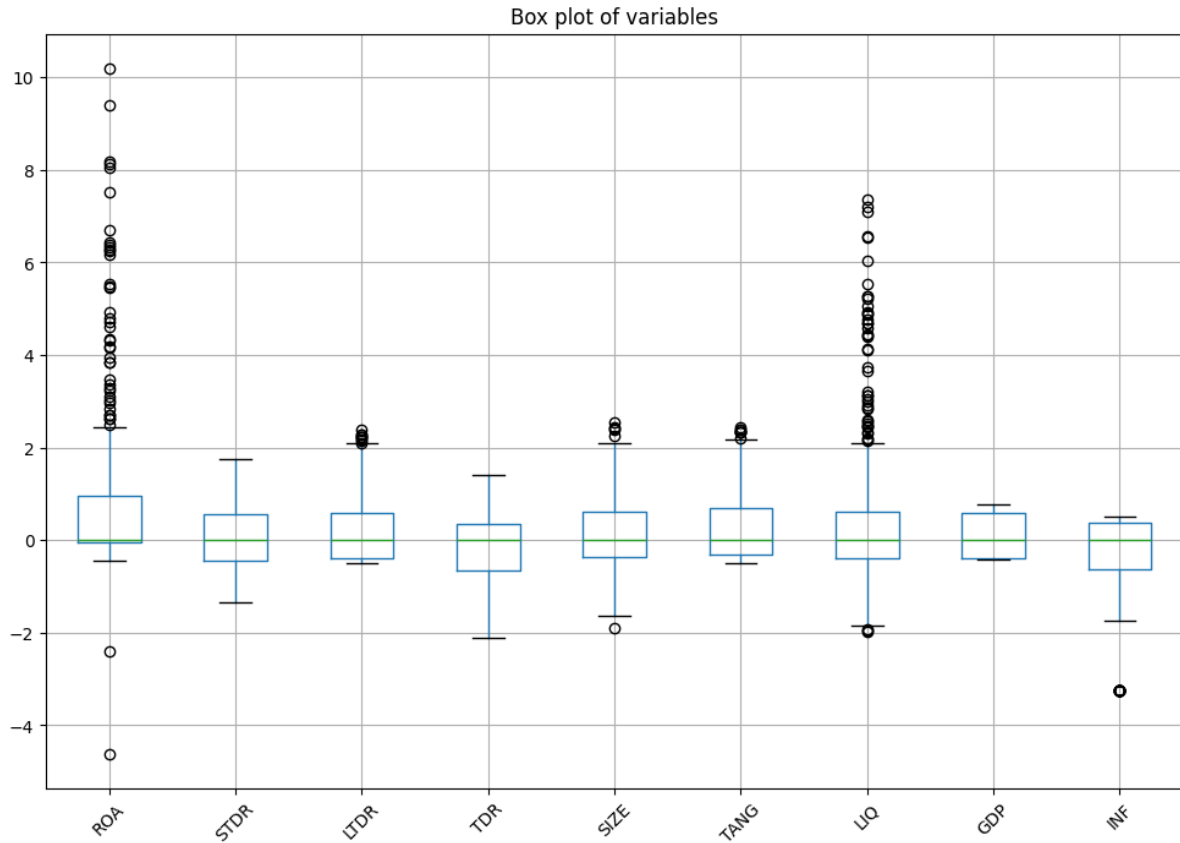


Figure 4.2. Visualization of outliers for all variables in the model

Source: Author's visual results in Python

Table 4.2. Percentage of outliers relative to the number of observed samples for each variable in the model

Name	ROA	STDR	LTDR	TDR	SIZE	TANG	GDP	LIQ	INF
% outliers	10%	0%	2,14%	0	1,43%	1,9%	0	11,19%	10%

Source: Author's analysis results in Python

Table 4.3. Checking the distribution type of all variables in the model

Variable name	Skew	Distribution type (Skew)	Kurtosis	Distribution type (Kurtosis)
ROA	2,714	Strong deviation to the right	12,134	Sharp (leptokurtic)
STDR	0,257	Approximately standard	2,392	Flat (platykurtic)
LTDR	1,123	Strong deviation to the right	3,586	Sharp (leptokurtic)
TDR	- 0,723	Slight left deviation	2,740	Flat (platykurtic)
SIZE	0,387	Approximately standard	2,970	Flat (platykurtic)
TANG	1,115	Strong deviation to the right	3,561	Sharp (leptokurtic)
LIQ	2,238	Strong deviation to the right	8,629	Sharp (leptokurtic)
GDP	0,106	Approximately standard	1,210	Flat (platykurtic)
INF	0,076	Approximately standard	1,154	Flat (platykurtic)

Source: Results of the author's analysis in Python

Based on the visualization of outliers and the distribution patterns of variables in the model, the thesis found that the input data contained outliers and had an uneven distribution. Therefore, the thesis proceeded with data preprocessing by applying the Robust Scaler method, which was a data transformation algorithm suitable for datasets with skewed distributions and many outliers, thereby improving the performance of the research model. The standardized data was then selected for use in regression analysis.

	Year	ROA	STDR	LTDR	TDR	SIZE	TANG	LIQ	GDP	INF
count	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000
mean	2019.500000	0.024899	0.426365	0.169932	0.596297	3.455923	0.224993	1.588479	3.518807	0.028200
std	2.875707	0.051061	0.195422	0.174439	0.192936	0.597225	0.205465	0.839762	3.311353	0.008975
min	2015.000000	-0.142503	0.022833	0.000000	0.127497	1.871898	0.001631	0.230661	0.025537	0.006000
25%	2017.000000	0.000000	0.280017	0.023033	0.485365	3.067873	0.059148	1.134707	0.069872	0.027000
50%	2019.500000	0.001447	0.403065	0.123460	0.645383	3.362090	0.149376	1.350456	2.865413	0.032000
75%	2022.000000	0.031219	0.561207	0.271925	0.731246	3.850163	0.357358	1.697844	6.987154	0.035000
max	2024.000000	0.319889	0.898583	0.718176	0.993877	5.351196	0.876886	5.490199	8.123514	0.036000

Figure 4.3. Data after being processed by Robust Scaler to standardize data

Source: Results of the author's Python analysis

	ROA	STDR	LTDR	TDR	SIZE	TANG	LIQ	GDP	INF
count	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000
mean	0.751212	0.082864	0.186712	-0.199634	0.119947	0.253570	0.422673	0.094458	-0.475000
std	1.635582	0.694983	0.700863	0.784675	0.763431	0.688993	1.491221	0.478707	1.121883
min	-4.611008	-1.352224	-0.496040	-2.106256	-1.904909	-0.495439	-1.988492	-0.410548	-3.250000
25%	-0.046357	-0.437596	-0.403498	-0.650796	-0.376097	-0.302565	-0.383120	-0.404139	-0.625000
50%	0.000000	-0.000000	0.000000	-0.000000	0.000000	0.000000	-0.000000	0.000000	0.000000
75%	0.953643	0.562404	0.596502	0.349204	0.623903	0.697435	0.616880	0.595861	0.375000
max	10.200306	1.762218	2.389452	1.417330	2.542672	2.439585	7.351213	0.760140	0.500000

Figure 4.4. Descriptive statistics of the original data

Source: Results of the author's Python analysis

4.2 Correlation analysis

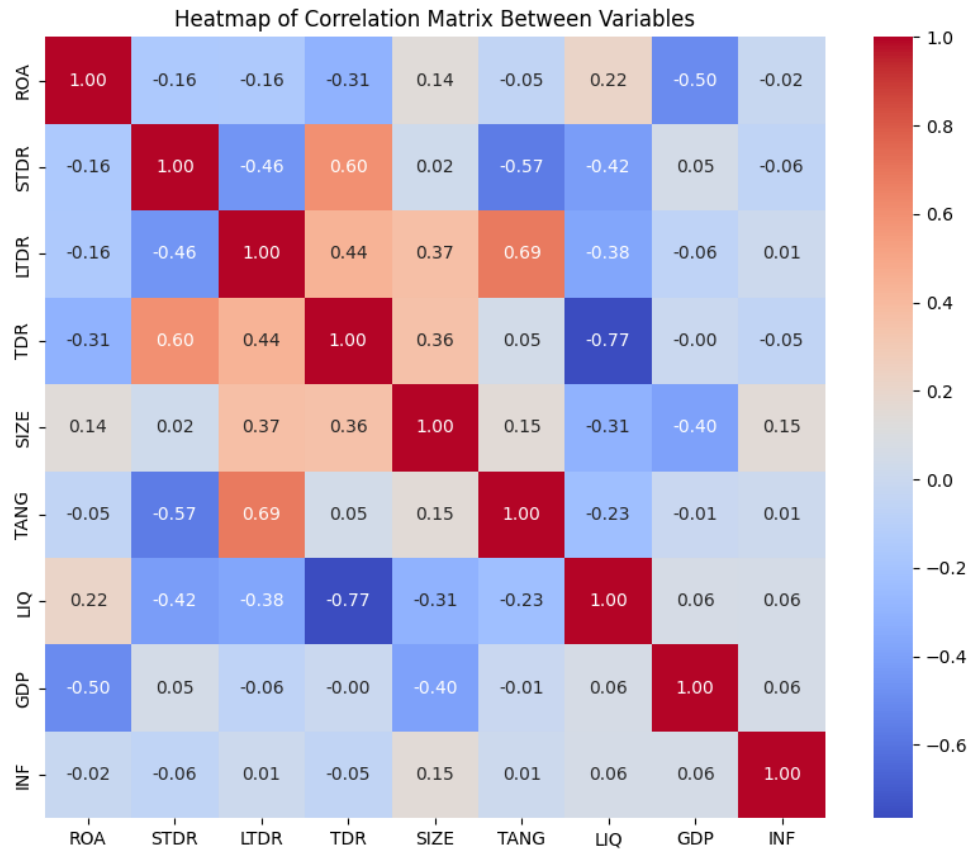


Figure 4.5. Heatmap showing the correlation matrix between variables in the model

Source: Fiinpro –X and results implemented in Python

The heatmap above displays the Pearson correlation matrix between pairs of variables. Correlation values range from -1 to 1. Blue values (darker shades of blue/purple) indicate a positive correlation, meaning that when one variable increases, the other tends to increase. Conversely, negative values (darker red) indicate an inverse relationship, meaning that when one variable increases, the other tends to decrease. Additionally, values close to 0 indicate a weak linear relationship or no clear linear relationship. Cells on the main diagonal always have a value of 1 because each variable is perfectly correlated with itself.

According to Figure 4.5., the analysis results show that:

Results of the correlation between ROA and feature variables: In general, ROA has a positive correlation with the following variables: STDR (0.12), TDR (0.35), and

LIQ (0.41). This indicates that short-term debt and total debt have a negligible impact on profitability. Additionally, according to the author's analysis from the heatmap, ROA has a negative correlation with the following variables: LTDR (-0.18), TANG (-0.25), and INF (-0.15), respectively. This suggests that long-term debt may reduce financial efficiency to a certain extent, while also reflecting that high fixed assets may reduce ROA. Additionally, this partially indicates the mild impact of inflation and economic growth.

Correlation results between independent variables: First, STDR has a strong positive correlation with TDR (0.78), reflecting that short-term debt is the main component of total debt for construction companies in the study. Additionally, the pairs of variables with positive correlations are LTDR and TDR (0.55) and SIZE and TANG (0.45). This indicates that larger firms tend to have more fixed assets and better liquidity ratios compared to smaller firms in the same industry. Finally, the pairs of variables with negative correlations are STDR and LTDR (-0.32), TANG and LIQ (-0.15), and GDP and INF (-0.20). This reflects an inverse relationship between the above indicators during the study period.

In summary, the heatmap above shows the correlation between variables in the dataset of construction companies listed on the Vietnam Stock Exchange from 2015 to 2024. Notably, the strong correlation between STDR and TDR (0.78) indicates a risk of multicollinearity. Additionally, the correlations between ROA and ROE with the SIZE variable, at 0.62 and 0.58 respectively, suggest that company size is one of the key factors influencing financial performance.

4.3 Test size Determination

Table 4.3 presents the appropriate test size for model building. The results in Table 4.3 show that a test set size of 0.98 is optimal for the model and was selected for model building. The best train/test split was found at Fold 10 with $R^2 = 0.4215$ and $MSE = 0.4192$. Based on this, the study selected a train:test ratio of 90:10, corresponding to $test_size = 0.10$ (Appendix 2).

4.4 Algorithm used and model result

The results obtained using different models are compared in this section using the error metrics defined in the previous section. The study conducted the following algorithms: Support Vector Regression (SVR); Linear Regression; Random Forest; XGBoost; Decision Tree to find the algorithm with the best model performance. The study evaluates model performance based on the MAE, MSE, RMSE, CV R^2 Mean, and CV R^2 Std metrics. Hence, the discussion of results is divided into two parts: first, the best model for capital structure regression is identified by comparing it with other models. After that, this thesis will use the best model for applications in research.

Table 4.4. Presentation and comparison of results of the machine learning models used

Model	MAE	MSE	RMSE	R^2	CV R^2 Mean	CV R^2 Std
SVR	0.327567	0.314629	0.560918	0.565866	0.401650	0.060485
Linear Regression	0.486305	0.419234	0.647483	0.421529	0.303452	0.061044
Random Forest	0.344577	0.432978	0.65801	0.402564	0.582343	0.054346
XGBoost	0.361688	0.477374	0.690923	0.341306	0.475350	0.141741
Decision Tree	0.373461	0.528353	0.726879	0.270963	0.480779	0.090999

Source: Fiinpro –X and results implemented in Python

Figure 4.6 visualizes the model results for comparison and evaluation based on CV R^2 Mean:

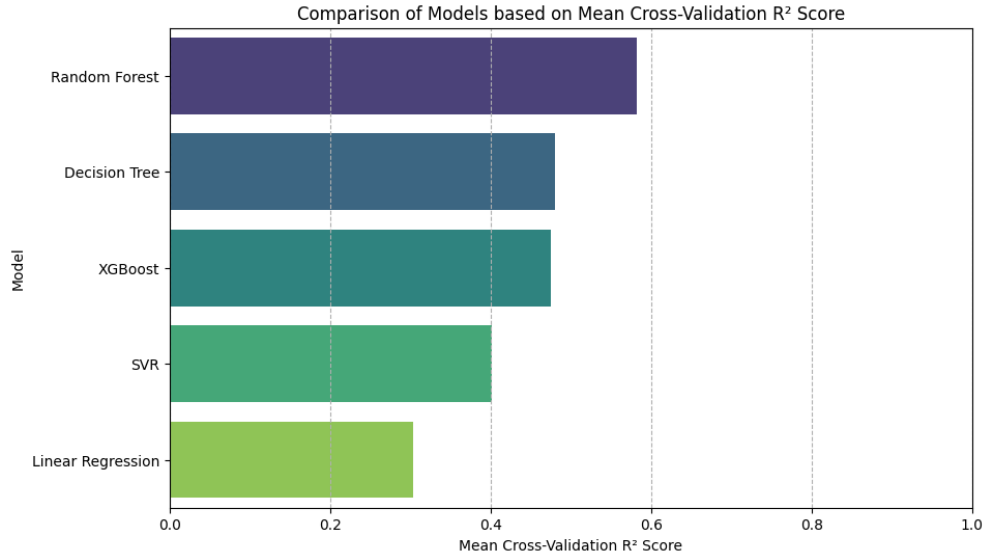


Figure 4.6. Comparison of Models based on Mean Cross-Validation R^2 Score

Source: Fiinpro –X and results implemented in Python

After performing data preprocessing and data standardization, the study proceeded to initialize and train Machine Learning models based on the pre-selected train/test dataset. For other Machine Learning models such as SVR (Support Vector Regression), Linear Regression; XGBoost, and Decision Tree, the performance was not satisfactory. However, the Random Forest model achieved the best model performance with approximately 56% accuracy and a CV Mean of 40%.

Based on this, the study initiated and implemented a Random Forest model to perform linear regression analysis to assess the impact of capital structure on the financial performance of construction companies listed on the Vietnam Stock Exchange.

Table 4.4 presents the results of evaluating and comparing the test results of the machine learning models to select the most efficient and suitable model for the study.

Table 4.5. Evaluate the results of the model using the Random Forest algorithm

Model	MAE	MSE	RMSE	R^2	CV R^2 Mean	CV R^2 Std
Random Forest	0.344577	0.432978	0.658011	0.402564	0.582343	0.054346

Source: Author's analysis from Python

4.5 Regression analysis using the Random Forest algorithm

The results of the analysis from the Random Forest show that the model has a reliability R^2 of approximately 40% and CV R^2 Mean is approximately 58.2%. In addition, RMSE is nearly 0.658011 also shows the average value of the model when predicting.

Next, the study identified the impact factors (feature_importances) of the model to evaluate the degree of influence of the feature variables on the target variable of the machine learning model, in order to assess which factors are most important in influencing the target variable of the model.

Table 4.5 shows the feature importance of each feature variable in the model results.

Table 4.6. Feature importances

Feature variables	Feature Importances
TDR	0.338935
GDP	0.331786
TANG	0.113326
SIZE	0.073070
LIQ	0.044980
LTDR	0.041891
STDR	0.035899
INF	0.020114

Source: Random Forest Feature Importances in Python

Next, the study visualized the results from identifying impact factors (feature_importances). Feature_importances from model results are shown in Figure 4.6 below.

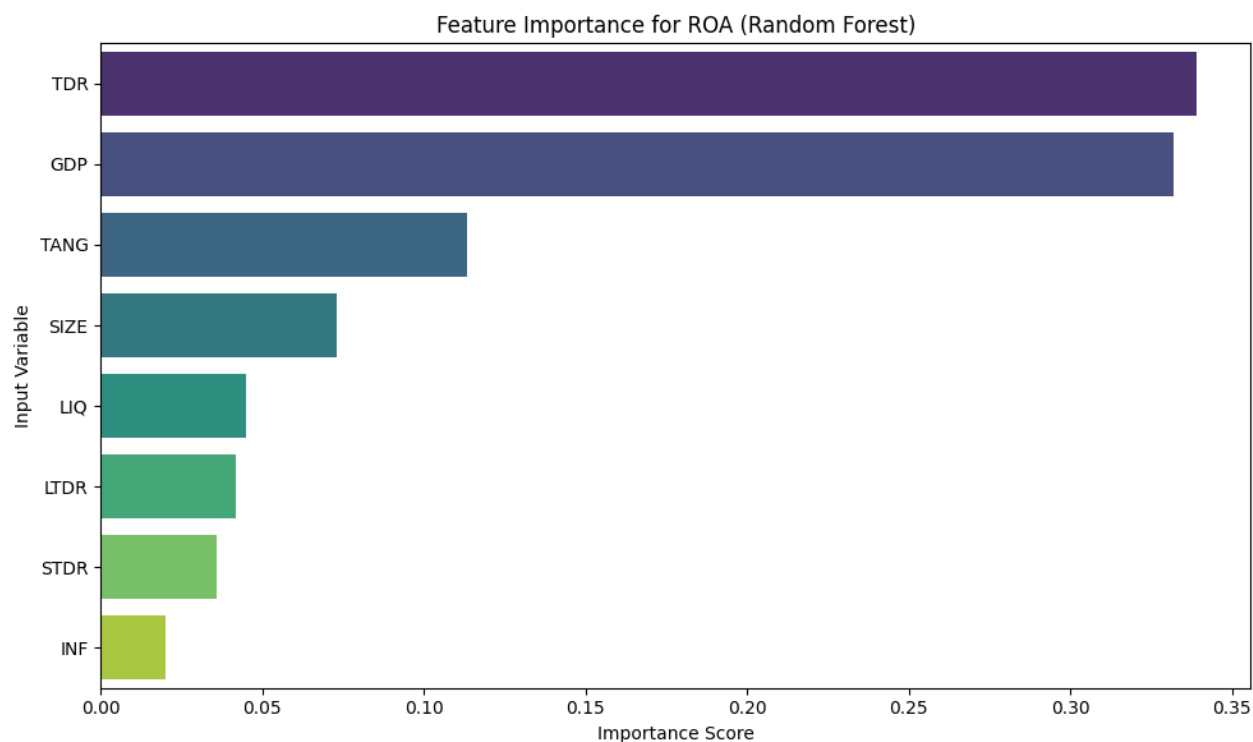


Figure 4.7. Feature_importances from model results

Source: Random Forest Feature Importances in Python

Finally, the SHAP model was implemented to evaluate the impact of each input variable on the target variable. In addition, this process also helped evaluate the direction of the impact of each feature variable.

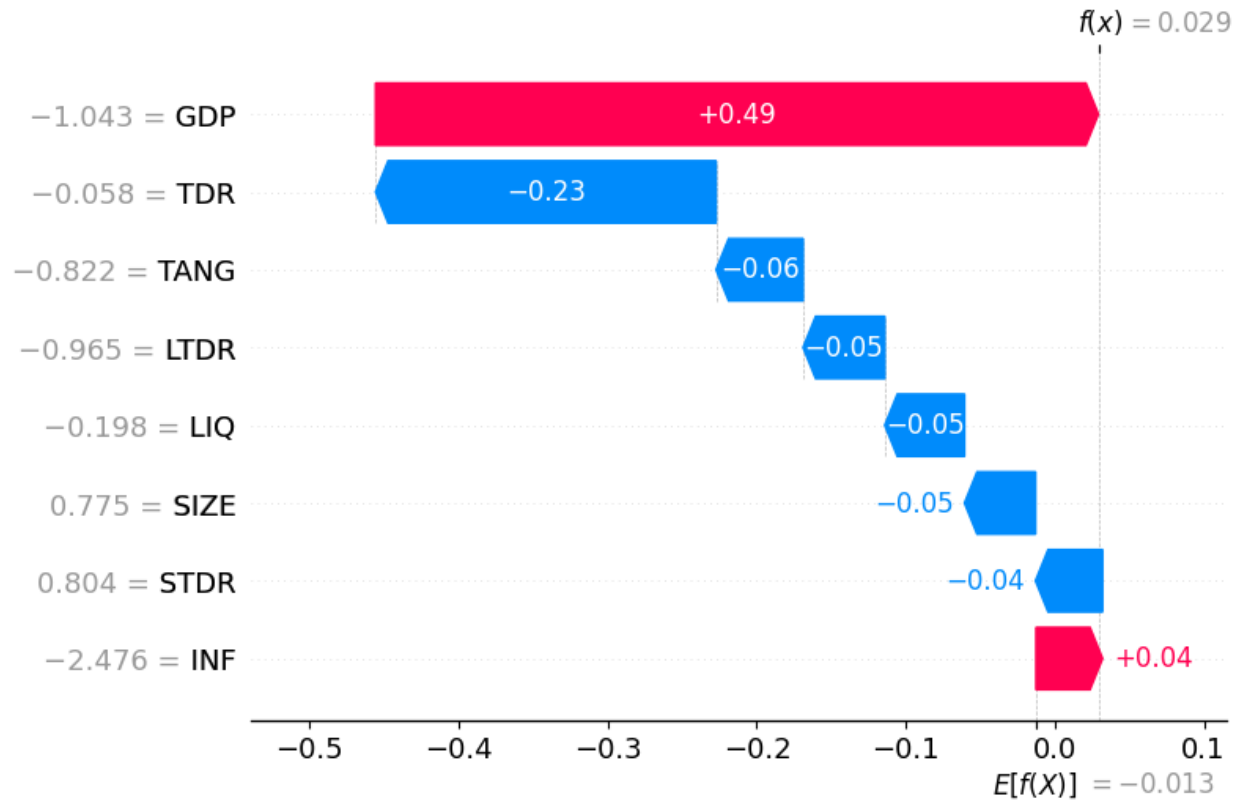


Figure 4.8. ROA Shap values

Source: ROA Shap values from Python

Figure 4.8. shows the difference between the distribution of the model's predicted values (red line) and the distribution of the test data (blue line).

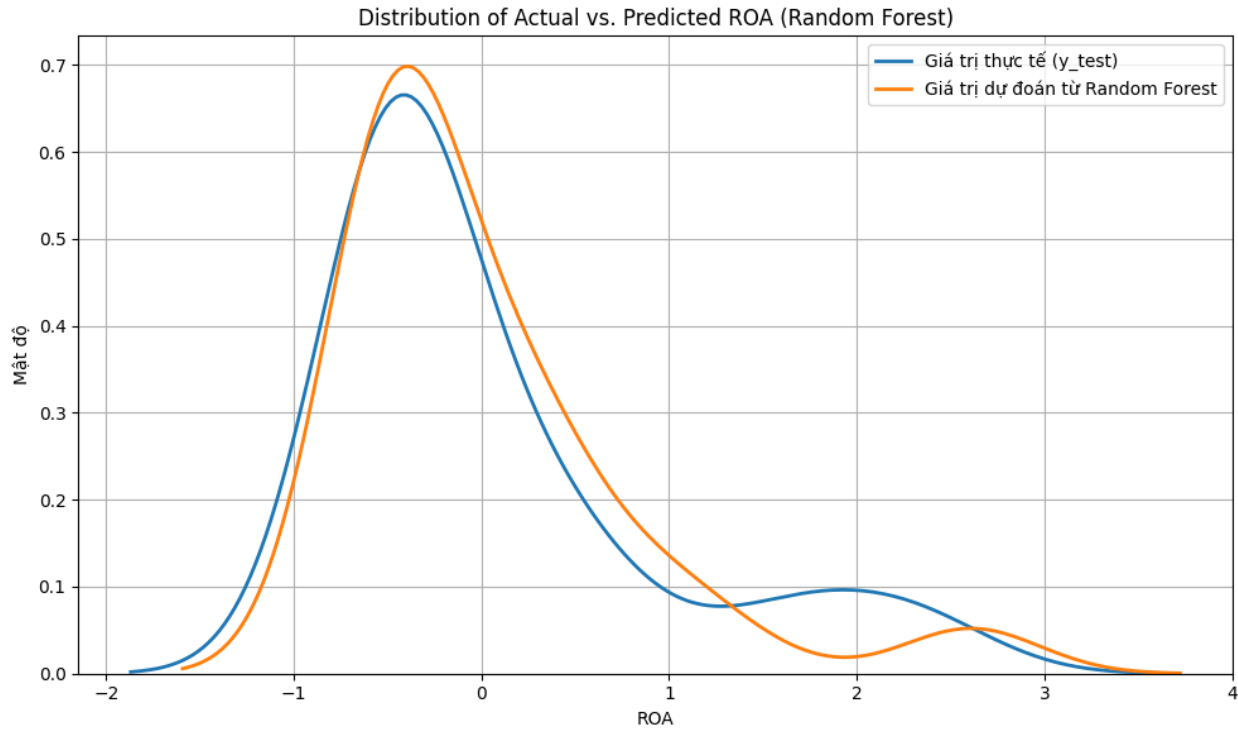


Figure 4.9. Comparison of predicted values of the model and test data for the model with ROA as the dependent variable

Source: Author's visual results in Python

4.6 General discussion of research result

Short-term debt ratio (STDR) and ROA

The short-term debt ratio (STDR) according to the analysis results from Feature Importances showed a slight impact (with $\text{feature_importance} = 0.0359$) and SHAP value -0.4 showed a low negative impact of this variable on the target variable ROA, reflecting an inverse relationship between the short-term debt ratio and the financial performance of construction companies. Specifically, companies with a higher total debt ratio have lower financial performance. Furthermore, this result is fully consistent with Hypothesis H1, supported by the conclusions of the authors: Olaniyi et al. (2015), Vu et al. (2020), Hang et al. (2022), Hashim and Hassan (2017).

The reasons for this result are as follows:

(i) According to the Pecking Order Theory, in an imperfect capital market with information asymmetry, businesses often prioritize the use of internal capital. When internal capital is insufficient, they are forced to raise debt, especially short-term debt, due to its quick and easy accessibility. However, for construction firms—which have the characteristic of long project cycles and slow cash flow recovery—the use of a high proportion of short-term debt increases financial costs and liquidity risks, thereby negatively impacting profitability (ROA).

(ii) According to the Trade-Off Theory, businesses can leverage the low interest rates of short-term debt to reduce capital costs. However, in the construction industry—where construction projects often have slow disbursement schedules and unstable cash flows—overreliance on short-term debt can lead to cash flow imbalances, continuous maturity pressures, and the risk of financial distress. Although initial capital costs are low, the frequent pressure to repay short-term debt often nullifies the potential benefits of borrowing, thereby reducing the company's financial efficiency.

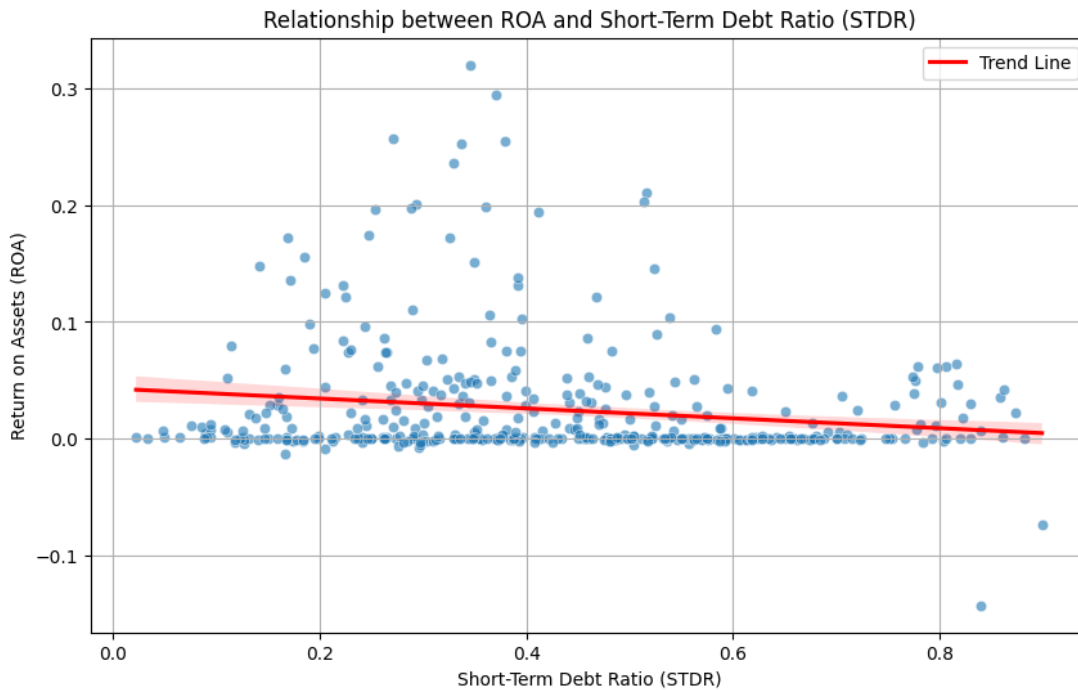


Figure 4.10. Relationship between ROA and Short-Term Debt Ratio (STDR)

Source: Author's compilation

Long-term debt ratio (LTDR) and ROA

The regression analysis results show that the long-term debt ratio (LTDR) according to the analysis results from Feature Importances showed a slight impact (with $\text{feature_importance} = 0.04419$) and SHAP value ≈ -0.05 showed a low negative impact of this variable on the target variable ROA, indicating an inverse relationship between total debt and the financial performance of construction companies.

This result reflects the inverse relationship between long-term debt and financial performance. According to the Trade-Off Theory, while long-term debt may provide tax shield benefits due to interest expense being tax-deductible, when debt levels exceed the optimal threshold, companies will face increasing financial distress costs, including high interest rates, default risk, and prolonged financial pressure. For construction firms, which have long project cycles and high capital requirements, excessive reliance on long-term debt can reduce financial flexibility, making it difficult for firms to adapt to market fluctuations or adjust investment strategies. Furthermore, the burden of long-term debt repayment may limit the ability to capitalize on new investment opportunities, thereby reducing operational efficiency and profitability.

These results are consistent with the hypothesis H2 proposed in this thesis and align with the empirical research findings of previous authors: Hang et al. (2022), Chinaemerem and Anthony (2012), Nguyen and Nguyen (2020), Abor (2005).

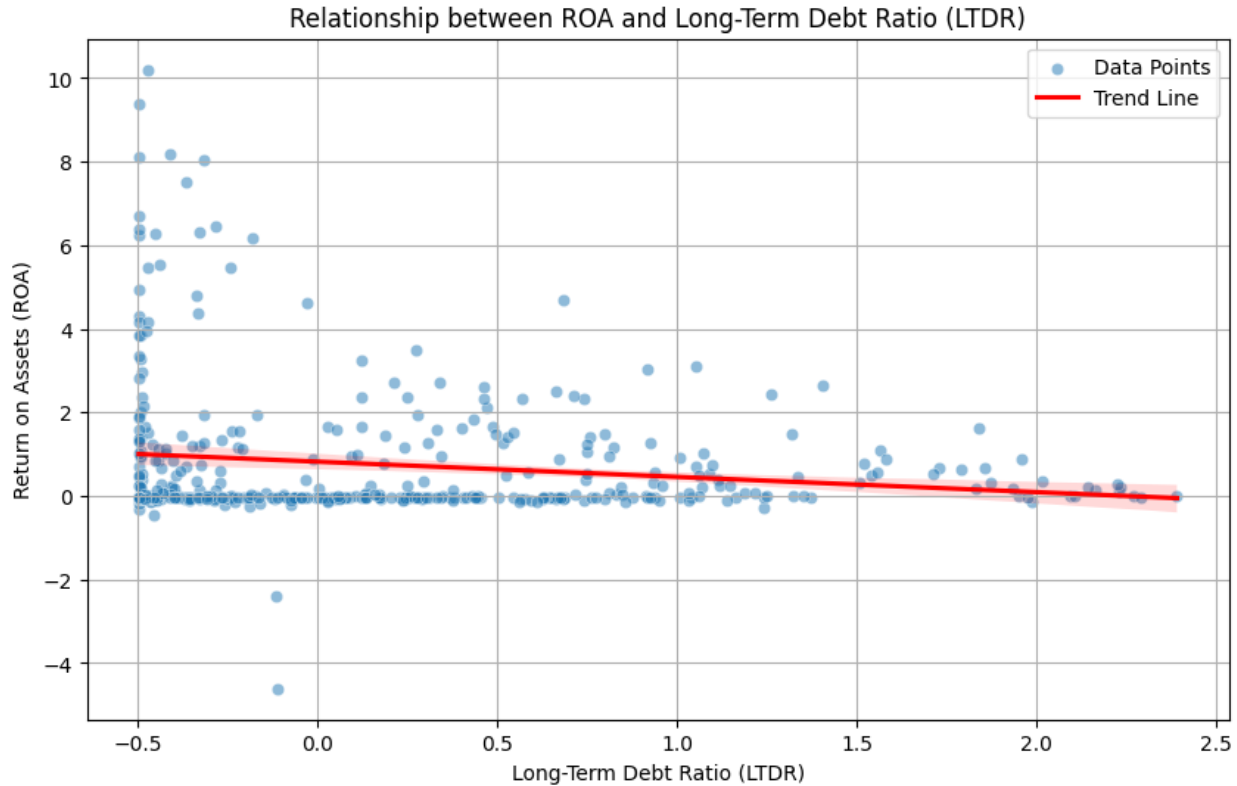


Figure 4.11. Relationship between ROA and Long-Term Debt Ratio (LTDR)

Source: Author's compilation

Total debt ratio (TDR) and ROA

The final component in the capital structure variables of the thesis is the Debt-to-Total Assets Ratio (TDR). According to the analysis results from Feature Importances, it showed a strong impact (with `feature_importance` = 0.3389) and SHAP value ≈ -0.23 , indicating a high negative impact of this variable on the target variable ROA. This indicates that the total debt-to-total assets ratio has an inverse effect on the financial performance of construction companies.

The research results are fully consistent with the previously mentioned hypothesis H3 and with the results from the author's previous studies: Hang et al. (2022), Chinaemerem and Anthony (2012), Nguyen and Nguyen (2020), Abor (2005), Gill et al. (2011).

This result can be explained by extending the Modigliani and Miller (1963) theory to imperfect markets, where increasing debt only increases firm value when tax

conditions and risk control capabilities are ensured—which is difficult to achieve for construction firms with incomplete internal control mechanisms.

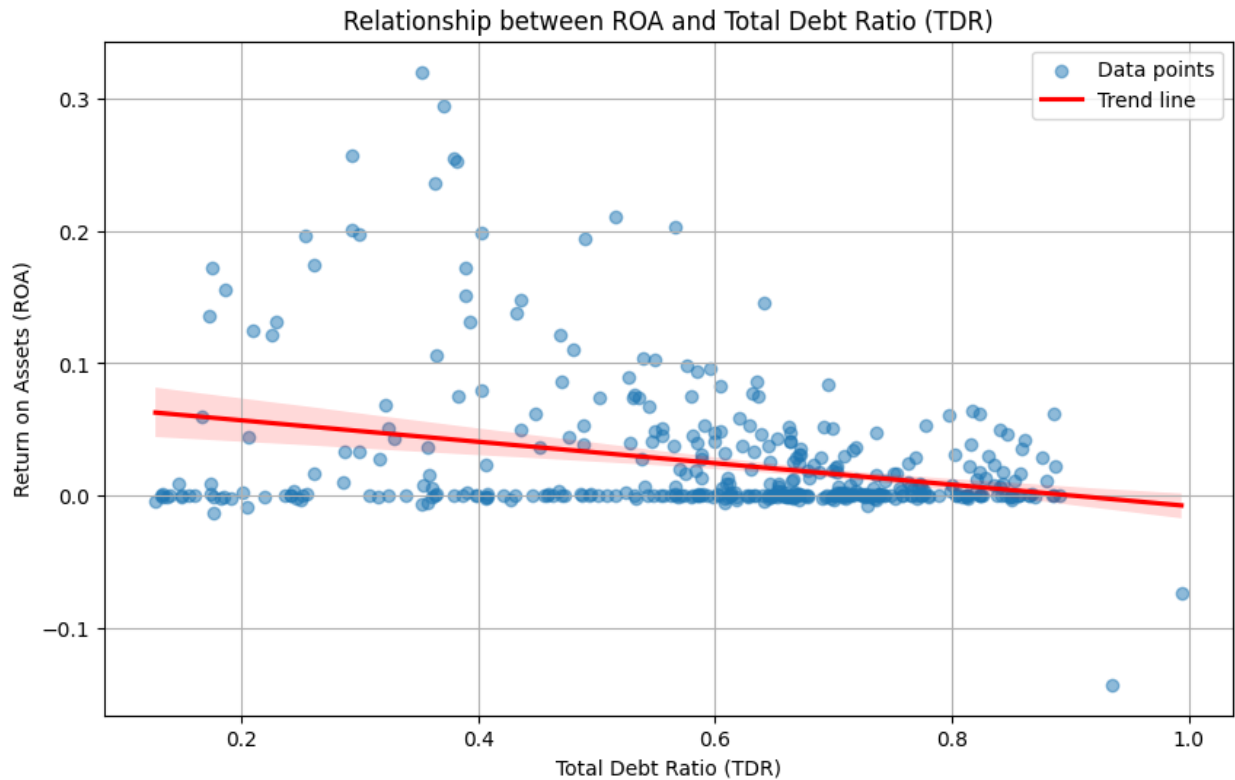


Figure 4.12. Relationship between ROA and Total Debt Ratio (TDR)

Source: Author's compilation

Firms size (SIZE) and ROA

Firms size (SIZE) according to the analysis results from Feature Importances showed a slight impact (with feature_importance = 0.0731) and SHAP value ≈ -0.05 showed a slight negative impact of this variable on the target variable ROA. This result is consistent with previous studies by Hang et al. (2022), Chinaemerem and Anthony (2012), Nguyen and Nguyen (2020), Abor (2005). Larger firms typically have higher financial performance due to advantages in resources, cost optimization, and economies of scale. Large firms have better bargaining power with suppliers and customers, easier access to capital at lower costs, and stronger management capabilities and risk-bearing capacity. These factors help firms maintain stable operations and enhance profitability.

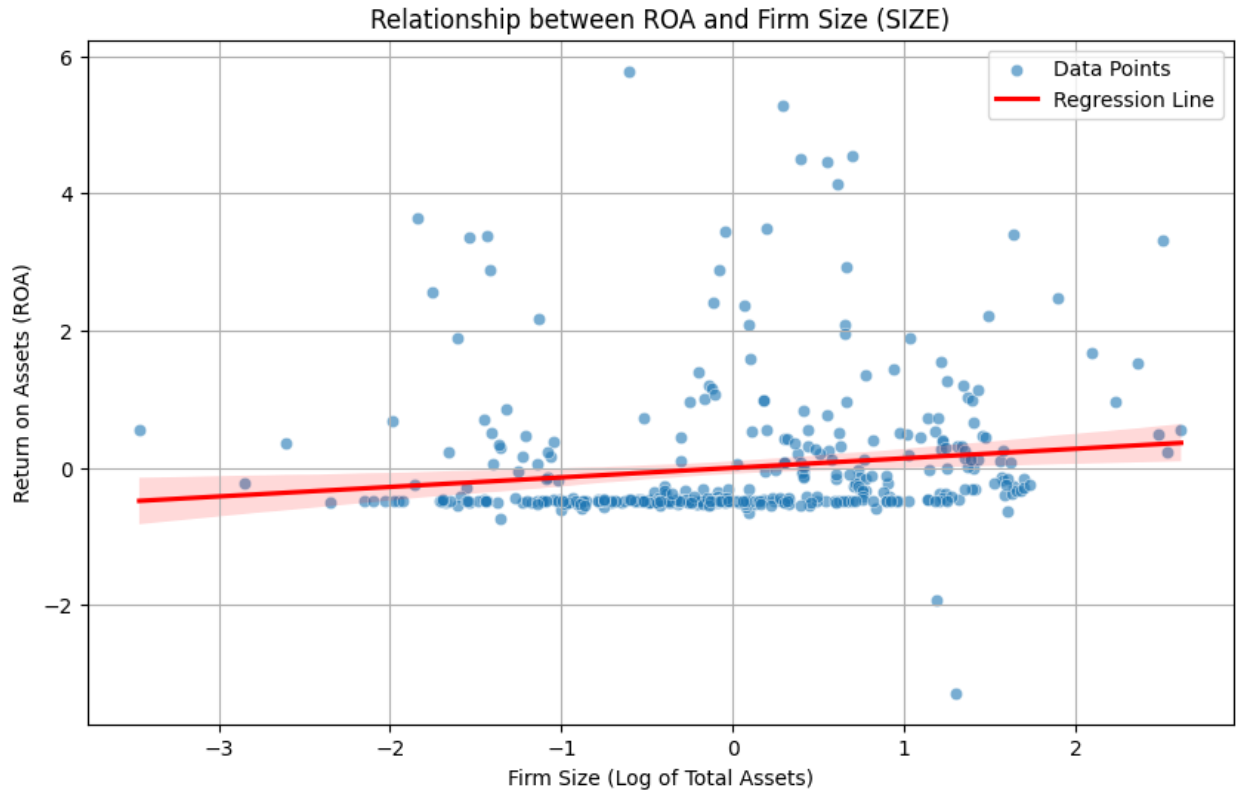


Figure 4.13. Relationship between ROA and Firm Size (SIZE)

Source: Author's compilation

Fixed asset ratio (TANG) and ROA

The fixed asset ratio (TANG) according to the analysis results from Feature Importances showed a slight impact (with $\text{feature_importance} = 0.1133$) and SHAP value ≈ -0.06 , indicating a slight negative influence of this variable on the target variable ROA.

This result is fully consistent with previous studies reviewed by Chinaemerem and Anthony (2012), Nguyen and Nguyen (2020). Accordingly, a high proportion of fixed assets is often associated with positive financial performance, especially in industries that use a lot of machinery and equipment, such as construction or manufacturing. Companies with large fixed assets usually demonstrate a high level of long-term investment, better control over production processes, and higher output quality. Additionally, fixed assets can be used as collateral to facilitate access to loans under favorable terms. When utilized effectively, fixed assets contribute to increased productivity, reduced operating costs, and ultimately improved profitability for businesses.

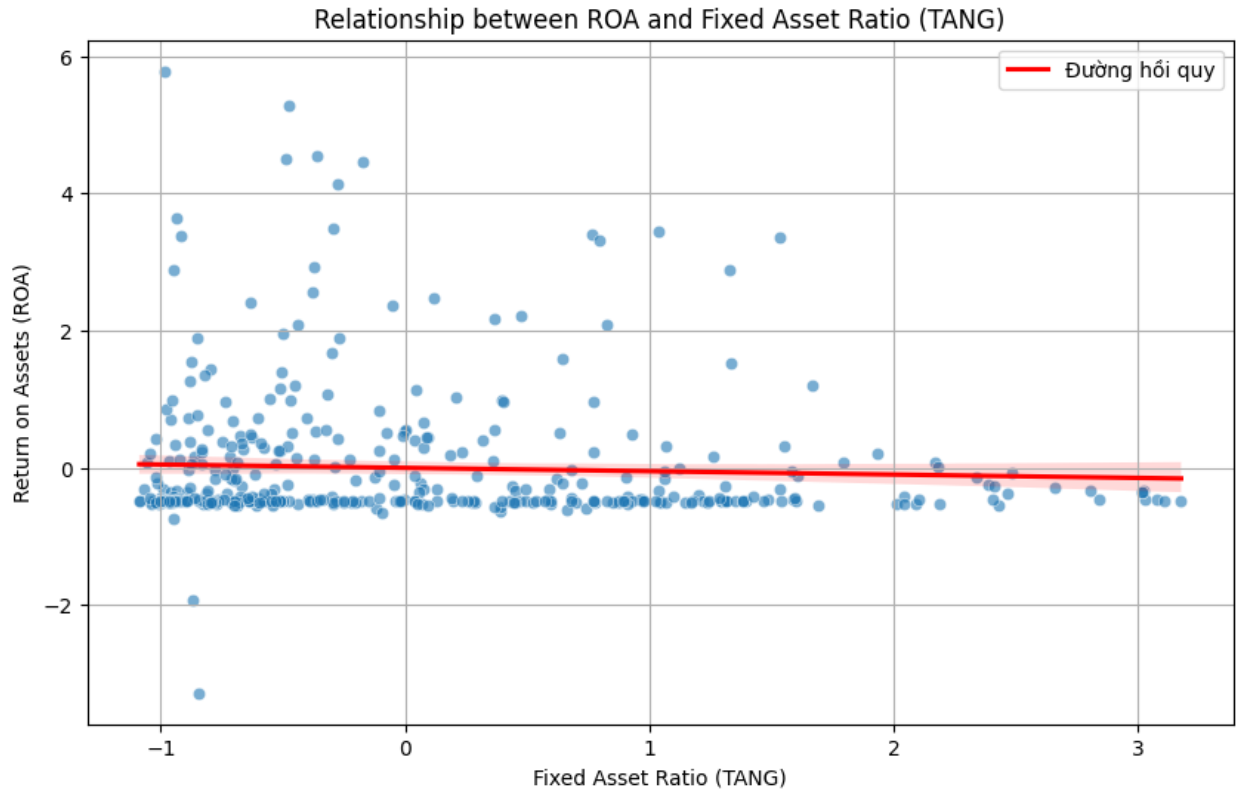


Figure 4.14. Relationship between ROA and Fixed Asset Ratio (TANG)

Source: Author's compilation

Liquidity Ratio (LIQ) and ROA

The Liquidity Ratio (LIQ) according to the analysis results from Feature Importances showed a slight impact (with $\text{feature_importance} = 0.0450$) and SHAP value ≈ -0.05 , indicating a slight negative impact of this variable on the target variable ROA. This result is consistent with the studies by Chinaemerem and Anthony (2012), Nguyen and Nguyen (2020), Abor (2005), Olaniyi et al. (2015). Accordingly, liquidity ratios reflect a company's ability to meet its short-term debt obligations using current assets. This means that maintaining good liquidity levels indicates better working capital management, thereby improving asset utilization efficiency.

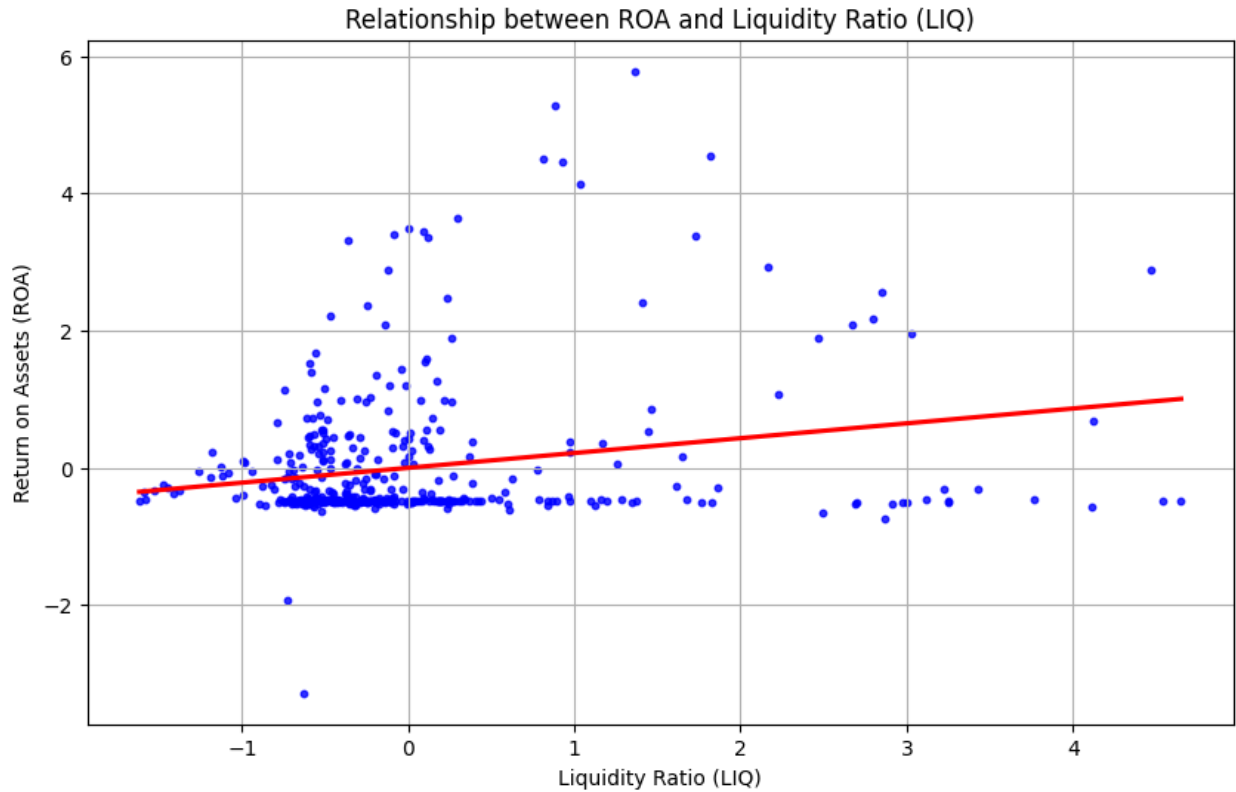


Figure 4.15. Relationship between ROA and Liquidity Ratio (LIQ)

Source: Author's compilation

GDP growth rate and ROA

The GDP growth rate according to the analysis results from Feature Importances showed a strong impact (with `feature_importance` = 0.3318) and SHAP value $\approx +0.49$, indicating a strong positive influence of this variable on the target variable ROA. Although GDP growth is expected to boost revenue and profits, it may have adverse effects on the construction industry. This is because when GDP grows, prices of raw materials or wages paid to workers may increase faster than revenue, leading to a decline in profit margins. Additionally, GDP growth may also lead to faster increases in borrowing costs due to tight monetary policy, which affects profits. This result contradicts previous empirical studies by other authors. The reason may stem from the fact that economic cycles affect different sectors and investment timings differently: Chinaemerem and Anthony (2012), Nguyen and Nguyen (2020), Abor (2005), Olaniyi et al. (2015).

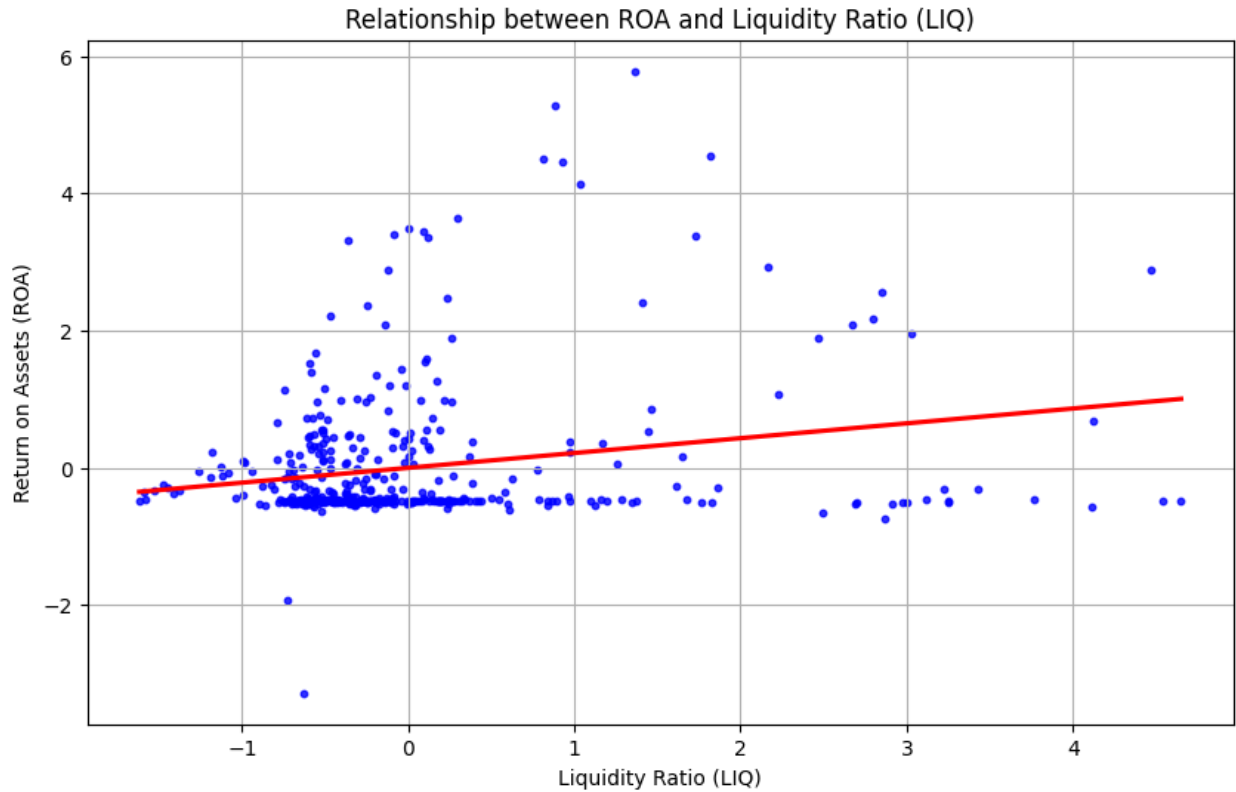


Figure 4.16. Relationship between ROA and GDP growth rate (GDP)

Source: Author's compilation

Inflation (INF) and ROA

The Inflation rate (INF) according to the analysis results from Feature Importances showed a slight impact (with `feature_importance` = 0.0201) and SHAP value $\approx +0.04$, indicating a slight positive influence of this variable on the target variable ROA, indicating an inverse relationship between the macroeconomic factor INF and financial performance. For construction companies, rising inflation may bring about various disadvantages for businesses. In the construction industry, high inflation increases input costs, including raw materials and labor, which negatively impacts profit margins. Additionally, rapid inflation may lead to reduced investment due to difficulties in forecasting long-term cash flows and valuing projects, particularly in industries with long project implementation periods, such as construction.

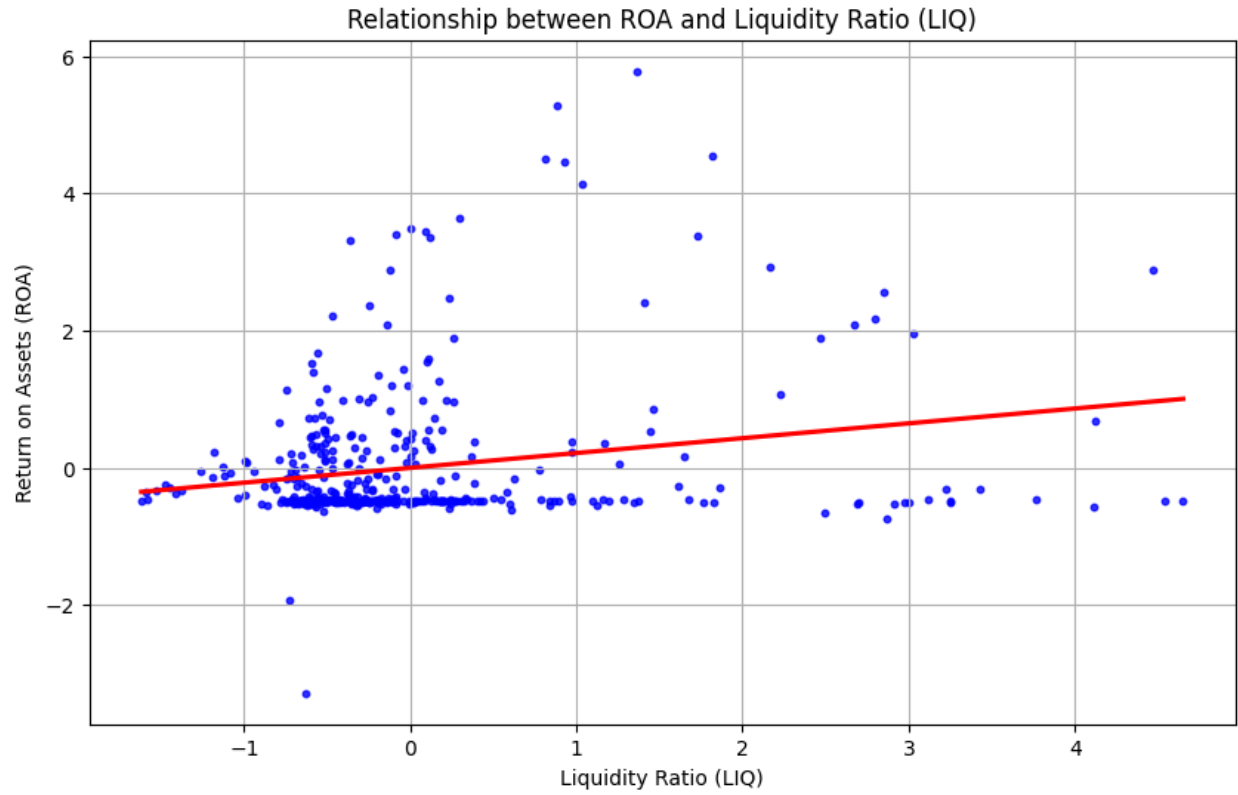


Figure 4.17. Relationship between ROA and Inflation (INF)

Source: Author's compilation

SUMMARY OF CHAPTER 4

In this chapter, the author presents and discusses the empirical results of the impact of capital structure on the financial performance of construction companies listed on the Vietnam Stock Exchange from 2015 to 2024. The research findings reveal the relationship between capital structure, its components, and financial performance of construction companies listed on the Vietnam Stock Exchange during the period from 2015 to 2024. Specifically:

- The components of capital structure include: Short-term debt ratio (STDR), Long-term debt ratio (LTDR), and Total debt to total assets ratio (TDR), which have an inverse relationship with the financial performance of construction companies listed on the Vietnam Stock Exchange. Additionally, the control variables that have an inverse relationship with financial performance are Liquidity Ratio (LIQ).*

- The remaining control variables include: Firm Size (SIZE); Fixed Asset Ratio (TANG); macroeconomic variables such as GDP Growth Rate (GDP) and Inflation Rate (INF) have a positive correlation with the financial performance of construction companies listed on the Vietnam Stock Exchange.*

- In addition to the independent variables related to capital structure, the regression results of the control variables were also analyzed and clarified by the author in the content of the chapter.*

Based on the research findings of Chapter 4, the author will use these results as a foundation to propose management implications for effectively managing and utilizing capital structure to enhance the financial performance of listed construction companies on the Vietnam Stock Exchange in the subsequent Chapter 5.

CHAPTER 5: CONCLUSION AND MANAGEMENT IMPLICATIONS

In Chapter 5, the author will propose management implications based on the research results of Chapter 4, followed by limitations and directions for future research.

5.1 Conclusion

This thesis aims to analyze and evaluate the impact of capital structure on the financial performance of construction companies listed on the Vietnam Stock Exchange during the period from 2015 to 2024. To achieve this, the thesis was conducted following the research process outlined below: First, the foundation for the research topic was established in Chapter 1; Next, the theoretical framework was synthesized and developed, and research gaps were identified through a review of relevant empirical research methods in Chapter 2; Then, a model is developed and research methods are identified in Chapter 3; In the next step, the research presents the research results in Chapter 4. Additionally, to achieve the overall objective, the thesis sets out three specific objectives as follows:

First specific objective : *Identify the capital structure and components of the capital structure that affect the financial performance of construction companies listed on the Vietnam Stock Exchange.*

Based on a review of relevant empirical studies, this thesis has identified the general concept of capital structure and determined its components. The components of capital structure discussed in this study are Total Debt Ratio (TDR), Short-Term Debt Ratio (STDR), and Long-Term Debt Ratio (LTDR). Based on empirical evidence from previous studies, the thesis finds that there is a relationship between capital structure and its components and the financial performance of construction companies listed on the Vietnam Stock Exchange. Specifically, the author determines that the effects vary depending on the variable, with both positive and negative effects emerging.

Second specific objective: *Measure the extent and determine the direction of the impact of capital structure and its components on the financial performance of construction companies listed on the Vietnam Stock Exchange.*

The thesis applied machine learning models and panel data regression analysis to quantify the relationship between intellectual capital and financial performance of construction companies listed on the Vietnam Stock Exchange. The research findings clearly affirm the crucial role of capital structure in managing and enhancing financial performance of businesses. Among these, the short-term debt ratio (STDR); long-term debt ratio (LTDR); total debt-to-total assets ratio (TDR); and liquidity ratio (LIQ) showed an inverse relationship with corporate financial performance. Meanwhile, variables such as firm size (SIZE); fixed asset ratio (TANG); GDP growth rate (GDP); and inflation rate (INF) have shown a positive relationship with firm financial performance.

Third specific objective : *Propose a number of management implications for the effective use and management of capital structure to improve the financial performance of construction companies listed on the Vietnam Stock Exchange.*

Based on the research findings presented in Chapter 4, this thesis will propose several management implications related to the effective use of capital structure to enhance the financial performance of construction companies listed on the Vietnam Stock Exchange in Chapter 5.

5.2 Management implications

Between 2015 and 2024, Vietnam's construction sector has remained a key contributor to national economic growth. However, despite progress in capital market integration and enterprise restructuring, many firms continue to struggle with inefficiencies in financial structure and internal management systems.

According to Vu et al. (2018), the equitization of state-owned enterprises has allowed construction firms to access capital more efficiently and encouraged them to improve governance practices. Nevertheless, issues such as concentrated ownership and overly stable boards still limit firm performance. Specifically, the study finds that CEO ownership and ownership concentration positively influence Return on Assets (ROA), while excessive board stability negatively affects both ROA and Return on Equity (ROE).

According to Pham et al. (2018), capital structure remains one of the most critical internal factors impacting financial performance. Their analysis of 30 listed construction-material companies shows that the average debt-to-total-assets ratio was 56.24%, indicating a heavy reliance on debt financing. The study demonstrates that a 1% increase in this ratio leads to a 0.1% decrease in ROA, highlighting the detrimental effect of high leverage on profitability.

Furthermore, according to Pham et al. (2018), inefficient fixed asset investment and poor receivables management also contribute to reduced financial performance. A high proportion of fixed assets increases depreciation expenses, while prolonged accounts receivable periods reduce liquidity and raise credit risk. These internal inefficiencies are particularly damaging in a capital-intensive industry like construction.

Interestingly, the same study finds that business risk has a positive effect on profitability, especially during market recovery phases. According to Pham et al. (2018), firms that accept higher levels of business risk tend to achieve better financial outcomes, suggesting that strategic risk-taking, if well-managed, can lead to competitive advantage.

In summary, according to both Vu et al. (2018) and Pham et al. (2018), Vietnam's construction sector in the 2015–2024 period continues to face structural challenges. While there has been progress in market participation and access to financing, limitations in capital structure, asset utilization, and governance still hinder overall performance. These findings provide a foundation for proposing targeted managerial implications in the subsequent chapter.

For short-term debt ratio (STDR)

The research findings revealed a negative correlation between ROA with respect to STDR. This implies that a higher short-term debt ratio leads to a decrease in the financial efficiency of the company. From this, the author suggests that companies should limit the use of short-term debt in their financial performance improvement strategies to reduce financial pressure and liquidity risk, especially in cases of long project cycles and slow cash flow recovery. Additionally, according to the Pecking Order theory, managers should prioritize internal capital or long-term debt with stable interest rates,

recommending the use of equity before borrowing. Furthermore, management should simultaneously establish a rigorous cash flow management plan to avoid financial imbalances in the company.

For the long-term debt ratio (LTDR)

The research results have shown a negative correlation between ROA with respect to LTDR. This reflects the negative impact when the long-term debt ratio exceeds the optimal level. This means that excessive use of long-term debt can reduce a company's financial efficiency, primarily due to increased costs and reduced flexibility in capital structure. Managers should determine the optimal debt threshold based on liquidity and market conditions, while diversifying capital sources to avoid overreliance on long-term debt.

Debt-to-total assets ratio

The research results have shown a negative correlation between ROA for TDR. This indicates that high total debt reduces financial efficiency. This means that businesses need to control total debt levels to avoid exceeding the optimal threshold, which can lead to financial pressure and reduced profitability.

5.3 Limitations and suggestions for further research

After conducting the research, the thesis identified the following limitations:

First, the study has not yet conducted a comprehensive assessment of the impact of capital structure on the financial performance of the entire construction industry in Vietnam. Therefore, the research findings on the impact of capital structure on the financial performance of listed construction companies on the Vietnam Stock Exchange only provide a relatively limited perspective and do not offer a comprehensive view of the entire industry.

Second, the data used in this study covers the period from 2015 to 2024, so the results do not assess the impact of capital structure on financial performance in the long term.

Third, the assessment of financial performance is based solely one indicator (ROA), which does not fully reflect the overall financial performance of the company and its relationship with capital structure.

Fourth, the Random Forest model may overlook other complex nonlinear factors in the analysis and evaluation of the extent to which capital structure affects the financial performance of construction companies.

From this, the thesis suggests the following directions for further research:

1) Future studies related to the topic of capital structure and financial performance should expand the scope of data collection to provide a more comprehensive overview and draw comparative conclusions between different types of domestic and international enterprises.

2) Future studies should expand the timeframe of the data to enhance the accuracy and comprehensiveness of research on the impact of capital structure on the financial performance of construction companies.

3) Future studies should consider diversifying financial performance indicators to optimize and enhance the accuracy of assessing the impact of capital structure on the financial performance of construction companies.

4) Future studies should incorporate other machine learning models such as SVR or XGBoost in the analysis and evaluation of the impact of capital structure on the financial performance of construction firms to enhance accuracy and comprehensiveness.

SUMMARY OF CHAPTER 5

In Chapter 5, the author achieved the research objectives set out in Chapter 1 and summarized the research results and process. Finally, the study proposes managerial implications for developing strategies to effectively utilize and manage capital structure in order to enhance the financial performance of construction companies listed on the Vietnam Stock Exchange. The limitations of the study are also discussed, and based on these, the study suggests some directions for future related research.

CONCLUSION

The study “The Impact of Capital Structure on the Financial Performance of Construction Companies Listed on the Vietnam Stock Exchange from 2015 to 2024” analyzed the impact of capital structure on the factors controlling the financial performance of the companies mentioned in the study.

The results indicated that financial performance is significantly influenced by capital structure management and macroeconomic conditions, with variables such as short-term debt, long-term debt, and total debt having a negative impact, while firm size and fixed assets yield positive benefits. Excess liquidity ratio was identified as a hindering factor, while GDP growth and inflation played a supportive role in a volatile economic environment. The study utilized the Random Forest model to analyze the relationship between capital structure and financial performance. The optimized Random Forest model achieved R^2 values of 40.2% for ROA ($CV R^2 = 58.2\%$) confirming high reliability in predicting financial performance, thereby enhancing the reliability of the study.

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APPENDIX

APPENDIX 1: List of 42 construction companies listed on the Vietnam Stock Exchange in the research sample of the thesis

No	Name	Company Name
1	VCG	Tổng Công ty Cổ phần Xuất nhập khẩu và xây dựng Việt Nam
2	CII	Công ty Cổ phần Đầu tư Hạ tầng Kỹ thuật T.P Hồ Chí Minh
3	HHV	Công ty cổ phần Đầu tư Hạ tầng Giao thông Đèo Cả
4	VGC	Tổng Công ty Viglacera - CTCP
5	HT1	Công ty Cổ phần xi măng VICEM Hà Tiên
6	PC1	Công ty Cổ phần Xây lắp điện 1
7	LCG	Công ty Cổ phần LIZEN
8	VCS	Công ty Cổ phần VICOSTONE
9	FCN	Công ty Cổ phần FECON
10	DNP	Công ty Cổ phần DNP Holding
11	CTR	Tổng công ty cổ phần công trình Viettel
12	CTD	Công ty Cổ phần Xây dựng Coteccons
13	S99	Công ty Cổ phần SCI
14	VNE	Tổng Công ty Cổ phần Xây dựng điện Việt Nam
15	DC4	Công ty cổ phần Xây dựng DIC Holdings
16	BMP	Công ty Cổ phần nhựa Bình Minh
17	VC2	Công ty Cổ phần Đầu tư và Xây dựng VINA2
18	CTI	Công ty Cổ phần Đầu tư Phát triển Cường Thuận IDICO
19	DPG	Công ty Cổ phần Đạt Phương
20	CRC	Công ty Cổ phần Create Capital Việt Nam
21	CDC	Công ty Cổ phần Chương Dương
22	HVH	Công ty cổ phần Đầu tư và Công nghệ HVC
23	TTL	Tổng Công ty Thăng Long - Công ty Cổ phần
24	LBM	Công ty Cổ phần Khoáng sản và Vật liệu Xây dựng Lâm Đồng

25	L18	Công ty Cổ phần Đầu tư và Xây dựng số 18
26	CSC	Công ty cổ phần Tập đoàn COTANA
27	C47	Công ty Cổ phần Xây dựng 47
28	SD9	Công ty Cổ phần Sông Đà 9
29	VTV	Công ty Cổ phần Năng lượng và Môi trường VINCEM
30	L14	Công ty Cổ phần Licogi 14
31	HUB	Công ty Cổ phần Xây lắp Thừa Thiên Huế
32	THG	Công ty Cổ phần Đầu tư và Xây dựng Tiền Giang
33	VHL	Công ty Cổ phần Viglacera Hạ Long
34	HTI	Công ty Cổ phần Đầu tư Phát triển Hạ tầng IDICO
35	SJE	Công ty Cổ phần Sông Đà 11
36	NNC	Công ty Cổ phần Đá Núi Nhỏ
37	HBC	Công ty Cổ phần Tập đoàn Xây dựng Hòa Bình
38	DHC	CTCP Đông Hải Bến Tre
39	HTN	CTCP Hưng Thịnh Incons
40	DTD	CTCP Đầu tư Phát triển Thành Đạt
41	CC1	Tổng Công ty xây dựng số 1
42	C4G	Công ty cổ phần Tập đoàn CIENCO4

Source: Compilation by the author

Appendix 2: Research data of 42 Construction Enterprises listed on the Vietnam Stock Exchange during the period from 2015 to 2024.

Name	Year	ROA	STDR	LTDR	TDR	SIZE	TANG	LIQ	GDP	INF
C4G	2015	0.01102	0.52486	0.35548	0.88034	3.84696	0.31503	1.01829	0.06987	0.00600
C4G	2016	0.02349	0.44973	0.38577	0.83550	3.86181	0.36504	1.00783	0.06690	0.02700
C4G	2017	0.01809	0.44845	0.39413	0.84259	3.86909	0.41087	1.01812	0.06940	0.03500
C4G	2018	0.01737	0.47030	0.38699	0.85730	3.89518	0.35267	0.99786	0.07465	0.03500
C4G	2019	0.01304	0.42795	0.40061	0.82857	3.84967	0.37384	1.04127	0.07359	0.02800
C4G	2020	0.00870	0.44160	0.40178	0.84338	3.86966	0.34599	1.08706	0.02865	0.03200
C4G	2021	0.00791	0.44667	0.38936	0.83603	3.89264	0.31792	1.11950	0.02554	0.01800
C4G	2022	0.01871	0.34117	0.35738	0.69854	3.91756	0.28470	1.51415	0.08124	0.03200
C4G	2023	0.01336	0.30136	0.30898	0.61034	3.98251	0.23723	1.91045	0.05046	0.03300
C4G	2024	0.01899	0.31366	0.26972	0.58337	3.97355	0.23262	1.81075	0.07090	0.03600
CII	2015	0.05254	0.11045	0.58148	0.69193	4.17799	0.14997	2.80317	0.06987	0.00600
CII	2016	0.09846	0.19096	0.38534	0.57630	4.00609	0.06082	1.55519	0.06690	0.02700
CII	2017	0.07787	0.19419	0.43734	0.63152	4.31616	0.26718	1.39941	0.06940	0.03500
CII	2018	0.00966	0.29195	0.36171	0.65367	4.34775	0.24092	0.90757	0.07465	0.03500
CII	2019	0.01784	0.39222	0.31012	0.70235	4.46611	0.19938	0.97520	0.07359	0.02800
CII	2020	0.01598	0.28067	0.45583	0.73650	4.47051	0.18333	1.32124	0.02865	0.03200
CII	2021	-0.00784	0.29669	0.43189	0.72858	4.48954	0.30498	1.14617	0.02554	0.01800
CII	2022	0.03013	0.33511	0.37424	0.70934	4.45575	0.29848	0.75511	0.08124	0.03200
CII	2023	0.01115	0.24542	0.49827	0.74369	4.52093	0.49422	0.84855	0.05046	0.03300
CII	2024	0.01684	0.24403	0.50656	0.75058	4.56493	0.44287	0.96308	0.07090	0.03600
CTD	2015	0.09377	0.58338	0.00172	0.58509	3.89293	0.05627	1.42261	0.06987	0.00600
CTD	2016	0.12113	0.46804	0.00103	0.46907	4.06970	0.04986	1.80948	0.06690	0.02700
CTD	2017	0.10409	0.53912	0.00069	0.53980	4.20078	0.04571	1.67331	0.06940	0.03500
CTD	2018	0.08978	0.52615	0.00055	0.52669	4.22591	0.04421	1.73120	0.07465	0.03500
CTD	2019	0.04389	0.47598	0.00117	0.47715	4.20948	0.04349	1.91010	0.07359	0.02800

CTD	2020	0.02363	0.40637	0.00039	0.40677	4.15098	0.04265	2.23664	0.02865	0.03200
CTD	2021	0.00173	0.40752	0.00018	0.40770	4.14378	0.03814	2.24708	0.02554	0.01800
CTD	2022	0.00110	0.53918	0.02776	0.56694	4.27800	0.02953	1.70780	0.08124	0.03200
CTD	2023	0.00868	0.58866	0.02304	0.61170	4.33550	0.02192	1.56048	0.05046	0.03300
CTD	2024	0.01372	0.67689	0.00222	0.67911	4.43260	0.01675	1.35512	0.07090	0.03600
CTI	2015	0.02236	0.23043	0.58474	0.81517	3.51602	0.44310	0.52860	0.06987	0.00600
CTI	2016	0.02933	0.15925	0.61042	0.76966	3.58516	0.67053	0.76515	0.06690	0.02700
CTI	2017	0.03577	0.16003	0.51248	0.67250	3.63370	0.62177	0.98590	0.06940	0.03500
CTI	2018	0.02900	0.15160	0.51657	0.66817	3.64643	0.59249	0.99342	0.07465	0.03500
CTI	2019	0.01855	0.16787	0.51033	0.67821	3.65578	0.55420	0.64794	0.07359	0.02800
CTI	2020	0.02236	0.14873	0.55352	0.70225	3.65622	0.54908	0.98839	0.02865	0.03200
CTI	2021	-0.00277	0.11787	0.61821	0.73607	3.67718	0.72410	0.86427	0.02554	0.01800
CTI	2022	0.02152	0.13222	0.56887	0.70110	3.65287	0.73453	0.68132	0.08124	0.03200
CTI	2023	0.01772	0.13840	0.54922	0.68762	3.65473	0.70514	0.58696	0.05046	0.03300
CTI	2024	0.02547	0.16422	0.50167	0.66589	3.65909	0.67224	0.64310	0.07090	0.03600
DHC	2015	0.19641	0.25423	0.00000	0.25423	2.60872	0.53921	1.68724	0.06987	0.00600
DHC	2016	0.13615	0.17226	0.00000	0.17226	2.79909	0.30025	3.93356	0.06690	0.02700
DHC	2017	0.06222	0.25587	0.19285	0.44872	3.11057	0.14274	1.70792	0.06940	0.03500
DHC	2018	0.07401	0.22732	0.30808	0.53540	3.25821	0.07361	1.37041	0.07465	0.03500
DHC	2019	0.08593	0.26260	0.20821	0.47080	3.32496	0.56663	1.57727	0.07359	0.02800
DHC	2020	0.17157	0.32587	0.06348	0.38934	3.35876	0.49709	1.48399	0.02865	0.03200
DHC	2021	0.20045	0.29325	0.00000	0.29325	3.38044	0.43791	1.66220	0.02554	0.01800
DHC	2022	0.13164	0.39241	0.00000	0.39241	3.45977	0.39486	1.47000	0.08124	0.03200
DHC	2023	0.10612	0.36448	0.00024	0.36472	3.46465	0.35734	1.68032	0.05046	0.03300
DHC	2024	0.07464	0.38096	0.00170	0.38265	3.51117	0.30647	1.76868	0.07090	0.03600
DTD	2015	0.00000	0.23731	0.29493	0.53223	2.42892	0.14667	2.51111	0.06987	0.00600
DTD	2016	0.02768	0.52405	0.01364	0.53770	2.67489	0.11595	1.60987	0.06690	0.02700
DTD	2017	0.04030	0.51883	0.00898	0.52781	2.69237	0.10574	1.50506	0.06940	0.03500
DTD	2018	0.04921	0.54307	0.00634	0.54941	2.76293	0.08677	1.28647	0.07465	0.03500

DTD	2019	0.04748	0.28391	0.45209	0.73600	3.23033	0.09642	1.34654	0.07359	0.02800
DTD	2020	0.08351	0.22296	0.47269	0.69565	3.33213	0.11951	1.16604	0.02865	0.03200
DTD	2021	0.09557	0.24412	0.35171	0.59583	3.28867	0.12131	1.09469	0.02554	0.01800
DTD	2022	0.07586	0.23064	0.30087	0.53152	3.30872	0.11163	1.33031	0.08124	0.03200
DTD	2023	0.14822	0.14182	0.29393	0.43575	3.33947	0.09547	2.76729	0.05046	0.03300
DTD	2024	0.07959	0.11454	0.28884	0.40338	3.34514	0.15967	3.45545	0.07090	0.03600
FCN	2015	0.05241	0.43910	0.22335	0.66245	3.46980	0.22380	1.51112	0.06987	0.00600
FCN	2016	0.05272	0.38513	0.24445	0.62958	3.52275	0.22560	1.75020	0.06690	0.02700
FCN	2017	0.04626	0.46857	0.17001	0.63858	3.58409	0.16830	1.58855	0.06940	0.03500
FCN	2018	0.05265	0.45989	0.13060	0.59049	3.67424	0.15815	1.68069	0.07465	0.03500
FCN	2019	0.03746	0.49647	0.06868	0.56514	3.75185	0.12009	1.60552	0.07359	0.02800
FCN	2020	0.01970	0.57463	0.05621	0.63084	3.83124	0.09927	1.42641	0.02865	0.03200
FCN	2021	0.00944	0.44899	0.16023	0.60922	3.87481	0.25165	1.50428	0.02554	0.01800
FCN	2022	0.00681	0.41579	0.12473	0.54052	3.87973	0.23700	1.66092	0.08124	0.03200
FCN	2023	-0.00490	0.50331	0.10494	0.60825	3.93356	0.20000	1.42080	0.05046	0.03300
FCN	2024	0.00310	0.49543	0.15757	0.65300	3.98704	0.16297	1.32462	0.07090	0.03600
HBC	2015	0.01136	0.79626	0.05638	0.85264	3.86280	0.08754	1.05868	0.06987	0.00600
HBC	2016	0.04961	0.77605	0.06411	0.84016	4.05880	0.09487	1.11416	0.06690	0.02700
HBC	2017	0.06147	0.77903	0.04474	0.82377	4.14608	0.10146	1.07225	0.06940	0.03500
HBC	2018	0.03900	0.77439	0.04179	0.81618	4.20141	0.08933	1.09823	0.07465	0.03500
HBC	2019	0.02426	0.72028	0.04291	0.76319	4.22327	0.08062	1.19752	0.07359	0.02800
HBC	2020	0.00538	0.69107	0.04222	0.73328	4.19179	0.07464	1.26616	0.02865	0.03200
HBC	2021	0.00585	0.70246	0.05282	0.75529	4.21950	0.05816	1.27763	0.02554	0.01800
HBC	2022	-0.14250	0.83952	0.09543	0.93495	4.26278	0.05170	1.06021	0.08124	0.03200
HBC	2023	-0.07314	0.89858	0.09529	0.99388	4.18327	0.04678	0.98147	0.05046	0.03300
HBC	2024	0.06248	0.80527	0.08131	0.88659	4.18785	0.04283	1.10669	0.07090	0.03600
HHV	2015	0.01364	0.47395	0.11501	0.58896	2.08002	0.35741	1.27144	0.06987	0.00600
HHV	2016	0.01195	0.78127	0.03976	0.82103	2.47115	0.21574	0.89320	0.06690	0.02700
HHV	2017	0.02243	0.87281	0.01498	0.88779	2.74297	0.20256	0.80149	0.06940	0.03500

HHV	2018	0.03655	0.70517	0.01335	0.71852	2.55680	0.27226	0.59552	0.07465	0.03500
HHV	2019	0.00517	0.08962	0.67763	0.76725	4.47718	0.47187	0.75531	0.07359	0.02800
HHV	2020	0.00542	0.11034	0.66186	0.77220	4.51077	0.73086	0.40364	0.02865	0.03200
HHV	2021	0.00856	0.09524	0.67875	0.77399	4.53101	0.84513	0.42945	0.02554	0.01800
HHV	2022	0.00834	0.10819	0.65690	0.76508	4.55210	0.80110	0.30691	0.08124	0.03200
HHV	2023	0.00991	0.08548	0.67709	0.76256	4.56561	0.77022	0.37405	0.05046	0.03300
HHV	2024	0.01273	0.09456	0.62452	0.71908	4.59002	0.71462	0.34873	0.07090	0.03600
HPG	2015	0.13739	0.39177	0.04106	0.43283	4.40666	0.32193	1.19237	0.06987	0.00600
HPG	2016	0.19882	0.36071	0.04187	0.40258	4.52149	0.38134	1.51713	0.06690	0.02700
HPG	2017	0.15116	0.34928	0.03970	0.38898	4.72446	0.24891	1.78556	0.06940	0.03500
HPG	2018	0.10995	0.28938	0.19130	0.48068	4.89333	0.16341	1.11807	0.07465	0.03500
HPG	2019	0.07446	0.26513	0.26534	0.53047	5.00765	0.30704	1.12795	0.07359	0.02800
HPG	2020	0.10270	0.39521	0.15448	0.54970	5.11896	0.49852	1.09181	0.02865	0.03200
HPG	2021	0.19368	0.41215	0.07853	0.49067	5.25100	0.38870	1.28173	0.02554	0.01800
HPG	2022	0.04958	0.36625	0.06949	0.43574	5.23131	0.41584	1.29060	0.08124	0.03200
HPG	2023	0.03621	0.38083	0.07153	0.45236	5.27366	0.38341	1.15666	0.05046	0.03300
HPG	2024	0.05354	0.33509	0.15420	0.48930	5.35120	0.30036	1.15220	0.07090	0.03600
HTN	2015	0.01799	0.82144	0.00195	0.82339	2.82414	0.01595	1.14831	0.06987	0.00600
HTN	2016	0.02991	0.82957	0.00107	0.83064	3.23011	0.02699	1.15552	0.06690	0.02700
HTN	2017	0.02842	0.75585	0.12021	0.87605	3.58674	0.00738	1.27693	0.06940	0.03500
HTN	2018	0.04614	0.81719	0.02961	0.84680	3.59734	0.01576	1.17412	0.07465	0.03500
HTN	2019	0.03586	0.85813	0.00111	0.85924	3.71743	0.01063	1.13675	0.07359	0.02800
HTN	2020	0.06450	0.81564	0.00116	0.81680	3.74370	0.04989	1.13937	0.02865	0.03200
HTN	2021	0.03135	0.80140	0.00076	0.80217	3.88582	0.03575	1.15354	0.02554	0.01800
HTN	2022	0.00698	0.83962	0.00082	0.84044	3.96257	0.02809	1.11368	0.08124	0.03200
HTN	2023	0.00791	0.77756	0.02302	0.80059	3.88208	0.03291	1.18831	0.05046	0.03300
HTN	2024	0.00335	0.71046	0.08814	0.79860	3.88402	0.03035	1.30910	0.07090	0.03600
HVH	2015	0.05304	0.77372	0.00411	0.77783	1.87190	0.05890	1.15600	0.06987	0.00600
HVH	2016	0.04349	0.32924	0.00000	0.32924	2.16881	0.10351	2.56892	0.06690	0.02700

HVH	2017	0.05985	0.16697	0.00000	0.16697	2.41458	0.08063	5.04982	0.06940	0.03500
HVH	2018	0.15535	0.18607	0.00000	0.18607	2.51623	0.14715	3.98247	0.07465	0.03500
HVH	2019	0.12141	0.22584	0.00000	0.22584	2.58066	0.16953	3.65832	0.07359	0.02800
HVH	2020	0.05057	0.32400	0.00000	0.32400	2.67149	0.13015	1.59654	0.02865	0.03200
HVH	2021	0.01600	0.35873	0.00000	0.35873	2.85598	0.08074	1.73959	0.02554	0.01800
HVH	2022	0.03303	0.29910	0.00000	0.29910	2.83267	0.07793	1.89399	0.08124	0.03200
HVH	2023	0.01678	0.26191	0.00000	0.26191	2.82031	0.07941	2.11149	0.05046	0.03300
HVH	2024	0.04404	0.20548	0.00000	0.20548	2.84133	0.07114	2.40517	0.07090	0.03600
L14	2015	0.04162	0.86151	0.00000	0.86151	2.69049	0.03101	1.09092	0.06987	0.00600
L14	2016	0.06052	0.79752	0.00000	0.79752	2.64860	0.02738	1.18221	0.06690	0.02700
L14	2017	0.21047	0.51572	0.00000	0.51572	2.47623	0.03270	1.83360	0.06940	0.03500
L14	2018	0.19707	0.28831	0.01084	0.29915	2.65963	0.03654	3.03594	0.07465	0.03500
L14	2019	0.17235	0.16893	0.00593	0.17486	2.66654	0.03082	5.33812	0.07359	0.02800
L14	2020	0.06867	0.31852	0.00323	0.32175	2.70857	0.02418	2.81175	0.02865	0.03200
L14	2021	0.31989	0.34562	0.00664	0.35226	3.06548	0.02344	2.73249	0.02554	0.01800
L14	2022	0.03350	0.26818	0.01877	0.28695	2.75345	0.04797	2.96874	0.08124	0.03200
L14	2023	0.03637	0.33802	0.01874	0.35676	2.82347	0.05422	2.40266	0.05046	0.03300
L14	2024	0.02755	0.30491	0.01143	0.31634	2.79449	0.05336	2.64199	0.07090	0.03600
LCG	2015	0.00670	0.54196	0.02444	0.56640	3.35459	0.04148	1.20328	0.06987	0.00600
LCG	2016	0.02741	0.56455	0.02383	0.58839	3.41717	0.04522	1.27124	0.06690	0.02700
LCG	2017	0.02393	0.65052	0.03306	0.68358	3.55197	0.06550	1.12974	0.06940	0.03500
LCG	2018	0.04322	0.59474	0.05712	0.65186	3.62630	0.08738	1.15919	0.07465	0.03500
LCG	2019	0.04106	0.61822	0.04538	0.66361	3.67382	0.07846	1.11474	0.07359	0.02800
LCG	2020	0.05083	0.56208	0.13684	0.69892	3.78764	0.20921	1.15480	0.02865	0.03200
LCG	2021	0.03115	0.44181	0.14669	0.58849	3.76377	0.21498	1.43015	0.02554	0.01800
LCG	2022	0.03849	0.45177	0.03653	0.48829	3.70311	0.05360	1.69686	0.08124	0.03200
LCG	2023	0.01699	0.54986	0.02563	0.57549	3.77664	0.06479	1.46597	0.05046	0.03300
LCG	2024	0.02001	0.54080	0.02878	0.56957	3.77959	0.07428	1.41520	0.07090	0.03600
PC1	2015	0.07526	0.39445	0.18563	0.58008	3.51368	0.02959	1.64748	0.06987	0.00600

PC1	2016	0.06725	0.30392	0.24069	0.54462	3.65606	0.20306	1.48603	0.06690	0.02700
PC1	2017	0.04125	0.29496	0.25218	0.54714	3.79336	0.44343	1.68769	0.06940	0.03500
PC1	2018	0.07449	0.26391	0.23901	0.50291	3.81934	0.38360	1.80935	0.07465	0.03500
PC1	2019	0.04527	0.30029	0.25537	0.55566	3.91988	0.29031	1.66314	0.07359	0.02800
PC1	2020	0.05075	0.34918	0.20597	0.55515	4.03026	0.35477	1.52658	0.02865	0.03200
PC1	2021	0.04090	0.31031	0.35359	0.66389	4.27154	0.54371	1.13636	0.02554	0.01800
PC1	2022	0.02468	0.27352	0.39676	0.67029	4.33755	0.45524	1.32991	0.08124	0.03200
PC1	2023	0.00900	0.23681	0.40891	0.64572	4.30482	0.53929	1.39889	0.05046	0.03300
PC1	2024	0.03383	0.24196	0.39040	0.63235	4.32196	0.48346	1.59231	0.07090	0.03600
VCG	2015	0.02526	0.47695	0.16954	0.64649	4.31659	0.16576	1.05424	0.06987	0.00600
VCG	2016	0.03119	0.46324	0.20933	0.67257	4.35880	0.14883	1.19720	0.06690	0.02700
VCG	2017	0.07532	0.48246	0.15423	0.63669	4.33504	0.12877	1.24462	0.06940	0.03500
VCG	2018	0.03170	0.45713	0.15088	0.60801	4.30288	0.13302	1.34725	0.07465	0.03500
VCG	2019	0.04072	0.39976	0.19967	0.59943	4.28597	0.12640	1.56100	0.07359	0.02800
VCG	2020	0.08620	0.45857	0.17615	0.63472	4.29248	0.13169	1.48983	0.02865	0.03200
VCG	2021	0.01679	0.49953	0.25417	0.75370	4.49093	0.08280	1.37659	0.02554	0.01800
VCG	2022	0.02909	0.39886	0.29080	0.68966	4.50514	0.08389	1.56500	0.08124	0.03200
VCG	2023	0.01292	0.46986	0.19649	0.66635	4.48706	0.12584	1.29528	0.05046	0.03300
VCG	2024	0.03765	0.43858	0.18384	0.62242	4.46895	0.11817	1.36993	0.07090	0.03600
VCS	2015	0.14547	0.52416	0.11682	0.64099	3.44428	0.21473	1.37928	0.06987	0.00600
VCS	2016	0.20252	0.51355	0.05291	0.56646	3.52348	0.16429	1.59206	0.06690	0.02700
VCS	2017	0.29445	0.37100	0.00012	0.37112	3.58090	0.12698	2.32864	0.06940	0.03500
VCS	2018	0.25510	0.37995	0.00008	0.38003	3.64388	0.12457	2.26748	0.07465	0.03500
VCS	2019	0.25254	0.33708	0.04529	0.38237	3.74693	0.18952	2.36765	0.07359	0.02800
VCS	2020	0.23589	0.32994	0.03296	0.36291	3.78214	0.16814	2.45281	0.02865	0.03200
VCS	2021	0.25708	0.27164	0.02122	0.29287	3.83840	0.15040	3.11314	0.02554	0.01800
VCS	2022	0.17431	0.24705	0.01413	0.26118	3.81888	0.14800	3.40349	0.08124	0.03200
VCS	2023	0.13085	0.22274	0.00645	0.22919	3.81079	0.13414	3.82853	0.05046	0.03300
VCS	2024	0.12481	0.20514	0.00475	0.20989	3.81070	0.12218	4.12579	0.07090	0.03600

VGC	2015	0.03485	0.40722	0.30966	0.71688	4.06682	0.26287	1.03870	0.06987	0.00600
VGC	2016	0.04779	0.34126	0.32253	0.66380	4.11359	0.24409	1.21000	0.06690	0.02700
VGC	2017	0.04503	0.26936	0.31155	0.58091	4.20505	0.23308	1.56953	0.06940	0.03500
VGC	2018	0.04036	0.27396	0.31040	0.58436	4.21836	0.24040	1.30946	0.07465	0.03500
VGC	2019	0.03818	0.31726	0.32799	0.64526	4.29859	0.20269	1.19827	0.07359	0.02800
VGC	2020	0.03129	0.34485	0.32574	0.67059	4.32885	0.17816	0.92693	0.02865	0.03200
VGC	2021	0.05815	0.38875	0.23135	0.62009	4.34236	0.24042	0.92869	0.02554	0.01800
VGC	2022	0.08332	0.36547	0.23881	0.60427	4.36095	0.23447	0.96630	0.08124	0.03200
VGC	2023	0.04823	0.34594	0.25886	0.60480	4.38202	0.22346	1.09207	0.05046	0.03300
VGC	2024	0.04783	0.35228	0.24683	0.59911	4.39493	0.24250	1.08210	0.07090	0.03600
DNP	2015	-0.00013	0.59424	0.03470	0.62894	2.85541	0.18858	1.15311	6.98715	0.00600
DNP	2016	0.00056	0.34735	0.43240	0.77974	3.40114	0.47557	1.02706	6.69001	0.02700
DNP	2017	0.00057	0.33525	0.41338	0.74863	3.58576	0.43886	1.09153	6.94019	0.03500
DNP	2018	0.00082	0.27359	0.46446	0.73805	3.82420	0.32116	1.29363	7.46501	0.03500
DNP	2019	0.00097	0.29252	0.43480	0.72732	3.88026	0.34463	1.07592	7.35926	0.02800
DNP	2020	0.00097	0.26453	0.45239	0.71692	3.98819	0.55085	1.21559	2.86541	0.03200
DNP	2021	0.00143	0.26081	0.46026	0.72107	4.14737	0.52960	1.29999	2.55373	0.01800
DNP	2022	0.00153	0.37376	0.35031	0.72407	4.21824	0.42196	1.23083	8.12351	0.03200
DNP	2023	0.00234	0.37936	0.33088	0.71023	4.20620	0.36440	1.36212	5.04643	0.03300
DNP	2024	0.00363	0.33056	0.32568	0.65625	4.24732	0.41195	1.51638	7.09000	0.03600
BMP	2015	0.00934	0.17336	0.00014	0.17350	3.38709	0.11531	4.46239	6.98715	0.00600
BMP	2016	-0.00797	0.20530	0.00006	0.20536	3.46106	0.20541	3.68291	6.69001	0.02700
BMP	2017	0.00928	0.14733	0.00000	0.14733	3.45822	0.32543	4.28971	6.94019	0.03500
BMP	2018	-0.00448	0.12750	0.00000	0.12750	3.44905	0.30480	5.03893	7.46501	0.03500
BMP	2019	0.00110	0.12546	0.00810	0.13356	3.45483	0.22462	4.20040	7.35926	0.02800
BMP	2020	-0.00205	0.17459	0.00777	0.18236	3.48040	0.16469	4.03403	2.86541	0.03200
BMP	2021	-0.00200	0.18418	0.00770	0.19188	3.45302	0.13460	3.84170	2.55373	0.01800
BMP	2022	-0.00004	0.13251	0.00657	0.13908	3.48356	0.12078	5.49020	8.12351	0.03200
BMP	2023	0.00172	0.16779	0.00587	0.17366	3.51255	0.08780	4.74942	5.04643	0.03300

BMP	2024	0.00014	0.15029	0.00556	0.15585	3.50522	0.08286	5.39733	7.09000	0.03600
VC2	2015	0.00002	0.70059	0.12252	0.82312	3.19434	0.02565	1.23937	6.98715	0.00600
VC2	2016	0.00001	0.88175	0.00314	0.88489	3.40470	0.01962	1.03042	6.69001	0.02700
VC2	2017	0.00003	0.79240	0.07231	0.86471	3.35406	0.02636	1.08887	6.94019	0.03500
VC2	2018	0.00133	0.86043	0.00665	0.86708	3.35841	0.02486	0.99156	7.46501	0.03500
VC2	2019	0.00000	0.82898	0.00908	0.83805	3.29353	0.05699	1.04544	7.35926	0.02800
VC2	2020	-0.00165	0.80388	0.00980	0.81367	3.22305	0.05494	1.07725	2.86541	0.03200
VC2	2021	0.00282	0.68044	0.02185	0.70228	3.34032	0.03545	1.13772	2.55373	0.01800
VC2	2022	0.00157	0.64120	0.06450	0.70570	3.36066	0.03124	1.15840	8.12351	0.03200
VC2	2023	-0.00046	0.63637	0.04223	0.67860	3.44306	0.02076	1.25315	5.04643	0.03300
VC2	2024	-0.00221	0.53572	0.13050	0.66623	3.44556	0.01773	1.44351	7.09000	0.03600
CC1	2015	0.00000	0.41423	0.37441	0.78864	4.02011	0.31248	1.07968	6.98715	0.00600
CC1	2016	0.00000	0.45499	0.35679	0.81177	3.97309	0.05195	1.20312	6.69001	0.02700
CC1	2017	0.00002	0.42153	0.38267	0.80420	3.92152	0.14422	1.42004	6.94019	0.03500
CC1	2018	0.00013	0.51913	0.28469	0.80382	3.97506	0.10287	1.24532	7.46501	0.03500
CC1	2019	-0.00007	0.58797	0.23643	0.82440	4.01867	0.11858	1.18033	7.35926	0.02800
CC1	2020	0.00000	0.57748	0.23553	0.81301	3.99501	0.11699	1.16204	2.86541	0.03200
CC1	2021	0.00082	0.48939	0.35184	0.84123	4.06697	0.03089	1.43191	2.55373	0.01800
CC1	2022	0.00033	0.39122	0.34170	0.73292	4.19265	0.02486	1.72053	8.12351	0.03200
CC1	2023	0.00000	0.49134	0.22456	0.71590	4.17465	0.01067	1.34871	5.04643	0.03300
CC1	2024	0.00004	0.50038	0.22676	0.72714	4.22344	0.01231	1.24036	7.09000	0.03600
HUB	2015	0.00067	0.35423	0.23451	0.58874	2.81294	0.16713	1.80391	6.98715	0.00600
HUB	2016	0.00098	0.35493	0.23439	0.58932	2.85974	0.14694	1.87131	6.69001	0.02700
HUB	2017	0.00053	0.34741	0.14636	0.49377	2.81754	0.12741	1.75527	6.94019	0.03500
HUB	2018	0.00008	0.34362	0.12675	0.47037	2.84948	0.11849	1.78694	7.46501	0.03500
HUB	2019	0.00029	0.31646	0.11467	0.43113	2.86438	0.10964	1.84619	7.35926	0.02800
HUB	2020	0.00007	0.29039	0.10736	0.39776	2.88027	0.12204	1.92962	2.86541	0.03200
HUB	2021	0.00216	0.29131	0.09975	0.39106	2.89866	0.10178	1.86563	2.55373	0.01800
HUB	2022	0.00085	0.23948	0.12440	0.36387	2.92770	0.08262	2.29484	8.12351	0.03200

HUB	2023	0.00088	0.21512	0.14994	0.36506	2.97047	0.06798	2.55977	5.04643	0.03300
HUB	2024	-0.00046	0.17830	0.18218	0.36047	2.98275	0.17906	3.06811	7.09000	0.03600
NNC	2015	0.01073	0.27119	0.01513	0.28632	2.60196	0.04377	3.15356	6.98715	0.00600
NNC	2016	-0.01261	0.16637	0.01050	0.17686	2.69412	0.02986	3.99328	6.69001	0.02700
NNC	2017	-0.00064	0.16759	0.00872	0.17631	2.75536	0.02892	4.31454	6.94019	0.03500
NNC	2018	0.00116	0.24160	0.01339	0.25499	2.65029	0.03697	2.66377	7.46501	0.03500
NNC	2019	0.00309	0.22806	0.01598	0.24404	2.58766	0.08617	2.39353	7.35926	0.02800
NNC	2020	-0.00253	0.24185	0.00787	0.24972	2.57934	0.08355	2.28921	2.86541	0.03200
NNC	2021	-0.00064	0.14020	0.00913	0.14933	2.54961	0.07859	3.84883	2.55373	0.01800
NNC	2022	-0.00055	0.12752	0.00912	0.13663	2.60944	0.08464	4.08614	8.12351	0.03200
NNC	2023	0.00004	0.12323	0.00905	0.13228	2.63491	0.11995	4.31438	5.04643	0.03300
NNC	2024	-0.00098	0.11865	0.01494	0.13359	2.68653	0.11371	4.10649	7.09000	0.03600
SJE	2015	0.00026	0.47881	0.21503	0.69384	3.13489	0.32664	1.14437	6.98715	0.00600
SJE	2016	0.00156	0.47809	0.23082	0.70891	3.16320	0.42450	1.11740	6.69001	0.02700
SJE	2017	-0.00090	0.37370	0.28826	0.66195	3.16614	0.43939	1.38445	6.94019	0.03500
SJE	2018	0.00795	0.38008	0.33391	0.71399	3.24523	0.43804	1.27309	7.46501	0.03500
SJE	2019	0.00346	0.30030	0.41885	0.71915	3.35574	0.64440	1.07477	7.35926	0.02800
SJE	2020	-0.00163	0.30336	0.40661	0.70997	3.33579	0.65438	1.02559	2.86541	0.03200
SJE	2021	-0.00171	0.30996	0.38003	0.68998	3.32808	0.63741	1.05371	2.55373	0.01800
SJE	2022	-0.00227	0.28518	0.35946	0.64464	3.28300	0.67362	1.00335	8.12351	0.03200
SJE	2023	-0.00189	0.30014	0.30813	0.60827	3.28283	0.64365	1.08163	5.04643	0.03300
SJE	2024	-0.00112	0.47712	0.18368	0.66080	3.42672	0.44466	0.96117	7.09000	0.03600
HTI	2015	0.00140	0.06445	0.64392	0.70837	3.10282	0.85678	0.94416	6.98715	0.00600
HTI	2016	0.00146	0.09501	0.60872	0.70373	3.12986	0.71870	0.26465	6.69001	0.02700
HTI	2017	0.00091	0.09078	0.64733	0.73811	3.22491	0.65677	1.04252	6.94019	0.03500
HTI	2018	0.00180	0.02283	0.71818	0.74101	3.24273	0.80789	2.99736	7.46501	0.03500
HTI	2019	0.00099	0.04995	0.68853	0.73847	3.24432	0.84707	1.26934	7.35926	0.02800
HTI	2020	-0.00022	0.03333	0.69317	0.72651	3.21969	0.86287	1.40847	2.86541	0.03200
HTI	2021	-0.00006	0.08839	0.61446	0.70285	3.19202	0.87689	0.23066	2.55373	0.01800

HTI	2022	0.00726	0.12523	0.57891	0.70414	3.17894	0.84352	0.25731	8.12351	0.03200
HTI	2023	0.00745	0.04849	0.60510	0.65360	3.14384	0.84437	2.07773	5.04643	0.03300
HTI	2024	0.01171	0.07641	0.58886	0.66527	3.17822	0.72010	2.94453	7.09000	0.03600
VHL	2015	0.00014	0.35522	0.20086	0.55608	3.02016	0.49075	1.14470	6.98715	0.00600
VHL	2016	-0.00010	0.36988	0.19682	0.56670	3.09161	0.49009	1.14505	6.69001	0.02700
VHL	2017	0.00009	0.33143	0.18391	0.51534	3.08674	0.49377	1.20583	6.94019	0.03500
VHL	2018	-0.00009	0.33169	0.14074	0.47243	3.09505	0.49395	1.20212	7.46501	0.03500
VHL	2019	-0.00021	0.33000	0.13286	0.46286	3.10815	0.46642	1.30755	7.35926	0.02800
VHL	2020	0.00121	0.39661	0.06309	0.45969	3.09587	0.42264	1.21155	2.86541	0.03200
VHL	2021	-0.00108	0.37251	0.03427	0.40678	3.04215	0.40461	1.32356	2.55373	0.01800
VHL	2022	-0.00105	0.35644	0.05101	0.40745	3.01134	0.36806	1.50193	8.12351	0.03200
VHL	2023	-0.00524	0.29739	0.05945	0.35684	2.92305	0.37677	1.78578	5.04643	0.03300
VHL	2024	-0.00604	0.27641	0.07627	0.35268	2.86269	0.36118	2.09632	7.09000	0.03600
THG	2015	-0.00048	0.60696	0.05278	0.65974	2.77975	0.17362	1.23364	6.98715	0.00600
THG	2016	0.00172	0.52214	0.08041	0.60255	2.84106	0.14391	1.20475	6.69001	0.02700
THG	2017	-0.00011	0.49062	0.13577	0.62638	2.93498	0.21608	1.20782	6.94019	0.03500
THG	2018	-0.00359	0.55744	0.08330	0.64074	2.98787	0.29967	1.10902	7.46501	0.03500
THG	2019	0.00053	0.53440	0.14423	0.67863	3.09412	0.26537	1.11392	7.35926	0.02800
THG	2020	0.00200	0.50339	0.15622	0.65961	3.16101	0.26388	1.12088	2.86541	0.03200
THG	2021	0.00003	0.50188	0.15705	0.65894	3.19191	0.23342	1.20700	2.55373	0.01800
THG	2022	-0.00080	0.48587	0.16698	0.65285	3.26684	0.18336	1.37792	8.12351	0.03200
THG	2023	-0.00059	0.53484	0.09346	0.62830	3.26769	0.16478	1.32536	5.04643	0.03300
THG	2024	0.00075	0.55686	0.10793	0.66479	3.33880	0.14993	1.27950	7.09000	0.03600
VTV	2015	0.00020	0.65100	0.02638	0.67737	3.10952	0.00163	1.18792	6.98715	0.00600
VTV	2016	0.00037	0.70754	0.03037	0.73791	3.21655	0.00202	1.05685	6.69001	0.02700
VTV	2017	0.00027	0.67713	0.03251	0.70964	3.22099	0.01708	0.97781	6.94019	0.03500
VTV	2018	-0.00092	0.66162	0.03164	0.69326	3.15571	0.02007	1.21617	7.46501	0.03500
VTV	2019	0.00000	0.58886	0.04511	0.63397	3.05440	0.02407	1.28037	7.35926	0.02800
VTV	2020	-0.00001	0.52095	0.05225	0.57320	2.97369	0.02594	1.37065	2.86541	0.03200

VTV	2021	0.00002	0.51106	0.05188	0.56294	2.97617	0.02266	1.44080	2.55373	0.01800
VTV	2022	-0.00001	0.67651	0.03269	0.70919	3.16585	0.01250	1.23905	8.12351	0.03200
VTV	2023	0.00000	0.56944	0.04418	0.61362	3.02769	0.01471	1.38481	5.04643	0.03300
VTV	2024	0.00000	0.53684	0.04793	0.58477	2.97953	0.01426	1.44429	7.09000	0.03600
SD9	2015	-0.00100	0.38436	0.18470	0.56906	3.26895	0.46037	1.34389	6.98715	0.00600
SD9	2016	-0.00014	0.39404	0.15391	0.54795	3.25944	0.41602	1.34710	6.69001	0.02700
SD9	2017	0.00008	0.38678	0.15641	0.54319	3.26941	0.36459	1.41982	6.94019	0.03500
SD9	2018	-0.00032	0.47507	0.13604	0.61111	3.33683	0.28088	1.09845	7.46501	0.03500
SD9	2019	0.00062	0.48938	0.16730	0.65668	3.37873	0.23231	0.98640	7.35926	0.02800
SD9	2020	-0.00011	0.48675	0.18060	0.66736	3.39121	0.51054	0.95360	2.86541	0.03200
SD9	2021	-0.00011	0.46082	0.18469	0.64551	3.36671	0.49498	1.07987	2.55373	0.01800
SD9	2022	-0.00012	0.35458	0.24894	0.60352	3.32403	0.50285	1.37401	8.12351	0.03200
SD9	2023	-0.00014	0.26299	0.29130	0.55429	3.27301	0.52612	1.74576	5.04643	0.03300
SD9	2024	-0.00015	0.24997	0.25887	0.50884	3.23703	0.54081	1.75496	7.09000	0.03600
C47	2015	0.00019	0.60074	0.28496	0.88570	3.35902	0.30662	1.08284	6.98715	0.00600
C47	2016	-0.00070	0.57631	0.29370	0.87000	3.35384	0.38447	1.00753	6.69001	0.02700
C47	2017	-0.00002	0.55227	0.31256	0.86483	3.34497	0.38258	1.06109	6.94019	0.03500
C47	2018	-0.00046	0.58398	0.26672	0.85070	3.33532	0.33167	1.03780	7.46501	0.03500
C47	2019	-0.00063	0.56989	0.27976	0.84965	3.28334	0.31758	1.16310	7.35926	0.02800
C47	2020	-0.00089	0.61247	0.19587	0.80834	3.20983	0.28331	1.09156	2.86541	0.03200
C47	2021	-0.00034	0.51614	0.26438	0.78052	3.19652	0.31591	1.28220	2.55373	0.01800
C47	2022	0.00029	0.48097	0.28981	0.77078	3.26577	0.28304	1.43156	8.12351	0.03200
C47	2023	0.00004	0.57458	0.18352	0.75810	3.25719	0.27772	1.21857	5.04643	0.03300
C47	2024	-0.00004	0.53195	0.23326	0.76521	3.27259	0.25095	1.36331	7.09000	0.03600
CSC	2015	0.00044	0.44972	0.28035	0.73007	2.65761	0.09774	1.55623	6.98715	0.00600
CSC	2016	0.00024	0.56092	0.14246	0.70338	2.65221	0.09382	1.39386	6.69001	0.02700
CSC	2017	0.00010	0.45807	0.12843	0.58650	2.76566	0.08183	1.82227	6.94019	0.03500
CSC	2018	-0.00149	0.48036	0.05219	0.53255	2.87563	0.06597	1.79168	7.46501	0.03500
CSC	2019	0.00126	0.57177	0.09954	0.67130	2.98855	0.05161	1.18271	7.35926	0.02800

CSC	2020	0.00133	0.45794	0.21006	0.66800	3.06867	0.04120	2.04239	2.86541	0.03200
CSC	2021	0.00289	0.65403	0.16612	0.82015	3.37211	0.01929	1.47699	2.55373	0.01800
CSC	2022	-0.00105	0.60569	0.06084	0.66653	3.37896	0.01805	1.60489	8.12351	0.03200
CSC	2023	0.00030	0.59218	0.06426	0.65645	3.38666	0.01614	1.64222	5.04643	0.03300
CSC	2024	0.00042	0.49818	0.13514	0.63332	3.36352	0.01452	1.95451	7.09000	0.03600
L18	2015	-0.00006	0.80598	0.04993	0.85591	3.14953	0.09989	1.03237	6.98715	0.00600
L18	2016	-0.00351	0.78338	0.06642	0.84980	3.16324	0.09990	1.06731	6.69001	0.02700
L18	2017	0.00000	0.81905	0.03733	0.85638	3.21976	0.09588	1.01491	6.94019	0.03500
L18	2018	0.00000	0.75407	0.10219	0.85626	3.29738	0.10597	0.99258	7.46501	0.03500
L18	2019	0.00000	0.71509	0.10791	0.82300	3.34628	0.21964	1.02977	7.35926	0.02800
L18	2020	0.00020	0.72406	0.12616	0.85021	3.42098	0.18070	1.08100	2.86541	0.03200
L18	2021	0.00000	0.62827	0.18426	0.81253	3.49815	0.11990	1.34836	2.55373	0.01800
L18	2022	-0.00002	0.59343	0.23214	0.82558	3.54984	0.09645	1.46056	8.12351	0.03200
L18	2023	0.00003	0.75086	0.14004	0.89091	3.74170	0.06165	1.21871	5.04643	0.03300
L18	2024	0.00077	0.65849	0.18664	0.84514	3.70449	0.06765	1.36119	7.09000	0.03600
LBM	2015	-0.00100	0.18472	0.00000	0.18472	2.27049	0.47903	2.71986	6.98715	0.00600
LBM	2016	0.00039	0.23796	0.00394	0.24190	2.39848	0.40463	2.33480	6.69001	0.02700
LBM	2017	0.00008	0.33569	0.00282	0.33851	2.54372	0.38327	1.61270	6.94019	0.03500
LBM	2018	0.00078	0.23344	0.00289	0.23633	2.53300	0.47981	1.81269	7.46501	0.03500
LBM	2019	0.00071	0.14694	0.00247	0.14941	2.60152	0.55371	2.58731	7.35926	0.02800
LBM	2020	-0.00019	0.15884	0.00219	0.16103	2.65406	0.50867	2.73810	2.86541	0.03200
LBM	2021	0.00242	0.19944	0.00181	0.20125	2.73566	0.55177	2.00595	2.55373	0.01800
LBM	2022	0.00074	0.25037	0.00138	0.25176	2.85309	0.50871	1.76439	8.12351	0.03200
LBM	2023	-0.00107	0.16798	0.05194	0.21992	2.88642	0.39973	3.11703	5.04643	0.03300
LBM	2024	-0.00206	0.21249	0.03438	0.24688	2.91060	0.41167	1.57985	7.09000	0.03600
TTL	2015	0.00000	0.63444	0.07692	0.71136	3.29492	0.03117	1.34950	6.98715	0.00600
TTL	2016	0.00102	0.62128	0.05009	0.67137	3.25040	0.02014	1.35786	6.69001	0.02700
TTL	2017	-0.00049	0.54630	0.03392	0.58022	3.12605	0.02722	1.47294	6.94019	0.03500
TTL	2018	-0.00018	0.55079	0.03495	0.58573	3.10689	0.03356	1.46561	7.46501	0.03500

TTL	2019	-0.00029	0.56212	0.03567	0.59779	3.13176	0.02774	1.42860	7.35926	0.02800
TTL	2020	-0.00027	0.61324	0.03474	0.64799	3.19097	0.02710	1.30157	2.86541	0.03200
TTL	2021	-0.00005	0.61661	0.10690	0.72351	3.34203	0.19488	1.20170	2.55373	0.01800
TTL	2022	0.00000	0.68821	0.04568	0.73389	3.35729	0.17304	1.14288	8.12351	0.03200
TTL	2023	0.00000	0.74718	0.02746	0.77464	3.44995	0.12011	1.14986	5.04643	0.03300
TTL	2024	0.00000	0.76608	0.00778	0.77386	3.43831	0.09399	1.15395	7.09000	0.03600
CDC	2015	0.00003	0.48159	0.19300	0.67460	2.90649	0.01225	1.89812	6.98715	0.00600
CDC	2016	-0.00005	0.52111	0.09156	0.61267	2.86614	0.01963	1.73623	6.69001	0.02700
CDC	2017	0.00004	0.53467	0.17449	0.70915	2.97415	0.01527	1.72899	6.94019	0.03500
CDC	2018	0.00003	0.53744	0.19058	0.72803	3.01086	0.01235	1.73949	7.46501	0.03500
CDC	2019	-0.00001	0.49757	0.16647	0.66404	2.92989	0.01336	1.86262	7.35926	0.02800
CDC	2020	-0.00195	0.59254	0.10486	0.69741	2.91783	0.01166	1.51547	2.86541	0.03200
CDC	2021	-0.00011	0.70019	0.05811	0.75830	3.10958	0.09428	1.20246	2.55373	0.01800
CDC	2022	0.00102	0.61798	0.18734	0.80532	3.21760	0.07284	1.42204	8.12351	0.03200
CDC	2023	-0.00037	0.67355	0.13911	0.81266	3.25669	0.06564	1.32626	5.04643	0.03300
CDC	2024	0.00892	0.58654	0.18589	0.77243	3.30234	0.00613	1.36552	7.09000	0.03600
CRC	2015	0.00000	0.19971	0.04208	0.24179	2.34883	0.06194	1.51040	6.98715	0.00600
CRC	2016	0.00053	0.19592	0.11179	0.30771	2.41946	0.14491	1.69657	6.69001	0.02700
CRC	2017	0.00009	0.38959	0.06867	0.45826	2.55920	0.15103	1.61966	6.94019	0.03500
CRC	2018	-0.00011	0.43417	0.01204	0.44621	2.54116	0.05674	1.58229	7.46501	0.03500
CRC	2019	0.00006	0.56417	0.00708	0.57125	2.60714	0.05713	1.35904	7.35926	0.02800
CRC	2020	-0.00002	0.61836	0.09586	0.71421	2.79635	0.05512	1.20917	2.86541	0.03200
CRC	2021	0.00002	0.24673	0.20901	0.45573	2.80763	0.43533	1.65609	2.55373	0.01800
CRC	2022	-0.00004	0.26434	0.15748	0.42182	2.81022	0.42461	1.62759	8.12351	0.03200
CRC	2023	-0.00003	0.26228	0.11674	0.37902	2.81156	0.40981	1.60257	5.04643	0.03300
CRC	2024	0.00000	0.21222	0.11245	0.32467	3.08455	0.33475	2.31600	7.09000	0.03600
DPG	2015	-0.00140	0.48123	0.27093	0.75216	3.30352	0.34687	1.13665	6.98715	0.00600
DPG	2016	0.00013	0.41181	0.35363	0.76544	3.39393	0.27199	1.23020	6.69001	0.02700
DPG	2017	0.00149	0.38900	0.38789	0.77688	3.59625	0.15529	1.11584	6.94019	0.03500

DPG	2018	0.00369	0.35070	0.42634	0.77705	3.66719	0.51675	1.00255	7.46501	0.03500
DPG	2019	0.00312	0.37670	0.38007	0.75677	3.70623	0.45076	1.14843	7.35926	0.02800
DPG	2020	-0.00133	0.38645	0.32181	0.70826	3.68309	0.46492	0.93341	2.86541	0.03200
DPG	2021	0.00016	0.40875	0.29181	0.70057	3.77457	0.40795	1.08074	2.55373	0.01800
DPG	2022	0.00038	0.36145	0.27908	0.64053	3.78810	0.39446	1.48400	8.12351	0.03200
DPG	2023	0.00036	0.43049	0.21823	0.64872	3.82539	0.34734	1.30845	5.04643	0.03300
DPG	2024	0.00004	0.38660	0.20637	0.59297	3.80744	0.34268	1.42151	7.09000	0.03600
DC4	2015	0.00000	0.62432	0.02539	0.64971	2.37057	0.04019	1.52404	6.98715	0.00600
DC4	2016	0.00000	0.67746	0.02158	0.69904	2.44111	0.03637	1.39865	6.69001	0.02700
DC4	2017	0.00035	0.66770	0.07522	0.74292	2.53774	0.04067	1.41751	6.94019	0.03500
DC4	2018	0.00124	0.53928	0.04697	0.58625	2.54280	0.05950	1.66888	7.46501	0.03500
DC4	2019	0.00005	0.46333	0.02433	0.48765	2.81902	0.06282	1.74831	7.35926	0.02800
DC4	2020	-0.00199	0.39355	0.01273	0.40628	2.79402	0.05893	2.08875	2.86541	0.03200
DC4	2021	0.00005	0.37475	0.01206	0.38681	2.99198	0.03149	2.40636	2.55373	0.01800
DC4	2022	0.00090	0.48783	0.00694	0.49477	3.07100	0.02180	1.49396	8.12351	0.03200
DC4	2023	0.00086	0.57405	0.00754	0.58159	3.22476	0.01863	1.35871	5.04643	0.03300
DC4	2024	0.00233	0.50405	0.16277	0.66683	3.38382	0.01016	1.66286	7.09000	0.03600
VNE	2015	0.00363	0.45109	0.01658	0.46767	3.27262	0.09645	1.28598	6.98715	0.00600
VNE	2016	0.00624	0.34971	0.01188	0.36159	3.18522	0.11188	1.70081	6.69001	0.02700
VNE	2017	0.00028	0.38861	0.01027	0.39888	3.19271	0.10545	1.37433	6.94019	0.03500
VNE	2018	0.00750	0.35226	0.00122	0.35349	3.17757	0.05922	1.39460	7.46501	0.03500
VNE	2019	-0.00076	0.31245	0.00265	0.31510	3.15573	0.06309	1.60728	7.35926	0.02800
VNE	2020	0.00198	0.52076	0.00434	0.52510	3.32076	0.04790	1.08073	2.86541	0.03200
VNE	2021	0.00297	0.62802	0.07807	0.70609	3.54199	0.09239	0.71480	2.55373	0.01800
VNE	2022	-0.00022	0.54935	0.19275	0.74210	3.60321	0.22190	1.00660	8.12351	0.03200
VNE	2023	0.00057	0.53304	0.19885	0.73189	3.57454	0.22072	0.98359	5.04643	0.03300
VNE	2024	-0.00182	0.58728	0.18270	0.76999	3.50250	0.23947	0.83682	7.09000	0.03600
S99	2015	-0.00343	0.29680	0.13071	0.42751	2.93101	0.11313	2.52771	6.98715	0.00600
S99	2016	0.00188	0.34903	0.13790	0.48693	2.98886	0.15358	1.79171	6.69001	0.02700

S99	2017	0.00535	0.38901	0.21677	0.60578	3.19989	0.10632	1.21168	6.94019	0.03500
S99	2018	-0.00302	0.40759	0.26305	0.67064	3.32171	0.08114	1.10239	7.46501	0.03500
S99	2019	0.00080	0.34846	0.31565	0.66411	3.32916	0.42522	1.39792	7.35926	0.02800
S99	2020	-0.00228	0.51785	0.21720	0.73505	3.60344	0.23530	1.28913	2.86541	0.03200
S99	2021	-0.00291	0.42576	0.27492	0.70068	3.53660	0.24392	1.31908	2.55373	0.01800
S99	2022	0.00258	0.27805	0.33461	0.61266	3.56950	0.20301	1.51142	8.12351	0.03200
S99	2023	0.00264	0.31523	0.30197	0.61719	3.60474	0.48868	1.43891	5.04643	0.03300
S99	2024	-0.00272	0.28180	0.33639	0.61819	3.64360	0.57169	1.31715	7.09000	0.03600
CTR	2015	0.00082	0.65800	0.00000	0.65800	3.16384	0.02203	1.47785	6.98715	0.00600
CTR	2016	-0.00021	0.50150	0.00000	0.50150	3.11228	0.02207	1.94452	6.69001	0.02700
CTR	2017	-0.00037	0.61793	0.00000	0.61793	3.26286	0.01527	1.59098	6.94019	0.03500
CTR	2018	0.00005	0.66572	0.00000	0.66572	3.36565	0.01866	1.46182	7.46501	0.03500
CTR	2019	0.00050	0.64518	0.00000	0.64518	3.38804	0.11251	1.35141	7.35926	0.02800
CTR	2020	-0.00024	0.72199	0.00452	0.72652	3.58873	0.10204	1.09293	2.86541	0.03200
CTR	2021	0.00024	0.62784	0.04120	0.66905	3.59807	0.11641	1.21074	2.55373	0.01800
CTR	2022	-0.00028	0.68417	0.04686	0.73103	3.78111	0.06172	1.20827	8.12351	0.03200
CTR	2023	0.00010	0.65581	0.05674	0.71255	3.83984	0.04115	1.23776	5.04643	0.03300
CTR	2024	0.00021	0.65477	0.08278	0.73755	3.85164	0.02684	1.16516	7.09000	0.03600

Source: Compilation by the author

Appendix 3: Regression model results using Python

Import data from Excel file

```
# Import data from file Excel
from google.colab import files
import pandas as pd # Import pandas
uploaded = files.upload()
df = pd.read_excel("Python Trial.xlsx")
# EDA - shape, head, info, missing values
df_shape = df.shape
df_head = df.head(10)
df_info = df.dtypes
missing_values = df.isnull().sum()

df shape, df head, df info, missing values
```

Result of the process

Saving Python Trial.xlsx to Python Trial (2).xlsx

((420, 11),

	Name	Year	ROA	STDR	LTDR	TDR	SIZE	TANG
0	C4G	2015	0.011016	0.524858	0.355481	0.880339	3.846958	0.315030
1	C4G	2016	0.023485	0.449727	0.385768	0.835495	3.861813	0.365044
2	C4G	2017	0.018093	0.448455	0.394131	0.842586	3.869092	0.410867
3	C4G	2018	0.017371	0.470303	0.386995	0.857298	3.895182	0.352666
4	C4G	2019	0.013038	0.427952	0.400615	0.828567	3.849670	0.373842
5	C4G	2020	0.008698	0.441597	0.401780	0.843377	3.869659	0.345992
6	C4G	2021	0.007907	0.446668	0.389360	0.836027	3.892644	0.317920
7	C4G	2022	0.018712	0.341168	0.357377	0.698545	3.917564	0.284700
8	C4G	2023	0.013362	0.301358	0.308980	0.610338	3.982514	0.237228
9	C4G	2024	0.018991	0.313658	0.269715	0.583373	3.973553	0.232621

	LIQ	GDP	INF
0	1.018288	0.069872	0.006
1	1.007831	0.066900	0.027
2	1.018122	0.069402	0.035
3	0.997863	0.074650	0.035
4	1.041275	0.073593	0.028
5	1.087062	0.028654	0.032
6	1.119497	0.025537	0.018
7	1.514146	0.081235	0.032
8	1.910451	0.050464	0.033
9	1.810747	0.070900	0.036

Source: Implementation by the author using Python

Validate data information and review missing data

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
df_sub=df[['ROA', 'STDR', 'LTDR', 'TDR', 'SIZE',
           'TANG', 'GDP', 'LIQ', 'INF']]
df_sub.info()
```

Result of the process

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 420 entries, 0 to 419
Data columns (total 9 columns):
 #   Column  Non-Null Count  Dtype
---  -
 0    ROA      420 non-null    float64
 1   STDR      420 non-null    float64
 2   LTDR      420 non-null    float64
 3    TDR      420 non-null    float64
 4   SIZE      420 non-null    float64
 5   TANG      420 non-null    float64
 6    GDP      420 non-null    float64
 7   LIQ      420 non-null    float64
 8    INF      420 non-null    float64
dtypes: float64(9)
memory usage: 29.7 KB
```

Source: Implementation by the author using Python

Description of data

```
df.describe().style.background_gradient(cmap='Reds')
```

Result of the process

	Year	ROA	STDR	LTDR	TDR	SIZE	TANG	LIQ	GDP	INF
count	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000
mean	2019.500000	0.024899	0.426365	0.169932	0.596297	3.455923	0.224993	1.588479	3.518807	0.028200
std	2.875707	0.051061	0.195422	0.174439	0.192936	0.597225	0.205465	0.839762	3.311353	0.008975
min	2015.000000	-0.142503	0.022833	0.000000	0.127497	1.871898	0.001631	0.230661	0.025537	0.006000
25%	2017.000000	0.000000	0.280017	0.023033	0.485365	3.067873	0.059148	1.134707	0.069872	0.027000
50%	2019.500000	0.001447	0.403065	0.123460	0.645383	3.362090	0.149376	1.350456	2.865413	0.032000
75%	2022.000000	0.031219	0.561207	0.271925	0.731246	3.850163	0.357358	1.697844	6.987154	0.035000
max	2024.000000	0.319889	0.898583	0.718176	0.993877	5.351196	0.876886	5.490199	8.123514	0.036000

Source: Implementation by the author using Python

Outlier processing using the IQR method

```

## 2.4. Outlier processing using the IQR method

def detect_outliers_iqr(series, threshold=1.5):
    Q1 = series.quantile(0)
    Q3 = series.quantile(1)
    IQR = Q3 - Q1
    lower_bound = Q1 - threshold * IQR
    upper_bound = Q3 + threshold * IQR
    return (series < lower_bound) | (series > upper_bound)

def remove_outliers_iqr(df, columns, threshold=1.5):
    df_clean = df.copy()
    for col in columns:
        Q1 = df_clean[col].quantile(0)
        Q3 = df_clean[col].quantile(1)
        IQR = Q3 - Q1
        lower_bound = Q1 - threshold * IQR
        upper_bound = Q3 + threshold * IQR
        df_clean = df_clean[(df_clean[col] >= lower_bound) &
(df_clean[col] <= upper_bound)]
    return df_clean

numeric_cols = ['ROA', 'STDR', 'LTDR', 'TDR', 'SIZE', 'TANG', 'GDP',
'LIQ', 'INF']
df_clean = remove_outliers_iqr(df_imputed, numeric_cols)

print("\nNumber of observations after removing outliers:",
len(df_clean))

```

Result of the process

```
nNumber of observations after removing outliers: 420
```

Source: Implementation by the author using Python

Data validation

```
## 2.5. Data validation
scaler = MinMaxScaler()
df_scaled = df_clean.copy()
df_scaled['SIZE'] = np.log(df_scaled['SIZE'] + 1) # Log-transform SIZE
df_scaled[numeric_cols] = scaler.fit_transform(df_scaled[numeric_cols])

print("\nStandardized data (first 5 lines):")
print(df_scaled.head())
```

Result of the process

	Name	Year	ROA	STDR	LTDR	TDR	SIZE	TANG
0	C4G	2015	0.332011	0.573251	0.494978	0.868952	0.659440	0.358066
1	C4G	2016	0.358978	0.487461	0.537150	0.817192	0.663296	0.415209
2	C4G	2017	0.347317	0.486008	0.548794	0.825376	0.665181	0.467562
3	C4G	2018	0.345754	0.510956	0.538858	0.842357	0.671914	0.401066
4	C4G	2019	0.336385	0.462596	0.557822	0.809194	0.660145	0.425260

	LIQ	GDP	INF
0	0.149752	0.005475	0.000000
1	0.147764	0.005108	0.700000
2	0.149721	0.005417	0.966667
3	0.145869	0.006065	0.966667
4	0.154123	0.005934	0.733333

Source: Implementation by the author using Python

Data validation using Robust Scaler

```
from sklearn.preprocessing import RobustScaler
scaler = RobustScaler()

# Exclude non-numeric columns 'Name' and 'Year' from scaling
numeric_cols_for_scaling = [col for col in df.columns if col not in
['Name', 'Year']]

df_scaled =
pd.DataFrame(scaler.fit_transform(df[numeric_cols_for_scaling]),
columns=numeric_cols_for_scaling)

print("► Before:")
display(df.describe().style.background_gradient(cmap='OrRd'))

print("► After (RobustScaler):")
display(df_scaled.describe().style.background_gradient(cmap='BuGn'))
```

Before processing:

	Year	ROA	STDR	LTDR	TDR	SIZE	TANG	LIQ	GDP	INF
count	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000
mean	2019.500000	0.024899	0.426365	0.169932	0.596297	3.455923	0.224993	1.588479	3.518807	0.028200
std	2.875707	0.051061	0.195422	0.174439	0.192936	0.597225	0.205465	0.839762	3.311353	0.008975
min	2015.000000	-0.142503	0.022833	0.000000	0.127497	1.871898	0.001631	0.230661	0.025537	0.006000
25%	2017.000000	0.000000	0.280017	0.023033	0.485365	3.067873	0.059148	1.134707	0.069872	0.027000
50%	2019.500000	0.001447	0.403065	0.123460	0.645383	3.362090	0.149376	1.350456	2.865413	0.032000
75%	2022.000000	0.031219	0.561207	0.271925	0.731246	3.850163	0.357358	1.697844	6.987154	0.035000
max	2024.000000	0.319889	0.898583	0.718176	0.993877	5.351196	0.876886	5.490199	8.123514	0.036000

After processing:

	ROA	STDR	LTDR	TDR	SIZE	TANG	LIQ	GDP	INF
count	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000	420.000000
mean	0.751212	0.082864	0.186712	-0.199634	0.119947	0.253570	0.422673	0.094458	-0.475000
std	1.635582	0.694983	0.700863	0.784675	0.763431	0.688993	1.491221	0.478707	1.121883
min	-4.611008	-1.352224	-0.496040	-2.106256	-1.904909	-0.495439	-1.988492	-0.410548	-3.250000
25%	-0.046357	-0.437596	-0.403498	-0.650796	-0.376097	-0.302565	-0.383120	-0.404139	-0.625000
50%	0.000000	-0.000000	0.000000	-0.000000	0.000000	0.000000	-0.000000	0.000000	0.000000
75%	0.953643	0.562404	0.596502	0.349204	0.623903	0.697435	0.616880	0.595861	0.375000
max	10.200306	1.762218	2.389452	1.417330	2.542672	2.439585	7.351213	0.760140	0.500000

Source: Implementation by the author using Python

Distribution statistics of quantitative variables (Skewness và Kurtosis)

```

from scipy.stats import skew, kurtosis
import numpy as np
import pandas as pd

def interpret_skew(val):
    if val > 1: return "Lệch phải mạnh"
    elif val > 0.5: return "Lệch phải nhẹ"
    elif val < -1: return "Lệch trái mạnh"
    elif val < -0.5: return "Lệch trái nhẹ"
    else: return "Xấp xỉ chuẩn"

def interpret_kurt(val):
    if val > 3: return "Nhọn (leptokurtic)"
    elif val < 3: return "Dẹt (platykurtic)"
    else: return "Chuẩn (mesokurtic)"

distribution_stats = []

# Exclude non-numeric columns
numeric_cols = df.select_dtypes(include=np.number).columns.tolist()

for col in numeric_cols:
    s = skew(df[col].dropna())
    k = kurtosis(df[col].dropna(), fisher=False)
    distribution_stats.append({
        'Tên biến': col,
        'Skew': round(s, 3),
        'Kiểu phân phối (Skew)': interpret_skew(s),
        'Kurtosis': round(k, 3),
        'Kiểu phân phối (Kurtosis)': interpret_kurt(k)
    })

df_dist = pd.DataFrame(distribution_stats)
print(df_dist)

```

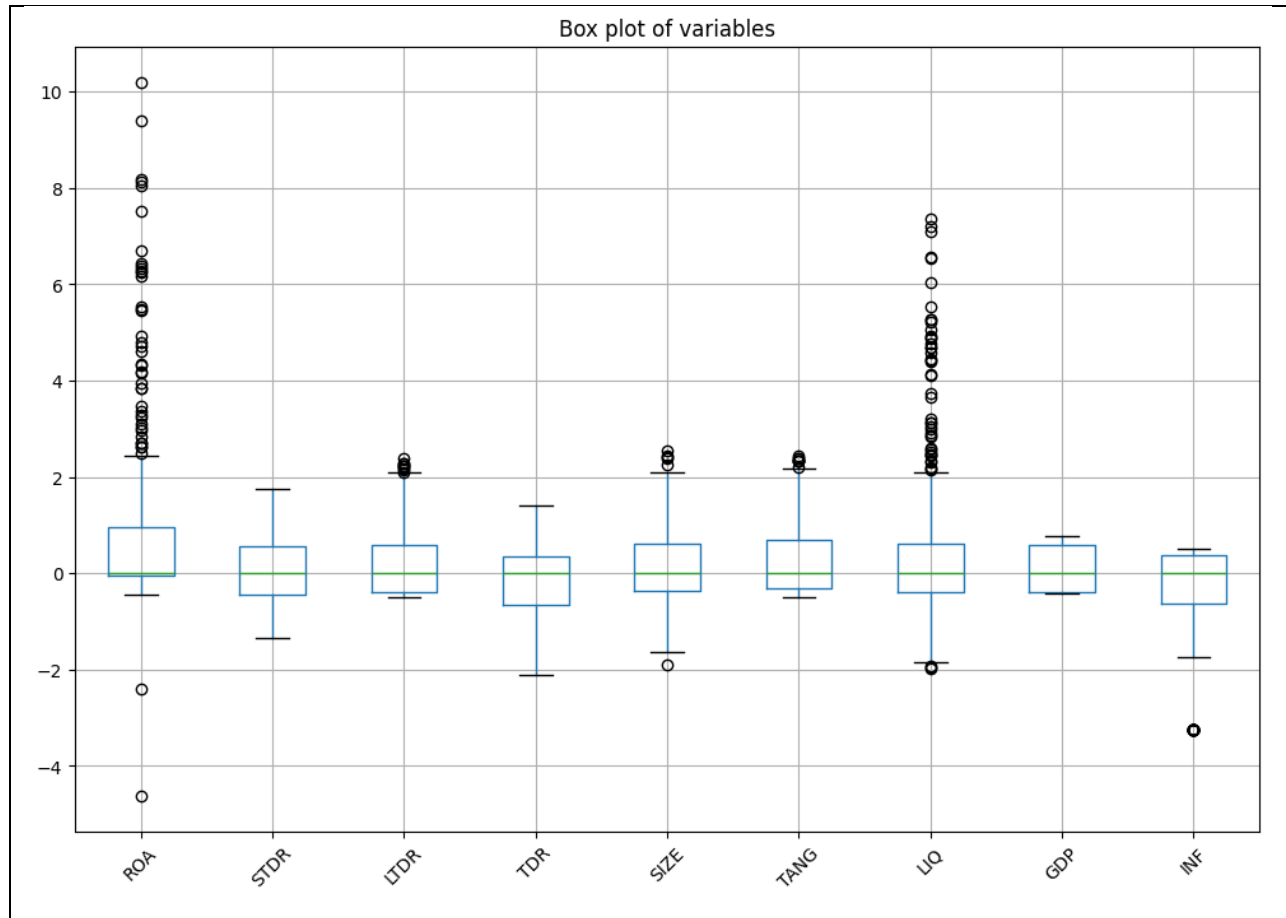
	Tên biến	Skew	Kiểu phân phối (Skew)	Kurtosis	Kiểu phân phối (Kurtosis)
0	Year	0.000	Xấp xỉ chuẩn	1.776	Dẹt (platykurtic)
1	ROA	2.714	Lệch phải mạnh	12.134	Nhọn (leptokurtic)
2	STDR	0.257	Xấp xỉ chuẩn	2.392	Dẹt (platykurtic)
3	LTDR	1.123	Lệch phải mạnh	3.586	Nhọn (leptokurtic)
4	TDR	-0.723	Lệch trái nhẹ	2.740	Dẹt (platykurtic)
5	SIZE	0.387	Xấp xỉ chuẩn	2.970	Dẹt (platykurtic)
6	TANG	1.115	Lệch phải mạnh	3.561	Nhọn (leptokurtic)
7	LIQ	2.238	Lệch phải mạnh	8.629	Nhọn (leptokurtic)
8	GDP	0.106	Xấp xỉ chuẩn	1.210	Dẹt (platykurtic)
9	INF	-1.483	Lệch trái mạnh	4.067	Nhọn (leptokurtic)

Source: Implementation by the author using Python

Boxplot of variable

```
# plt.figure(figsize=(12, 8))
# df_scaled[numeric_cols_for_scaling].boxplot()
# plt.title('Biểu đồ hộp của các biến')
# plt.xticks(rotation=45)
# plt.show()

# Box plot
plt.figure(figsize=(12, 8))
df_scaled[numeric_cols_for_scaling].boxplot()
plt.title('Box plot of variables')
plt.xticks(rotation=45)
plt.show()
```



Source: Implementation by the author using Python

Data validation and log-transform

```
# Data validation and log-transform

# Make a copy for processing
df_processed = df.copy()

# Applying log-transform cho biến SIZE (log(total_assets))
df_processed['SIZE'] = np.log(df_processed['SIZE'])

# Determind variables (trừ Name và Year)
features_to_scale = df_processed.columns.difference(['Name', 'Year'])
scaler = StandardScaler()
df_processed[features_to_scale] =
scaler.fit_transform(df_processed[features_to_scale])

# Print out data after log-transform
df_processed.head()
```

Result of the process

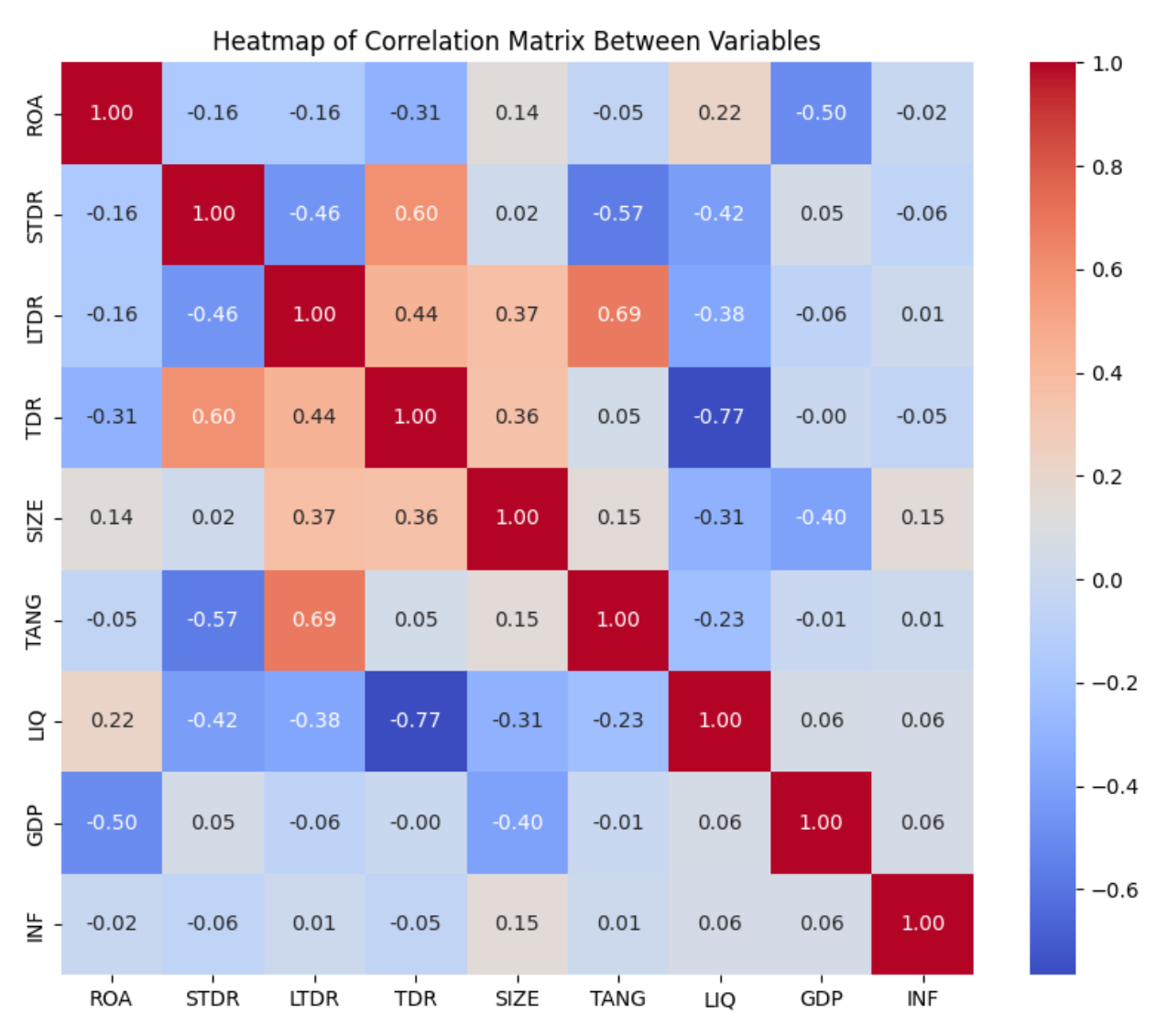
	Name	Year	ROA	STDR	LTDR	TDR	SIZE	TANG	LIQ	GDP	INF
0	C4G	2015	-0.272216	0.504598	1.064959	1.473962	0.706374	0.438732	-0.679801	-1.042791	-2.476470
1	C4G	2016	-0.027721	0.119689	1.238790	1.241258	0.728680	0.682443	-0.692268	-1.043689	-0.133863
2	C4G	2017	-0.133445	0.113169	1.286789	1.278052	0.739579	0.905728	-0.679998	-1.042933	0.758558
3	C4G	2018	-0.147616	0.225102	1.245831	1.354396	0.778476	0.622128	-0.704153	-1.041346	0.758558
4	C4G	2019	-0.232563	0.008127	1.324003	1.205303	0.710452	0.725312	-0.652395	-1.041666	-0.022311

Source: Implementation by the author using Python

Plot correlation heatmap

```
# Correlation matrix
# Calculate the correlation matrix (excluding 'Name' and 'Year' columns)
corr_matrix = df_processed.drop(columns=['Name', 'Year']).corr()
# Plot correlation heatmap
plt.figure(figsize=(10, 8))
sns.heatmap(corr_matrix, annot=True, fmt=".2f", cmap="coolwarm",
            square=True)
plt.title("Heatmap of Correlation Matrix Between Variables")
plt.show()
```

Result of the process



Source: Implementation by the author using Python

Find train-test set

```

X = df_processed.drop(columns=["ROA", "Name", "Year"])
y = df_processed["ROA"]
# Split train/test 80% / 20%
# X_train, X_test, y_train, y_test = train_test_split(
#     X, y, test_size=0.2, random_state=42
# )
# # Checking test size
# print("Train:", X_train.shape, y_train.shape)
# print("Test :", X_test.shape, y_test.shape)

# Repeat the process until find out the most suitable for the model
n_splits = 10 # number of split round
best_r2 = -np.inf
best_split = None
best_X_train, best_X_test, best_y_train, best_y_test = None, None, None,
None

kf = KFold(n_splits=n_splits, shuffle=True, random_state=42)

print(f"Chia dữ liệu thành {n_splits} tập train/test...")

for fold, (train_index, test_index) in enumerate(kf.split(X)):
    X_train_fold, X_test_fold = X.iloc[train_index], X.iloc[test_index]
    y_train_fold, y_test_fold = y.iloc[train_index], y.iloc[test_index]

    model = LinearRegression()
    model.fit(X_train_fold, y_train_fold)
    y_pred_fold = model.predict(X_test_fold)

    # Evaluate the model on the test set of the current fold
    r2_fold = r2_score(y_test_fold, y_pred_fold)
    mse_fold = mean_squared_error(y_test_fold, y_pred_fold)

    print(f"Fold {fold+1}: R2 = {r2_fold:.4f}, MSE = {mse_fold:.4f}")

    # Check if this is the best set based on R2
    if r2_fold > best_r2:
        best_r2 = r2_fold
        best_split = (fold, r2_fold, mse_fold)
        best_X_train = X_train_fold
        best_X_test = X_test_fold
        best_y_train = y_train_fold
        best_y_test = y_test_fold

print(f"\nTập train/test tốt nhất được tìm thấy ở Fold {best_split[0]+1}")

```

```

với R2 = {best_split[1]:.4f} và MSE = {best_split[2]:.4f}")

# Use the best train/test set for the next steps
X_train, X_test, y_train, y_test = best_X_train, best_X_test,
best_y_train, best_y_test

# Check the size of the best set
print("\nKích thước của tập train/test tốt nhất:")
print("Train:", X_train.shape, y_train.shape)
print("Test :", X_test.shape, y_test.shape)

```

Result of the process

Chia dữ liệu thành 10 tập train/test...

```

Fold 1: R2 = 0.1607, MSE = 0.4148
Fold 2: R2 = 0.3171, MSE = 0.8741
Fold 3: R2 = 0.2222, MSE = 0.9051
Fold 4: R2 = 0.3583, MSE = 0.5229
Fold 5: R2 = 0.3338, MSE = 1.2035
Fold 6: R2 = -0.5376, MSE = 0.2821
Fold 7: R2 = 0.3705, MSE = 0.5362
Fold 8: R2 = 0.3250, MSE = 0.8140
Fold 9: R2 = 0.3863, MSE = 0.7916
Fold 10: R2 = 0.4215, MSE = 0.4192

```

Tập train/test tốt nhất được tìm thấy ở Fold 10 với R2 = 0.4215 và MSE = 0.4192

Kích thước của tập train/test tốt nhất:

```

Train: (378, 8) (378,)
Test : (42, 8) (42,)

```

Source: Implementation by the author using Python

Linear Regression Model Training

```
# Linear Regression Model Training
# Model creation and training
model = LinearRegression()
model.fit(X_train, y_train)

# Prediction on the test set
y_pred = model.predict(X_test)

# Calculate evaluation indicators
mse = mean_squared_error(y_test, y_pred)
mae = mean_absolute_error(y_test, y_pred)
r2 = r2_score(y_test, y_pred)

print("Evaluating the Linear Regression Model:")
print(f"Mean Squared Error (MSE): {mse:.4f}")
print(f"Mean Absolute Error (MAE): {mae:.4f}")
print(f"R-squared (R2): {r2:.4f}")
```

Result of the process

```
Evaluating the Linear Regression Model:
Mean Squared Error (MSE): 0.4192
Mean Absolute Error (MAE): 0.4863
R-squared (R2): 0.4215
```

Source: Implementation by the author using Python

Applying Machine Learning to find the appropriate machine learning model

```
models = {
    "Linear Regression": LinearRegression(),
    "Random Forest": RandomForestRegressor(random_state=42),
    "Decision Tree": DecisionTreeRegressor(random_state=42),
    "XGBoost": XGBRegressor(random_state=42, verbosity=0),
    "SVR": SVR()
}
results = []
kf = KFold(n_splits=5, shuffle=True, random_state=42)
for name, model in models.items():
    # Huấn luyện mô hình trên tập train mới
    model.fit(X_train, y_train)
    # Dự đoán trên tập test mới
    y_pred = model.predict(X_test)
    mse = mean_squared_error(y_test, y_pred)
    mae = mean_absolute_error(y_test, y_pred)
    rmse = np.sqrt(mse)
    r2 = r2_score(y_test, y_pred)
    # Thực hiện Cross-validation trên tập train mới
    cv_r2 = cross_val_score(model, X_train, y_train, cv=kf, scoring='r2')
    results.append({
        "Model": name,
        "MAE": mae,
        "MSE": mse,
        "RMSE": rmse,
        "R2": r2,
        "CV R2 Mean": np.mean(cv_r2),
        "CV R2 Std": np.std(cv_r2)
    })
results_df = pd.DataFrame(results)
print(results_df.sort_values("R2", ascending=False))
```

Model	MAE	MSE	RMSE	R2	CV R2 Mean	\
SVR	0.327567	0.314629	0.560918	0.565866	0.401650	
Linear Regression	0.486305	0.419234	0.647483	0.421529	0.303452	
Random Forest	0.344577	0.432978	0.658011	0.402564	0.582343	
XGBoost	0.361688	0.477374	0.690923	0.341306	0.475350	
Decision Tree	0.373461	0.528353	0.726879	0.270963	0.480779	
CV R2 Std						
	0.060485					
	0.061044					
	0.054346					
	0.141741					
	0.090999					

Source: Implementation by the author using Python

Reprint the results of the best performing model based on CV R^2 Mean

```
results_df_sorted = results_df.sort_values("CV R2 Mean", ascending=False)

# Lấy dòng đầu tiên (mô hình có 'CV R2 Mean' cao nhất)
best_model_result = results_df_sorted.iloc[0]

print("\nKết quả của mô hình có hiệu suất tốt nhất dựa trên CV  $R^2$  Mean:")
best_model_result
```

1	
Model	Random Forest
MAE	0.344577
MSE	0.432978
RMSE	0.658011
R2	0.402564
CV R2 Mean	0.582343
CV R2 Std	0.054346

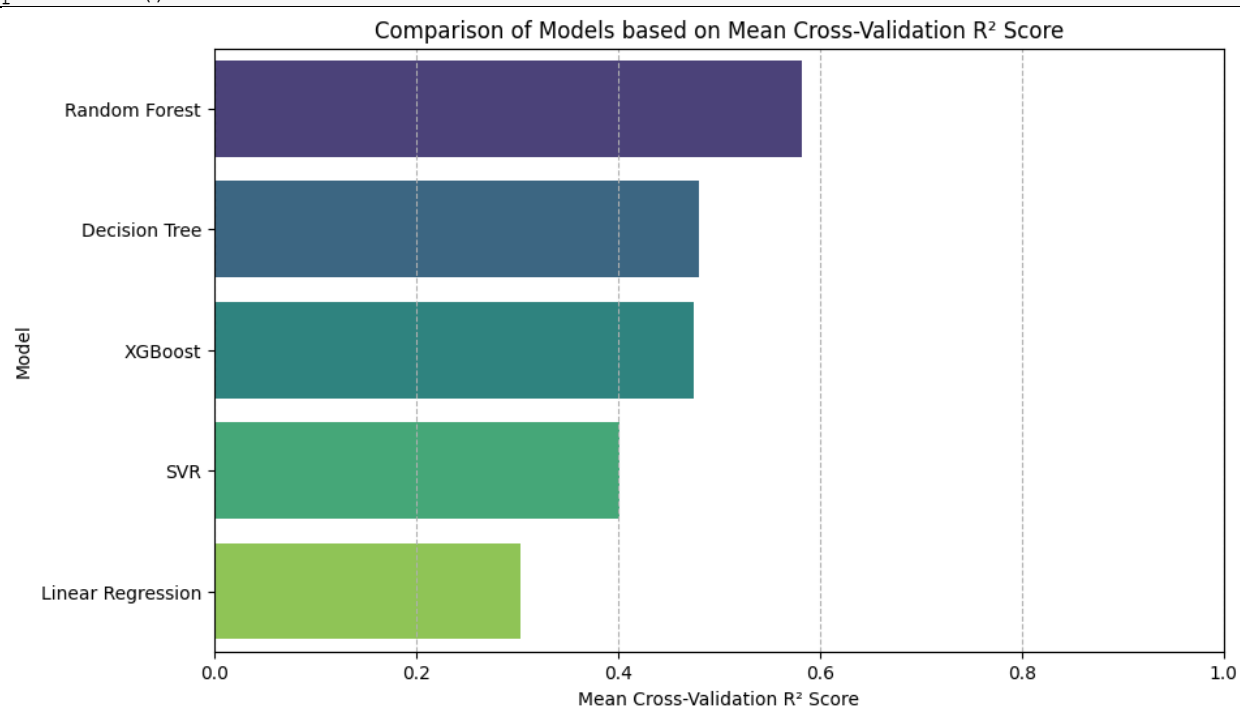
dtype: object

Source: Implementation by the author using Python

Visualize for comparison model based on based on CV R² Mean

```
models = results_df_sorted["Model"]
cv_r2_means = results_df_sorted["CV R2 Mean"]

# Create bar chart
plt.figure(figsize=(10, 6))
sns.barplot(x=cv_r2_means, y=models, palette='viridis')
plt.xlabel("Mean Cross-Validation R2 Score")
plt.ylabel("Model")
plt.title("Comparison of Models based on Mean Cross-Validation R2 Score")
plt.xlim([0, 1]) plt.grid(axis='x', linestyle='--')
plt.show()
```



Source: Implementation by the author using Python

Finding Feature Importance

```
best_model = RandomForestRegressor(random_state=42)
best_model.fit(X_train, y_train)

# Finding Feature Importance
importances = best_model.feature_importances_
importance_df = pd.DataFrame({
    "Feature variables": X.columns,
    "Feature Importance": importances
}).sort_values(by="Feature Importance", ascending=False)

print("=== Feature Importance (Random Forest - ROA) ===")
print(importance_df)
```

	Feature Variables	Fearture Importance
2	TDR	0.338935
6	GDP	0.331786
4	TANG	0.113326
3	SIZE	0.073070
5	LIQ	0.044980
1	LTDR	0.041891
0	STDR	0.035899
	7	INF 0.020114

Source: Implementation by the author using Python

Calculating SHAP Values

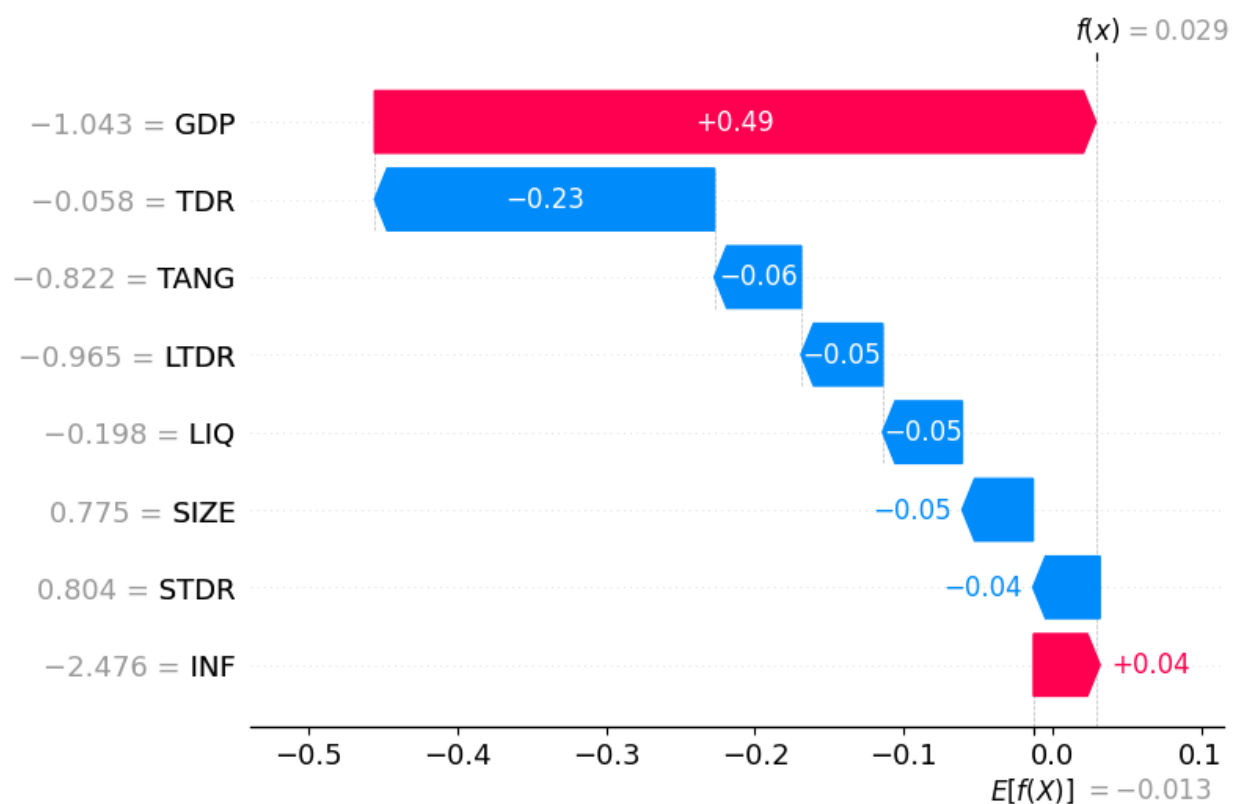
```

model = RandomForestRegressor(n_estimators=100, random_state=42)
model.fit(X_train, y_train)

explainer = shap.TreeExplainer(model)
shap_values = explainer.shap_values(X_test)

# Vẽ SHAP waterfall plot cho một quan sát
shap.plots.waterfall(
    shap.Explanation(
        values=shap_values[0],
        base_values=explainer.expected_value,
        data=X_test.iloc[0],
        feature_names=X_test.columns
    ),
    max_display=10
)

```



Source: Implementation by the author using Python